

EQUAL-MASS SUNSET INTEGRALS AND GENUS-ZERO LOCAL GROMOV–WITTEN THEORY OF CALABI–YAU n -FOLDS

by

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Abstract. — We prove an exact identity, as convergent logarithmic germs at large Euclidean momentum, between the equal-mass two-dimensional $(n-1)$ -loop sunset Feynman integral, with $n \geq 4$, and the genus-zero one-point descendant invariants of the local Calabi–Yau n -fold $X_n = \text{Tot}(K_{F_n})$, where $F_n \subset (\mathbb{P}^1)^n$ is a smooth $(1, \dots, 1)$ hypersurface. Writing s for the momentum variable and $y = -1/s$, the identity reads

$$I_{\odot}^{(n-1)}(s) = (-1)^{n-1} y A \left[n R^{n-1} - \theta_Q \mathcal{D}_0^{\Delta} R^{n-2} + \sum_{j=0}^{n-3} c_j R^j \right],$$

where $R = \log y + O(y)$ is the local mirror coordinate, $A = \theta_y R$, $Q = e^R$, $\mathcal{D}_0^{\Delta}(Q)$ is the primary one-point potential, and each c_j is an explicit linear combination of 1, the descendant series $\mathcal{D}_a^{\Delta}(Q)$ and their derivatives $\theta_Q \mathcal{D}_a^{\Delta}$, $0 \leq a \leq n-2$, with coefficients in the odd-zeta values $\mathbb{Q}[\zeta(3), \zeta(5), \dots]$ coming from the Gamma class. The descendant series are given explicitly by the ambient hypergeometric series and the inverse mirror map. We also prove global univalence of the principal mirror map, with explicit upper and lower bounds on the Q -Taylor radii, extract the full primary evaluation divisor and its integral cover transform, and reduce the ambient endpoint constants to a finite universal Gamma-class part and $\lfloor (n-1)/2 \rfloor$ real parameters.

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1. Introduction and statement of results

Feynman integrals are the basic building blocks of perturbative quantum field theory. They enter in the evaluation of scattering amplitudes and the comparison between the theoretical predictions and the experimental measurements in particle colliders (see [1–3] for instance), gravitational-waves observables [4–6] or in cosmology [7, 8]. Their analytic structure is rich and has been the subject of intense study in the last decades. It is now recognized that Feynman integrals are periods of algebraic varieties [9, 10], with physical observables expressed as relative period integrals [9–21].

Amongst the simplest Feynman integrals are the equal-mass two-dimensional $(n - 1)$ -loop sunset integrals, which are the subject of this paper. This family has a particular status since it is a natural seed of many quantum field theory computations, and it is directly connected to the geometry of Calabi–Yau n -folds [17, 22–25]. The two-loop case was studied in detail by Bloch–Kerr–Vanhove [17], who related the two-loop sunset integral to the genus-zero prepotential of a local Calabi–Yau threefold through a mirror/Legendre construction. In this paper we prove the corresponding one-point descendant statement for all $n \geq 4$.

For $n \geq 2$ the two-dimensional $(n - 1)$ -loop sunset integral with all masses equal to one has, for $\operatorname{Re} s < 0$, the convergent Schwinger representation

$$I_{\ominus}^{(n-1)}(s, 1) = \int_{\mathbb{R}_+^n} e^{-\sum_{i=1}^n x_i + sW(x)} W(x) \prod_{i=1}^n \frac{dx_i}{x_i}, \quad W(x) = \left(\sum_{i=1}^n x_i^{-1} \right)^{-1}, \quad (1.1)$$

with the momentum variable written s . This defines an analytic function on $\mathbb{C} \setminus [n^2, \infty)$. Set

$$\phi(x) = \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n x_i^{-1} \right), \quad (1.2)$$

a Laurent polynomial homogeneous of degree zero, hence a regular function on $(\mathbb{C}^*)^n / \mathbb{C}^* \cong (\mathbb{C}^*)^{n-1}$. $\phi(x)$ is the Landau–Ginzburg superpotential mirror to a smooth $(1, \dots, 1)$ hypersurface $F_n \subset (\mathbb{P}^1)^n$.

Since $W(x)\phi(x) = \sum_{i=1}^n x_i$, the exponent in (1.1) factors as

$$-\sum_{i=1}^n x_i + sW(x) = -W(x)(\phi(x) - s). \quad (1.3)$$

The projective integral representation in Proposition 9.1 has denominator $\phi(x) - s$, whose zero locus is the fiber $\{\phi(x) = s\}$. In [26, Section 2, equation (10)] Iritani equates that oscillatory integral with a $\hat{\Gamma}$ -pairing on F_n . Such an F_n is Fano with $K_{F_n} = \mathcal{O}_{F_n}(-H)$, and $X_n = \operatorname{Tot}(K_{F_n})$ is the local Calabi–Yau n -fold whose genus-zero invariants the theorem below produces. Any smooth member of the system may be taken as defined in Section 3.

Our main result, Theorem 10.1, expresses the amplitude through the genus-zero one-point descendant invariants of X_n . To state it, put $y = -1/s$ and $\theta_y = y d/dy$, and let

$$c_D = \sum_{|\mathbf{d}|=D} \binom{D}{d_1, \dots, d_n}^2, \quad R = \log y + \sum_{D \geq 1} \frac{(-1)^D c_D}{D} y^D, \quad A = \theta_y R, \quad Q = e^R$$

be the local mirror coordinate, its derivative and the Novikov variable (see eq. (6.7)). For $0 \leq a \leq n-2$ and a positive curve class β of degree $|\beta| = H \cdot \beta$, the invariants and their diagonal generating series are

$$N_{a,\beta} = \int_{[\overline{\mathcal{M}}_{0,1}(X_n, \beta)]^{\text{vir}}} \psi^a \text{ev}_{X_n}^* \varpi^* \delta_a, \quad \delta_a = \frac{H^{n-2-a}}{(n-2-a)!}, \quad \mathcal{D}_a^\Delta(Q) = \sum_{\beta > 0} N_{a,\beta} Q^{|\beta|},$$

where $\varpi: X_n \rightarrow F_n$ is the projection (see eq. (5.1)). Finally, the Gamma-class constants γ_k are defined by

$$\sum_{k \geq 0} \gamma_k t^k = \exp \left(\sum_{r \geq 1} \frac{2\zeta(2r+1)}{2r+1} t^{2r+1} \right),$$

so that $\gamma_0 = 1$, $\gamma_1 = \gamma_2 = 0$ and $\gamma_3 = \frac{2}{3}\zeta(3)$ (see eq. (9.7)).

Theorem A (Theorem 10.1). — *For every $n \geq 4$, as convergent logarithmic germs at $s = -\infty$,*

$$I_\ominus^{(n-1)}(s) = (-1)^{n-1} y A \sum_{k=0}^{n-1} (-1)^k \gamma_k \mathcal{T}_{n-1-k}(R, Q),$$

where $\theta_Q = Q d/dQ$ and

$$\mathcal{T}_j = \frac{n!}{j!} R^j - \sum_{a=0}^{j-2} \frac{(n-2-a)!}{(j-a-2)!} R^{j-a-2} \mathcal{D}_a^\Delta - \sum_{a=0}^{j-1} \frac{(n-2-a)!}{(j-a-1)!} R^{j-a-1} \theta_Q \mathcal{D}_a^\Delta.$$

In particular,

$$I_\ominus^{(n-1)}(s) = (-1)^{n-1} y A \left[n R^{n-1} - \theta_Q \mathcal{D}_0^\Delta R^{n-2} - ((n-2) \mathcal{D}_0^\Delta + \theta_Q \mathcal{D}_1^\Delta) R^{n-3} + \dots \right],$$

where the omitted terms have degree at most $n-4$ in R ; the odd zeta values first enter there, through the term $-\gamma_3 \mathcal{T}_{n-4}$.

The coefficient of R^{n-2} is thus $-\theta_Q$ applied to the primary one-point potential $\mathcal{D}_0^\Delta$, which for $n \geq 4$ plays the role that the prepotential of a local Calabi–Yau threefold plays at two loops [17]. The descendant series are themselves given in closed form. In the square-free algebra $\mathbb{Q}[p_1, \dots, p_n]/(p_1^2, \dots, p_n^2)$, with $H = p_1 + \dots + p_n$, put

$$I(y, z) = e^{H \log y/z} \sum_{\mathbf{d} \in \mathbb{N}^n} (-y)^{|\mathbf{d}|} \frac{\prod_{r=0}^{|\mathbf{d}|-1} (H + rz) \prod_{r=1}^{|\mathbf{d}|} (H + rz)}{\prod_{i=1}^n \prod_{r=1}^{d_i} (p_i + rz)^2}.$$

Then $I(y, z) = 1 + R(y)H/z + O(z^{-2})$. The sum is symmetric in the p_i , and symmetric elements of degree j in this algebra are multiples of H^j ; hence $e^{-HR/z}I(y, z) = 1 + \sum_{j=2}^n K_j(y)H^j/z^j$ for convergent power series K_j .

Theorem B (Theorem 6.9). — *The mirror map $y \mapsto Q = e^{R(y)} = y + O(y^2)$ is invertible near $y = 0$, and for $0 \leq a \leq n - 2$,*

$$\mathcal{D}_a^\Delta(Q) = -\frac{n!}{(n-2-a)!} K_{a+2}(y(Q)).$$

Corollary 11.2 continues the identity of Theorem A along the whole negative axis. Corollaries 10.2 and 10.3 specialize it to three and four loops; these are the Calabi–Yau fourfold and fivefold analogues of the two-loop theorem of [17], and they are the subject of the companion paper [27].

The proof is uniform in n and uses four independent parts.

1. *Geometry.* Stable maps to X_n factor through the zero section over an arbitrary base (Theorem 5.4), and the resulting virtual class carries the excess factor $c_{m-1}(\mathcal{V})$ with the obstruction morphism identified, not merely its K -class (Theorem 5.5).
2. *Mirror symmetry.* A hypergeometric modification with two independent characteristic classes, one Euler and one inverse Euler, puts the series (6.4) on the twisted Givental cone (Theorem 6.6); monic division by $\lambda - H$ and the ambient pairing then extract all of $N_{a,\beta}$ at once (Theorem 6.9). Theorem 15.1, proved from the universal hypersurface rather than from mirror symmetry, upgrades the comparison against ambient insertions to the full one-point cohomology classes, which matters when n is odd.
3. *Differential equations.* Definition 8.1 is a finite algorithm producing an order- $(n-1)$ operator P_{n-1} over $\mathbb{Q}$, and Theorem 8.2 certifies it against the complete ambient series, degree by degree in the Novikov variable $Q = \exp(R)$, with the nonzero source $H^{n-1}y^H$. Proposition 15.2 establishes that P_{n-1} is irreducible, and hence the minimal operator for the period A ; the proof uses the irreducibility of its symmetric-power seed and two Fourier transforms with checked extensions at zero.
4. *Analysis and arithmetic.* A positive Laplace transform converts the oscillatory Gamma identity of [26, Section 2, equation (10)] into the signed formula (9.8), whose degree factor $(-1)^D$ is derived in (9.11). Separately, a smooth compact Calabi–Yau double cover (Theorem 14.3) isolates the local sector on a target with trivial canonical bundle, which makes the published integrality statements applicable and gives $b_\beta \in \mathbb{Z}$ for every $n \geq 4$ (Corollary 14.10).

For every loop order, Theorem 4.1 of [28] establishes, by a Mellin–Barnes residue computation rather than by any geometric input, an exact convergent expansion

of the equal-mass sunset integral having the same shape as (10.2): the holomorphic period times the sum of a leading power of the flat coordinate (since $\log y = -\log s + i\pi$ and its holomorphic period agrees with Π_{n-1} to leading order at large Euclidean momentum; see Section 16 for the normalization dictionary) and an instanton series $\sum_{l \geq 1} e^{lR} \sum_r d_r(l) R^r$ with rational $d_r(l)$, plus a boundary combination $\sum_r \alpha_r \text{Frob}_r$ whose coefficients lie in $\mathbb{Q}[\zeta(3), \zeta(5), \dots]$ and are given there by an explicit generating function.

Theorem 10.1 supplies an enumerative identity for those coefficients, for every $n \geq 4$.

- The instanton coefficients $d_r(l)$, which [28] obtains as rational numbers from a residue calculation, are here the genus-zero one-point descendant invariants $\mathcal{D}_a^\Delta$ of the local Calabi–Yau n -fold $X_n = \text{Tot}(K_{F_n})$, with the descendant exponent a and the insertion δ_a specified. At two loops this identification is the theorem of [17]; for $n \geq 4$ it was not available.
- The boundary coefficients α_r are here derived from the $\widehat{\Gamma}$ -class of F_n rather than read off a Mellin residue. This is the point left open in [28, Remark 4.5], which records that the zeta pattern is consistent with a $\widehat{\Gamma}$ -class prediction and states that establishing it rigorously beyond two loops would extend the argument of [26] and is left for future work.
- The arithmetic statements have no counterpart on the analytic side. The coefficients produced by a residue computation are rational numbers with no predicted denominators; Corollary 14.10 and Theorem 14.9 prove integrality of the transformed primaries and express them through quantum K -theory.

The leading term alone has a longer history. It was observed numerically in [22] and [23], where it was matched against the modified $\widehat{\Gamma}$ -class of the degree- $(1, \dots, 1)$ Fano hypersurface, and proved in [26]; see also [29]. The present Theorem 10.1 determines every coefficient below the leading one, which is where the descendants enter. Those two sources also supply the Calabi–Yau operator framework in which (9.9) sits; no priority is claimed for the operators themselves, as recorded after Example 8.5.

Corollary 14.10 and Theorem 14.9 concern one specific observable, namely the highest nonconstant logarithmic coefficient of the expansion, transformed by the multiple-cover rule with weight r^{-2} for a single primary insertion. That weight is the one-marked specialization of $r^{\ell-3}$ for ℓ insertions, and it is not the r^{-3} of the unmarked genus-zero prepotential familiar from the Calabi–Yau threefold case used in [17]. The exponent is fixed by the number of markings, not by the codimension of the insertion or the dimension of the target. The two conventions are compared in the companion paper [27]. Three things are deliberately not asserted anywhere in this paper, and the reader should not supply them.

First, the resulting integers are not claimed to be unsigned counts of embedded rational curves, nor multiplicities of a specified sheaf-theoretic moduli problem, nor physical BPS-state degeneracies. They are the output of a fixed arithmetic transform applied to a fixed virtual invariant. Second, no multiple-cover law is asserted for the descendant series $\mathcal{D}_a^\Delta$ with $a > 0$; only the primary $a = 0$ sector is transformed, and the rational descendant description of the full integral is proved independently of any such law. Third, the ideal-geometry hypotheses under which higher-dimensional covering behaviour is usually discussed — unobstructedness of simple rational curves, and the moving-cover conditions of [30, Sections 0.5 and 1.1] — are not hypotheses here, and indeed fail for X_n , because a simple rational map $u : \mathbb{P}^1 \rightarrow F_n$ of H -degree $m \geq 2$ has $h^1(\mathbb{P}^1, u^*T_{X_n}) \geq m - 1 > 0$. The proofs below are algebraic and do not need them. This is discussed again where it is used, at the end of Section 14.

This paper is organised as follows. Section 3 fixes conventions and Section 4 the ambient state space. Sections 5–7 contain the geometric and mirror-theoretic input, i.e. the zero-section factorization and virtual reduction, the twisted mirror theorem and the extraction of all descendants, and the quantum-Serre coefficient identity. Section 8 gives the operator algorithm and its all-degree certificate. Section 9 states the single analytic input and derives the signed Gamma formula, and Section 10 assembles Theorem 10.1. Sections 11–13 give the real continuation, the global univalence of the mirror map with the resulting Q -Taylor radius, and the low-degree geometry, and Section 14 the compact Calabi–Yau cover and the primary integrality statements. Sections 15–17 record the ambientness and minimality statements, a specialization guide, the normalization dictionary, and the independent numerical verifications. Sections 18 and 19 then extract the full primary evaluation divisor and reduce the ambient endpoint constants. Appendix A implements the operator algorithm, Appendix B derives the string and divisor identities used with our twists, Appendix C proves ambientness of the one-point evaluation classes, and Appendix D supplies the extension and Fourier-transform lemmas used in the minimality argument.

Acknowledgements and use of AI

This work started as an application of the AI co-mathematician tool [31] developed by Google DeepMind, in which a human works with AI agents to develop a theory. Later, the output of the AI co-mathematician tool was also reviewed by different AI tools (first Gemini, later also Claude and ChatGPT) and the reviews were used to improve the theorems, sometimes in further collaboration with the AI co-mathematician.

The final paper is human written, using parts of the AI generated proofs, and also proofread by AI.

The work of HM was done at Google DeepMind.

The work of PV was funded by the Agence Nationale de la Recherche (ANR) under grant Observables (ANR-24-CE31-7996).

2. Index of notation

The main symbols, grouped by topic. The last column gives the equation, result or section where each is defined.

Symbol	Meaning	Defined
Variables and geometry		
n	Integer $n \geq 4$; the sunset has $n - 1$ loops, and $n = \dim_{\mathbb{C}} X_n$	§3
s, y, T	Momentum s , the variable $y = -1/s$, and the Euclidean momentum $T = -s = 1/y$	§1, Thm. 9.7
θ_y, θ_Q	Euler operators $y d/dy$ and $Q d/dQ$; $\theta_y = A \theta_Q$	§8, Thm. 10.1
Y_n, p_i, H	$Y_n = (\mathbb{P}^1)^n$, the hyperplane classes of its factors ($p_i^2 = 0$), and $H = \sum_i p_i$	§3
F_n	Smooth $(1, \dots, 1)$ hypersurface in Y_n ; Fano, with $K_{F_n} = \mathcal{O}_{F_n}(-H)$	(3.1), Prop. 4.1
X_n, ϖ	Local Calabi–Yau n -fold $\text{Tot}(K_{F_n})$ and its projection to F_n	§3
$\beta, m = \beta $	Curve class in $H_2(F_n, \mathbb{Z}) \cong \mathbb{Z}^n$ and its degree $H \cdot \beta$	Prop. 4.1
δ_a	Insertion $H^{n-2-a}/(n-2-a)!$ for the descendant exponent $0 \leq a \leq n-2$	Prop. 4.1
$N_{a,\beta}$	One-point descendant invariant	(5.1)
$\mathcal{D}_a^\Delta(Q)$	$\int_{[\overline{\mathcal{M}}_{0,1}(X_n, \beta)]^{\text{vir}}} \psi^a \text{ev}_{X_n}^* \varpi^* \delta_a$ Diagonal series $\sum_{\beta > 0} N_{a,\beta} Q^{ \beta }$; $\mathcal{D}_0^\Delta$ is the primary one-point potential	(5.1)
Mirror side		
R, A, c_D	Local mirror coordinate $R = \log y + \sum_{D \geq 1} (-1)^D c_D y^D / D$ and period $A = \theta_y R$, where $c_D = \sum_{ \mathbf{d} =D} \binom{D}{d_1, \dots, d_n}^2$	(6.7)
$Q, y(Q)$	Novikov variable $Q = e^R = y + O(y^2)$ and the inverse mirror map	(6.7), Thm. 6.9
$I(y, z)$	Diagonal ambient I -series $I_{X_n}^{0, \text{amb}}(y, z)$, equal to $1 + RH/z + O(z^{-2})$	§1, (6.4)

Symbol	Meaning	Defined
$K_j(y)$	$e^{-HR/z}I(y, z) = 1 + \sum_{j=2}^n K_j H^j / z^j$; Theorem B: $\mathcal{D}_a^\Delta = -\frac{n!}{(n-2-a)!} K_{a+2}(y(Q))$	(6.8), (6.9)
$\mathcal{I}^\sharp, W_j$	Signed ambient series and its coefficients, $\mathcal{I}^\sharp(y, 1) = \sum_j H^j W_j(y)$, with $W_0 = A$	(7.1)
P_{n-1}	Sunset operator of order $n-1$, with $P_{n-1}\mathcal{I}^\sharp(y, 1) = H^{n-1}y^H$; the minimal operator of A	(8.1), Thm. 8.2

The sunset integral

$I_\ominus^{(n-1)}(s)$	Equal-mass two-dimensional $(n-1)$ -loop sunset integral	(1.1)
ϕ	Landau–Ginzburg superpotential $(\sum_i x_i)(\sum_i x_i^{-1})$, mirror to F_n	(1.2)
$\widehat{\Gamma}_{X_n}, \gamma_k$	Gamma class, with $\widehat{\Gamma}_{X_n} _{F_n} = \sum_k \gamma_k H^k$ and $\sum_k \gamma_k t^k = \exp(\sum_{r \geq 1} \frac{2\zeta(2r+1)}{2r+1} t^{2r+1})$	(9.7)
$\Pi_{n-1}(s)$	Period prefactor $(-1)^{n-1}yA(y)$	§10
$\mathcal{T}_j(R, Q)$	Assembly polynomials; Theorem A: $I_\ominus^{(n-1)} = \Pi_{n-1} \sum_k (-1)^k \gamma_k \mathcal{T}_{n-1-k}$	(10.1), (10.2)
$h_j(Q)$	Coefficients of the zeta-free bracket $\mathcal{T}_{n-1} = nR^{n-1} + \sum_{j=0}^{n-2} h_j R^j$	(10.4)

Arithmetic and global results

b_β, n_m	Primary cover transform, $N_{0,\beta} = \sum_{k\alpha=\beta} b_\alpha/k^2$, and $n_m = \sum_{ \beta =m} b_\beta$, so that $\mathcal{D}_0^\Delta = \sum_m n_m \text{Li}_2(Q^m)$; both are integers	Thm. 14.6, Cor. 14.10
$\widetilde{Z}$	Smooth Calabi–Yau double cover of $Z = \mathbb{P}(\mathcal{O}_{F_n} \oplus K_{F_n})$ that isolates the local sector	Thm. 14.3
r_n, ρ_n	Bounds $r_n \leq R_y \leq \rho_n$ on the Taylor radius R_y of $y(Q)$; the $\mathcal{D}_a^\Delta$ converge on $ Q < r_n$	(11.3), (11.6)
D_β, B_β	Full primary evaluation class $\sum_i d_{\beta,i} p_i \in H^2(F_n, \mathbb{Q})$ and its integral cover transform $\sum_{r \beta} \mu(r) r^{-2} D_{\beta/r}$, with μ the Möbius function	§18
$G(H), b_j$	Universal Gamma factor $\sum_k \gamma_k H^k$ and the $\lfloor (n-1)/2 \rfloor$ real parameters of the endpoint constants	(19.1), Thm. 19.1

3. Scope and conventions

Throughout, work over $\mathbb{C}$, with $n \geq 4$. We write

$$Y_n = (\mathbb{P}^1)^n, \quad H = \sum_{i=1}^n p_i, \quad F_n \in |\mathcal{O}_{Y_n}(H)| \text{ smooth}, \quad X_n = \text{Tot}(K_{F_n}).$$

Thus F_n is a smooth multilinear hypersurface given by the zero locus in $(\mathbb{P}^1)^n$, with homogeneous coordinates $[u_i : v_i]$, of a section

$$\sum_{\epsilon \in \{0,1\}^n} a_\epsilon \prod_{i=1}^n u_i^{\epsilon_i} v_i^{1-\epsilon_i} \in H^0(Y_n, \mathcal{O}_{Y_n}(1, \dots, 1)) \quad (3.1)$$

whose vanishing locus is smooth, which is the generic member of the system. No individual member is singled out because all smooth members are deformation equivalent, so the genus-zero invariants used below do not depend on the choice, and none of the arguments refers to the coefficients a_ϵ . By adjunction $K_{Y_n} = \mathcal{O}_{Y_n}(-2, \dots, -2)$ gives $K_{F_n} = \mathcal{O}_{F_n}(-H)$, so F_n is Fano of index one with respect to H and $X_n = \text{Tot}(K_{F_n})$ is a local Calabi–Yau n -fold; this is Proposition 4.1. The connection with the Feynman integral is through the Landau–Ginzburg mirror of F_n rather than through any particular defining equation (3.1); see (1.3) and Section 9.

Write $i : F_n \hookrightarrow Y_n$, $\varpi : X_n \rightarrow F_n$, and $\iota : F_n \hookrightarrow X_n$ for the inclusions and projection. The symbol H also denotes its restriction when the target is clear. Cohomological degrees are real degrees $p_i \in H^2(Y_n, \mathbb{Z})$. Thus H^j has complex codimension j . The foundational stable-map statements are [32, Theorem 3.14] and [33, Propositions 2, 4, and 5]. The actual obstruction morphism is the unnumbered construction following Proposition 4 of [33], and Proposition 5 supplies global resolutions. All virtual classes are the standard stable-map virtual classes, in rational Chow groups or their cohomological realizations; integration on a proper Deligne–Mumford stack is not replaced by integration on its coarse space.

We use ℓ for the number of marked points, reserving n for the loop order plus one. A curve class is denoted β , and $m = |\beta| = H \cdot \beta$. The letter a denotes a descendant exponent. Only positive curve classes are included in local numerical potentials. The equivariant parameter λ is the first Chern class of the weight-one character acting on the fibers of K_{F_n} ; μ is a separate weight used for the positive hypersurface twist.

Notation. A superscript on H carrying a space as argument, as in $H^k(F_n, \mathbb{Q})$, denotes a cohomology group; a bare power H^k denotes the k -fold cup product of the divisor class H . The letters $E_\pm$ always denote the index bundles of Lemma 5.3 on a stable-map stack, while $\rho_\pm = \pm H$ and $\mathcal{B}_\pm = \mathcal{O}_{Y_n}(\pm H)$ denote the corresponding cohomology class and line bundle on Y_n . The operators T_k of Definition 8.1 carry a subscript and are unrelated to the tangent bundles $T_{F_n}, T_{Y_n}, T_{X_n}$, to the assembly polynomials $\mathcal{T}_j$ of (10.1), or to the positive variable $T = -s$ of Section 9. Similarly

P_{n-1} is the differential operator and $\bar{p}$ is a sum of passive divisor classes. Where an external theorem is quoted in full, it is stated at the point of use, in the notation of its source where this causes no conflict, and the paragraph following it fixes the translation.

Standing caveats. The following apply throughout and are not repeated at every occurrence. Rational numbers produced by a multiple-cover transform are not asserted to count curves, sheaves, or BPS states. Integrality, where proved, is an arithmetic statement about the specified observable. No multiple-cover law is asserted for the descendants $\psi^a \delta_a$ with $a > 0$. No finite computation of initial coefficients is used as a proof of an all-degree statement; the finite checks collected in Section 17 verify statements that are proved elsewhere in the paper.

Genus, dimension, and the primary observable. The loop number $n - 1$ changes the target dimension, not the genus of the stable maps, which is zero throughout the numerical theory considered here. The subscript 0 in $N_{0,\beta}$ denotes descendant exponent zero: it does not mean that there are no marked points. The primary invariant has one marking and the integral insertion δ_0 of complex codimension $n - 2$. Its r^{-2} cover transform and integrality are statements about this specified observable. They are separate from any multiple-cover formula for descendants or a sheaf-theoretic interpretation of the resulting integers. The comparison with the unmarked r^{-3} Calabi–Yau threefold prepotential is carried out in the companion paper [27].

4. Geometry and the ambient state space

Proposition 4.1. — *One has*

$$K_{F_n} = \mathcal{O}_{F_n}(-H), \quad K_{X_n} \cong \mathcal{O}_{X_n}, \quad \int_F H^{n-1} = n!, \quad H_2(F_n, \mathbb{Z}) \cong \mathbb{Z}^n.$$

Under the last isomorphism β corresponds to $(p_1 \cdot \beta, \dots, p_n \cdot \beta)$. Every effective class has nonnegative coordinates. In particular, at most $\binom{m+n-1}{n-1}$ effective classes have H -degree m . Moreover

$$\delta_a = \frac{H^{n-2-a}}{(n-2-a)!} = \sum_{|S|=n-2-a} \prod_{i \in S} p_i \in H^{2(n-2-a)}(F_n, \mathbb{Z}), \quad 0 \leq a \leq n-2.$$

Proof. — Adjunction gives $K_{F_n} = (-2H + H)|_{F_n} = -H$. For a line bundle E on a smooth base, the vertical tangent bundle on $\text{Tot}(E)$ is the pullback of E , so $K_{\text{Tot}(E)} = \varpi^*(K_{F_n} \otimes E^{-1})$; take $E = K_{F_n}$. Also $i_*[F_n] = H$, and $H^n = n!p_1 \cdots p_n$ on Y_n . The Segre embedding realizes $\mathcal{O}_{Y_n}(H)$ as a hyperplane bundle. Apply the integral homology statement of [34, Corollary 7.3] to this smooth projective n -fold and its smooth hyperplane section. Since $2 < n - 1$, it gives $H_2(F_n, \mathbb{Z}) = H_2(Y_n, \mathbb{Z}) = \mathbb{Z}^n$. The same theorem in degree zero makes F_n connected. Since

it is smooth, it is irreducible and reduced. Each p_i is nef and H is ample; hence effective nonzero classes have nonnegative coordinates with positive sum. Counting all nonnegative tuples gives the stated upper bound, without assuming that every tuple is represented by an irreducible rational curve. Finally, expand H^r using $p_i^2 = 0$, then each square-free monomial occurs $r!$ times. $\square$

Set

$$\mathcal{R} = H^*(Y_n, \mathbb{Q}) = \mathbb{Q}[p_1, \dots, p_n]/(p_1^2, \dots, p_n^2), \quad V_{\text{amb}} = i^*H^*(Y_n, \mathbb{Q}) \subset H^{\text{even}}(F_n, \mathbb{Q}).$$

Proposition 4.2 (Ambient pairings and diagonal symmetry)

The ordinary Poincaré pairing on V_{amb} is nondegenerate, and

$$\ker i^* = \ker(H \cup - : \mathcal{R} \rightarrow \mathcal{R}), \quad \mathcal{R}^{S_n} = \mathbb{Q}[H]/(H^{n+1}).$$

The radical of $\int_Y abH/(\lambda - H)$ over $\mathbb{Q}(\lambda)$ is also $\ker i^$. If n is even, all even cohomology of F_n is ambient. If n is odd, the only possible nonambient even group is $H^{n-1}(F_n, \mathbb{Q})$.*

Proof. — [34, Corollary 7.3], followed by [35, Theorem 3.2] over $\mathbb{Q}$, identifies cohomology below degree $n-1$ and is injective in degree $n-1$. Poincaré duality applies to the closed oriented real $(2n-2)$ -manifold $F_n(\mathbb{C})$ [35, Theorem 3.30]. For a degree $k > n-1$, the pairing of $i^*H^k(Y_n)$ with $H^{2n-2-k}(F_n) = i^*H^{2n-2-k}(Y_n)$ is represented on Y_n by multiplication by H . Lemma 4.3 below says this multiplication has the required full rank, so i^* is surjective in degree k . In degree $n-1$ the bilinear form $\int_Y abH$ on $H^{n-1}(Y_n)$ is nondegenerate because $H : H^{n-1}(Y_n) \rightarrow H^{n+1}(Y_n)$ is an isomorphism by that lemma ($\dim_{\mathbb{C}} Y_n = n$). These facts prove nondegeneracy on the ambient image. Since $i_*i^*a = Ha$, one inclusion of kernels is immediate; for the other, if $Ha = 0$, then i^*a pairs to zero with every ambient class and is therefore zero. Multiplication by $(\lambda - H)^{-1}$ is an automorphism of the ambient algebra, so does not change the radical. A square-free monomial basis of $\mathcal{R}$ is indexed by subsets of $\{1, \dots, n\}$. An invariant element has constant coefficients on subsets of each fixed size, whose sum is $H^j/j!$. This proves the invariant-ring assertion. Finally the only cohomology possibly not inherited from Y_n is the middle group, by the preceding Lefschetz and duality arguments. $\square$

We state a required Lefschetz linear algebra

Lemma 4.3. — *In $\mathcal{R}$, multiplication by H from square-free degree r to degree $r+1$ is injective for $2r < n$ and surjective for $2r \geq n-1$.*

Proof. — Give the monomial basis p_S the orthonormal positive-definite inner product over $\mathbb{R}$. Let U multiply by H and put $Dp_S = \sum_{i \in S} p_{S \setminus \{i\}}$. Then $D = U^*$ and direct expansion gives $DU - UD = (n-2r)\text{id}$ in degree r . If $Uv = 0$ and $2r < n$, then $0 = \|Dv\|^2 + (n-2r)\|v\|^2$, so $v = 0$. The complement involution $p_S \mapsto p_{S^c}$ interchanges U and D . Injectivity in complementary degree therefore gives surjectivity in the stated range. All maps are rational, so the rank conclusions hold over $\mathbb{Q}$. In particular the middle map is an isomorphism when its two graded pieces have the same dimension. $\square$

Remark 4.4 (The nonambient middle summand at $n = 5$)

For $n = 5$, $\chi(F_n) = 40$, $b_2 = b_6 = 5$, and $b_4 = 28$, whereas the ambient part of $H^4(F_n, \mathbb{Q})$ has dimension $\binom{5}{2} = 10$. Thus the nonambient middle summand has dimension 18. These numbers follow from

$$\chi(F_n) = n! \sum_{k=0}^{n-1} \frac{(-1)^{n-1-k} 2^k}{k!},$$

obtained by integrating the degree- $(n-1)$ part of $c(TF) = \prod_{i=1}^5 (1 + 2p_i)/(1 + H)$; the remaining Betti numbers are forced by Lefschetz and Poincaré duality. The full Lefschetz-primitive middle group has dimension $28 - 5 = 23$; it includes ambient primitive classes and is distinct from the 18-dimensional orthogonal nonambient summand. This 18-dimensional summand is the reason the ambientness statement of Theorem 15.1 is needed when n is odd. The four solutions of the differential operator P_4 given in eq. (8.5) correspond to the ambient classes $1, H, H^2, H^3$, and nothing in the operator sees the orthogonal complement.

Remark 4.5 (Relation to the Hulek–Verrill threefolds)

The root lattice A_4 enters this story, but on the mirror side rather than here. The superpotential of (1.2) is $\phi(x) = n + \sum_{i \neq j} x_i x_j^{-1}$, so its Newton polytope is the convex hull of the roots $e_i - e_j$, that is the reflexive polytope $\Delta_{A_{n-1}}$. At $n = 5$ the toric closures of the fibers $\{\phi = s\}$, with the small resolutions described below and at nonsingular parameters, give the Calabi–Yau threefolds of Hulek–Verrill [25], whose equation $(\sum_{i=1}^5 X_i)(\sum_{i=1}^5 a_i/X_i) = a_6$ is the equal-mass member $a_i = 1$, $a_6 = s$ of their five-parameter family. By eq. (1.3) this is the graph hypersurface of the four-loop sunset at momentum s .

The Fano fourfold F_n corresponds to the Landau–Ginzburg model $((\mathbb{C}^*)^5/\mathbb{C}^*, \phi)$, whereas a compactified fiber has dimension three. The Calabi–Yau counterpart of the latter is an anticanonical hypersurface in F_n , equivalently a complete intersection of two $(1, \dots, 1)$ hypersurfaces in $(\mathbb{P}^1)^5$; see [26, Remarks 3 and 9]. The numbers $h^{11} = 26$, $h^{21} = 16$ of [25, Proposition 3.1] belong to the general smooth member of the full twenty-parameter family, not to the sunset subfamily. For a general member of the five-parameter family, small resolution of its thirty nodes gives $h^{11} = 45$, $h^{12} = 5$. These smooth-family statements must be restricted to nonsingular parameters; in particular they do not apply without qualification at $s = 0, 1, 9, 25$. The period-generated subvariation is the one generated by the holomorphic residue form and its Gauss–Manin derivatives. Under the natural coordinate permutation action this form transforms by the sign character. The logarithmic volume form on $(\mathbb{C}^*)^n/\mathbb{C}^*$ transforms by the determinant of the standard permutation quotient, namely $\text{sgn}(\sigma)$, and ϕ is invariant. Thus this is not literally the invariant subvariation unless the action has first been twisted by the sign character.

5. Stable maps in families and virtual reduction

Lemma 5.1 (Convexity and concavity on a tree)

Let C be a connected nodal curve of arithmetic genus zero and $f : C \rightarrow F_n$ or $f : C \rightarrow Y_n$ a nonconstant map of total H -degree m . Then

$$H^0(C, f^*\mathcal{O}(-H)) = 0, \quad h^1(C, f^*\mathcal{O}(-H)) = m - 1, \quad H^1(C, f^*\mathcal{O}(H)) = 0.$$

The last line bundle is globally generated on C and has $h^0 = m + 1$. Its evaluation at any smooth marked point is surjective.

Proof. — Every irreducible component is $\mathbb{P}^1$ and the dual graph is a tree. For the negative bundle a section vanishes on each noncontracted component. On a contracted component it is constant, so matching values at nodes and connectedness propagate vanishing to all components. For the positive bundle all component degrees are nonnegative. Inductively remove a leaf. Sections on the remaining tree and the leaf can be matched at their common node since evaluation on the leaf is surjective. The normalization exact sequence then gives $H^1 = 0$; the same argument, with a prescribed value at a chosen point, gives global generation. Riemann–Roch on a genus-zero nodal curve gives $\chi(M) = \deg M + 1$, which proves both ranks. $\square$

Lemma 5.2 (Artinian vanishing without a simultaneous normalization)

Let A be a local Artinian $\mathbb{C}$ -algebra with residue field $\mathbb{C}$, $C_A \rightarrow \operatorname{Spec} A$ a flat proper family, and $\mathcal{L}$ an invertible sheaf with $H^0(C_0, \mathcal{L}_0) = 0$. For every finite A -module J , $H^0(C_A, \mathcal{L} \otimes_A J) = 0$.

Proof. — Choose a composition series for J with successive quotients $\mathbb{C}$. Flatness of $\mathcal{L}$ over A makes the induced sequences of sheaves exact, and their successive quotients are the pushforward of $\mathcal{L}_0$ from the closed fiber. Left exactness of H^0 proves vanishing by induction. Lemma 5.1 supplies the hypothesis for the negative target line bundle. For a square-zero extension $A' \twoheadrightarrow A$, the difference of two lifted fiber coordinates lies in this group with $J = \ker(A' \rightarrow A)$, so it is zero. Normalization is needed only on the closed fiber. A smoothing such as $xy = \epsilon$ over $\mathbb{C}[\epsilon]/(\epsilon^2)$ has no section reducing to the node and need not admit a componentwise normalization over its base; the argument applies to it without such an assumption. $\square$

Lemma 5.3 (Universal base change). — For a flat proper finitely presented family $\pi : C \rightarrow S$ of such curves over an arbitrary $\mathbb{C}$ -scheme, with a map f whose fibers have fixed positive degree m , the complexes associated to $M_{\pm} = f^*\mathcal{O}(\pm H)$ satisfy

$$R\pi_* M_- \simeq E_-[-1], \quad R\pi_* M_+ \simeq E_+,$$

where E_-, E_+ are locally free of ranks $m - 1, m + 1$ and commute with arbitrary base change. For a marked section there is an exact sequence

$$0 \longrightarrow E'_+ \longrightarrow E_+ \longrightarrow \operatorname{ev}^* \mathcal{O}(H) \longrightarrow 0, \quad \operatorname{rk} E'_+ = m.$$

Proof. — Cohomology and base change for a perfect object and a finitely presented flat proper-support complex, in [36, Tag 0A1H], applies with $E = M_{\pm}$ and $\mathcal{G}^{\bullet} = \mathcal{O}_C$. Thus $R\pi_* M_{\pm}$ is perfect and commutes with arbitrary base change. Its fibers have amplitude $[0, 1]$, so locally it can be represented by a two-term complex of finite locally free modules $N^0 \rightarrow N^1$. For M_- the fiber maps are injective by Lemma 5.1; an invertible maximal minor makes the map locally split injective. Its cokernel is locally free. For M_+ the fiber maps are surjective and are locally split surjective, giving a locally free kernel instead. This proves the derived assertions, including on nonreduced bases. The marked evaluation map is fiberwise surjective, hence locally split surjective, and its kernel has the indicated rank. The argument applies on atlases of the stable-map stacks and descends canonically. $\square$

Theorem 5.4 (Zero-section factorization over arbitrary bases)

For every $\beta \neq 0$ and every number $\ell \geq 0$ of markings,

$$\overline{\mathcal{M}}_{0,\ell}(F_n, \beta) \xrightarrow{\sim} \overline{\mathcal{M}}_{0,\ell}(X_n, \beta)$$

is an isomorphism of stacks, compatible with arbitrary base change. In particular the local stable-map stack is proper and Deligne–Mumford.

Proof. — A map $g : C \rightarrow \text{Tot}(K_{F_n})$ over a base S consists of $f = \varpi g : C \rightarrow F_n$ and a section of $f^* K_{F_n}$. By Lemma 5.3, $\pi_* f^* K_{F_n} = 0$, so this section is zero over S , not merely on its geometric fibers. Conversely composing a map to F_n with ι gives a map to X_n . These constructions are inverse and functorial. They preserve stability because exactly the same components are contracted; automorphism groups agree because ι is a monomorphism. Properness and the Deligne–Mumford property now follow from [32, Theorem 3.14], F_n is a smooth projective variety over the characteristic-zero field $\mathbb{C}$. $\square$

Theorem 5.5 (Virtual reduction). — On the stack in Theorem 5.4, put $\mathcal{V} = R^1 \pi_* f^* K_{F_n}$, of rank $m - 1$. Then

$$[\overline{\mathcal{M}}_{0,\ell}(X_n, \beta)]^{\text{vir}} = c_{m-1}(\mathcal{V}) \cap [\overline{\mathcal{M}}_{0,\ell}(F_n, \beta)]^{\text{vir}}.$$

With the fiber scaling action, the equivariant formula replaces c_{m-1} by $e_{\lambda}(\mathcal{V})$.

Proof. — Along the zero section the differential of ϖ and of ι give the canonical split tangent sequence $\iota^* T_{X_n} = T_{F_n} \oplus K_{F_n}$. Relative to the smooth Artin stack of prestable marked curves, the stable-map obstruction complex is therefore

$$E_{X_n} = E_{F_n} \oplus \mathcal{V}^{\vee}[1], \quad E_{F_n} = (R\pi_* f^* T_{F_n})^{\vee}.$$

The assertion includes the morphism of obstruction theories. The projection and zero section give $d\varpi \circ d\iota = 1$, so the splitting of tangent bundles induces a splitting after derived pushforward and duality, compatible with the tangential stable-map obstruction morphism. For the normal component, construct that morphism by the relative differential of the universal evaluation map and the trace map for the universal curve. In a local trivialization of K_{F_n} , its fiber coordinate pulls back to the zero section of $\mathcal{O}_C$ over the entire base, by Theorem 5.4. Its differential

in base directions is therefore zero as a morphism of complexes. A change of trivialization multiplies this coordinate by a unit and adds to its differential a term proportional to the coordinate, also zero. Hence the normal component is globally the zero morphism, including over nonreduced bases. Thus the resulting obstruction theory is $(\phi_{F_n}, 0) : E_{F_n} \oplus \mathcal{V}^\vee[1] \rightarrow \mathbb{L}$, with the specified morphism, not just a splitting of its underlying K -class. The intrinsic normal cone is embedded in the vector-bundle stack for E_{F_n} with zero component in $\mathcal{V}$. Factoring the zero-section Gysin map for the enlarged obstruction bundle stack through that of $\mathcal{V}$ gives the top Chern class. On a zero-locus chart $Z \subset U$ over an atlas S of the prestable-curve stack, the relative cone maps are explicitly

$$[C_{Z/U}/T_{U/S}|_Z] \hookrightarrow [E_+|_Z/T_{U/S}|_Z] \hookrightarrow [E_+|_Z/T_{U/S}|_Z] \oplus \mathcal{V},$$

where the second map has zero $\mathcal{V}$ coordinate. These are compatible with descent. Only one zero-section self-intersection in $\mathcal{V}$ occurs, producing $c_{m-1}(\mathcal{V})$. For precision, the cone and zero-section construction is that of [37, Sections 5 and 7, Proposition 5.3]. For a zero locus of a vector-bundle section on a smooth ambient stack, the normal cone maps into the restricted bundle and the ambient pushforward of the virtual class is the top Chern class of that bundle, with no regularity of the section or reducedness of its zero locus required. The argument just given proves the required split excess formula from that construction, rather than postulating a comparison from a K -class. It works equivariantly by applying the same construction on finite-dimensional approximations to the classifying space of the fiber-scaling torus. $\square$

Remark 5.6 (The corresponding virtual structure sheaf)

On the common stack of Theorem 5.4, the same split obstruction morphism gives

$$\mathcal{O}_{X_n}^{\text{vir}} = \lambda_{-1}(\mathcal{V}^\vee) \otimes \mathcal{O}_{F_n}^{\text{vir}}, \quad \lambda_{-1}(\mathcal{V}^\vee) = \sum_{j=0}^{m-1} (-1)^j [\Lambda^j \mathcal{V}^\vee].$$

Indeed the extra cone coordinate is zero in $\mathcal{V}$; the Koszul resolution for its zero section gives the displayed factor in the alternating Tor construction of the virtual structure sheaf [38, Section 2.3, Proposition 1]. The subscripts indicate the standard stable-map obstruction theories for the two targets, not different underlying stacks. Thus neither virtual classes nor virtual structure sheaves for F_n and X_n are identified merely from their stack isomorphism.

For one marking, $\text{vdim}_{\mathbb{C}} \overline{\mathcal{M}}_{0,1}(X_n, \beta) = n - 2$. Define, for $0 \leq a \leq n - 2$,

$$N_{a,\beta} = \int_{[\overline{\mathcal{M}}_{0,1}(X_n, \beta)]^{\text{vir}}} \psi^a \text{ev}_{X_n}^* \varpi^* \delta_a, \quad \mathcal{D}_a(\mathbf{Q}) = \sum_{\beta > 0} N_{a,\beta} \mathbf{Q}^\beta, \quad \mathcal{D}_a^\Delta(Q) = \sum_{\beta > 0} N_{a,\beta} Q^{|\beta|}. \quad (5.1)$$

The sums defining each coefficient of the last series are finite by Proposition 4.1. On F_n we also define proper evaluation classes

$$C_{a,\beta}^\lambda = (\mathrm{ev}_{F_n})_* (\psi^a e_\lambda(\mathcal{V}) \cap [\overline{\mathcal{M}}_{0,1}(F_n, \beta)]^{\mathrm{vir}}). \quad (5.2)$$

They are polynomial in λ and at $\lambda = 0$ have complex codimension $a + 1$ in F_n . Thus $N_{a,\beta} = \int_{F_n} \delta_a C_{a,\beta}^0$. The evaluation target in (5.2) is F_n , not X_n . Equivariantly $\iota^* \iota_* C = (\lambda - H)C$; overlooking this factor changes the descendant extraction.

6. The ambient twisted mirror calculation

6.1. Virtual comparison before any degenerate limit. — For positive β , let $M_{Y_n} = \overline{\mathcal{M}}_{0,1}(Y_n, i_*\beta)$, $M_{F_n} = \overline{\mathcal{M}}_{0,1}(F_n, \beta)$, and $j : M_{F_n} \rightarrow M_{Y_n}$. The class identification is unambiguous by Proposition 4.1. Use E_+, E'_+, E_- from Lemma 5.3 on M_{Y_n} .

Proposition 6.1. — *For every $a \geq 0$,*

$$j_* (e_\lambda(E_-|_{M_{F_n}})[M_{F_n}]^{\mathrm{vir}}) = e(E_+)e_\lambda(E_-)[M_{Y_n}]^{\mathrm{vir}}, \quad (6.1)$$

$$i_* C_{a,\beta}^\lambda = H E_{a,\beta}^{0,\lambda}, \quad (6.2)$$

where

$$E_{a,\beta}^{\mu,\lambda} = (\mathrm{ev}_{Y_n})_* (\psi^a e_\mu(E'_+) e_\lambda(E_-)[M_{Y_n}]^{\mathrm{vir}}).$$

In particular, for every $b \in H^*(Y_n)$,

$$\int_{F_n} i^* b C_{a,\beta}^\lambda = \int_F i^* b i^* E_{a,\beta}^{0,\lambda}. \quad (6.3)$$

The first identity is also valid with M_{F_n}, M_{Y_n} replaced by the corresponding stacks with any number $\ell \geq 0$ of markings, and with the bundles formed on those stacks. It remains valid after multiplication by products of cotangent-line classes and evaluation pullbacks. The evaluation identities (6.2)–(6.3) use the one-marked notation above.

Proof. — We prove the virtual comparison with an arbitrary number $\ell \geq 0$ of markings and then take $\ell = 1$ for the evaluation formulas. Write $M_V^{(\ell)} = \overline{\mathcal{M}}_{0,\ell}(V, \beta_V)$, where $\beta_{F_n} = \beta$ and $\beta_{Y_n} = i_*\beta$, and put $m = H \cdot \beta > 0$. The letter V is a placeholder for either target and is unrelated to the ambient subspace V_{amb} . All Chow groups in the argument have rational coefficients. The equivariant identities are identities with an indeterminate λ ; no limit of an equivariant pairing is involved.

1. Universal bundles and the zero stack. On $M_{Y_n}^{(\ell)}$, let $\pi_{Y_n} : \mathcal{C}_{Y_n} \rightarrow M_{Y_n}^{(\ell)}$ and $f_{Y_n} : \mathcal{C}_{Y_n} \rightarrow Y_n$ be the universal family. By Lemmas 5.1–5.3,

$$\begin{aligned} R\pi_{Y_n*} f_{Y_n}^* \mathcal{O}_{Y_n}(H) &= E_+, & R\pi_{Y_n*} f_{Y_n}^* \mathcal{O}_{Y_n}(-H) &= E_-[-1], \\ \mathrm{rk} E_+ &= m + 1, & \mathrm{rk} E_- &= m - 1. \end{aligned}$$

These are identifications of perfect complexes with arbitrary derived base change. In applying [36, Tag 0A1H] on an atlas, the morphism is the proper flat finitely presented curve family, the perfect object is the indicated line bundle, and the flat finitely presented complex with proper support is $\mathcal{O}_{C_{Y_n}}$ in degree zero. Thus the statement applies to nonreduced and non-Noetherian base changes as well.

Let $s \in H^0(Y_n, \mathcal{O}_{Y_n}(H))$ define F_n . Adjunction gives a section $\sigma = \pi_{Y_n*} f_{Y_n}^* s$ of E_+ . For any scheme $S \rightarrow M_{Y_n}^{(\ell)}$, the pullback of σ is zero if and only if $f_S^* s = 0$ as a section on C_S . This is precisely the condition that f_S factor through the closed subscheme F_n of Y_n . The factorization is unique and is compatible with isomorphisms and base change. Composition with i contracts exactly the same components, so it preserves stability and the automorphism group. The class on F_n is uniquely determined by its pushforward, by Proposition 4.1. Consequently $j : M_{F_n}^{(\ell)} \hookrightarrow M_{Y_n}^{(\ell)}$ is the zero stack of σ , as a closed immersion of stacks, not just an equality of geometric points.

2. The relative obstruction-theory morphisms. Let $\mathfrak{B} = \mathfrak{M}_{0,\ell}$ be the smooth Artin stack of connected prestable marked genus-zero curves. One can replace it with a finite-type open substack containing the images of the two stable-map stacks, since these images are quasi-compact. Its dimension is $\ell - 3$. For $V = F_n, Y_n$, the usual relative stable-map obstruction theory is

$$\phi_V : \mathbb{E}_V = (R\pi_{V*} f_V^* T_V)^\vee \longrightarrow \mathbb{L}_{M_V^{(\ell)}/\mathfrak{B}}.$$

The relative stable-map obstruction morphism is the construction following Proposition 4 of [33]; Propositions 2, 4, and 5 give the prestable base, the stable-map description, and global resolutions, respectively. We construct its compatibility here. Here no stabilization of the source is necessary after composition with i . Hence the universal curve over $M_{F_n}^{(\ell)}$ is the actual pullback of C_{Y_n} , and derived base change identifies the corresponding terms in the obstruction theories.

At this point we use the special convexity of the ambient target. Indeed, $T_{Y_n} = \bigoplus_{r=1}^n \mathrm{pr}_r^* \mathcal{O}_{\mathbb{P}^1}(2)$, and each summand has nonnegative degree on every component of every genus-zero source. Hence the normalization-sequence argument of Lemma 5.1 gives $H^1(C, f^* T_{Y_n}) = 0$. It follows that $M_{Y_n}^{(\ell)} \rightarrow \mathfrak{B}$ is smooth, with relative tangent bundle $\pi_{Y_n*} f_{Y_n}^* T_{Y_n}$, and $\mathbb{E}_{Y_n} = (\pi_{Y_n*} f_{Y_n}^* T_{Y_n})^\vee = \Omega_{M_{Y_n}^{(\ell)}/\mathfrak{B}}$. This convexity statement concerns the ambient target only and does not assert smoothness of $M_{F_n}^{(\ell)}$.

The normal sequence of the smooth hypersurface,

$$0 \longrightarrow T_{F_n} \longrightarrow i^* T_{Y_n} \xrightarrow{ds} \mathcal{O}_{F_n}(H) \longrightarrow 0,$$

gives, after pullback to the universal curve, derived pushforward and duality, the distinguished triangle

$$j^* \mathbb{E}_{Y_n} \longrightarrow \mathbb{E}_{F_n} \longrightarrow (j^* E_+)^\vee[1] \longrightarrow j^* \mathbb{E}_{Y_n}[1].$$

We specify its compatibility with the cotangent triangle. On the zero stack, the derivative of the universal section in a relative tangent direction $v \in \pi_* f^* T_{Y_n}$ is $\pi_*(ds(v))$. Indeed over a square-zero extension, locally writing a lift as $f + \epsilon v$, the section is $s(f + \epsilon v) = s(f) + \epsilon ds(v)$; changes of trivialization of $\mathcal{O}(H)$ contribute a term proportional to $s(f)$, which vanishes on $M_{F_n}^{(\ell)}$. Thus this differential is globally well defined and is exactly the map obtained from the normal sequence.

More explicitly, after a smooth base change from an atlas of $\mathfrak{B}$ and an étale chart on the resulting smooth ambient algebraic space, trivialize E_+ and write $\sigma = (\sigma_1, \dots, \sigma_{m+1})$. For the ideal $I = (\sigma_1, \dots, \sigma_{m+1})$, the zero-locus obstruction theory is the morphism

$$[(j^* E_+)^{\vee} \xrightarrow{d\sigma} j^* \Omega_{M_{Y_n}^{(\ell)}/\mathfrak{B}}] \longrightarrow \mathbb{L}_{M_{F_n}^{(\ell)}/\mathfrak{B}}.$$

The explicit two-term target on this chart is

$$[I/I^2 \longrightarrow \Omega_{U/S}|_Z] \simeq \tau_{\geq -1} \mathbb{L}_{Z/S},$$

where $U \rightarrow S$ is the smooth chart just described and $Z = Z(\sigma)$. It is the truncation, not an identification with the full cotangent complex unless further lci hypotheses hold. The canonical morphism to $\mathbb{L}_{Z/S}$ is supplied by the derived Koszul zero locus; the following component description is its truncation. Its degree -1 map sends the dual basis to the classes of σ_a in I/I^2 ; its degree zero map is the restriction of relative differentials. It is an obstruction theory without a regular-sequence hypothesis on the σ_a . The preceding computation of $d\sigma$ identifies this two-term complex with $(R\pi_* f^* T_{F_n})^{\vee}$ and the morphism with ϕ_{F_n} , since both are obtained by differentiating the same universal evaluation map subject to the relation $s \circ f = 0$. These maps and their commutativity are preserved on overlaps, since they are induced by the universal evaluation map and ds , rather than chosen quasi-isomorphisms.

In particular there is a morphism of distinguished triangles

$$\begin{array}{ccccc} j^* \mathbb{E}_{Y_n} & \longrightarrow & \mathbb{E}_{F_n} & \longrightarrow & (j^* E_+)^{\vee}[1] \\ \downarrow & & \downarrow & & \downarrow \\ j^* \mathbb{L}_{M_{Y_n}^{(\ell)}/\mathfrak{B}} & \longrightarrow & \mathbb{L}_{M_{F_n}^{(\ell)}/\mathfrak{B}} & \longrightarrow & \mathbb{L}_{M_{F_n}^{(\ell)}/M_{Y_n}^{(\ell)}}. \end{array}$$

This is the compatibility required in [39, Definition 4.5]; it is stronger than an equality of classes in K -theory. The relative theory for j is concentrated in degree -1 , namely $(j^* E_+)^{\vee}[1]$.

3. The virtual cycle and its pushforward. The morphisms to $\mathfrak{B}$ and the closed immersion j are of Deligne–Mumford type. The stable-map stacks are proper finite-type Deligne–Mumford stacks over $\mathbb{C}$ and satisfy the quotient-stratification hypothesis by [39, Remark 3.2]. The three relative theories above are perfect of amplitude $[-1, 0]$. Thus [39, Corollary 4.9], with the compatible

triple just constructed, applies and gives

$$[M_{F_n}^{(\ell)}]^\text{vir} = j_{E_+}^! [M_{Y_n}^{(\ell)}]^\text{vir}.$$

The virtual pullback is the one defined by the normal-cone embedding $C_{M_{F_n}^{(\ell)}/M_{Y_n}^{(\ell)}} \hookrightarrow j^*E_+$ associated with the section σ . In particular it is the refined pullback of the zero section of E_+ in the Cartesian zero-locus square. For this pullback the pushforward identity is

$$j_* j_{E_+}^! (\alpha) = c_{m+1}(E_+) \cap \alpha.$$

One can see the identity by deforming the graph of σ to the zero section through the graphs of $t\sigma$, or by zero-section self-intersection and compatibility of refined pullback with proper pushforward. No regularity of j is used. In the present smooth ambient case the same formula is $[M_{F_n}^{(\ell)}]^\text{vir} = 0_{j^*E_+}^! [C_{M_{F_n}^{(\ell)}/M_{Y_n}^{(\ell)}}]$; this is precisely the zero-section construction of [37, Section 5]. This proves $j_* [M_{F_n}^{(\ell)}]^\text{vir} = e(E_+) [M_{Y_n}^{(\ell)}]^\text{vir}$.

4. The negative twist and marked evaluation. The universal curves are Cartesian under j , and $i^* \mathcal{O}_{Y_n}(-H) = K_{F_n}$. Lemma 5.3 therefore gives the canonical identification $j^*E_- = R^1\pi_* f^* K_{F_n} = \mathcal{V}$. It respects the fiber character of weight λ . Multiplication by $e_\lambda(E_-)$ and the projection formula prove (6.1), including its assertion for arbitrary ℓ . The cotangent lines and evaluations pull back under j , since the universal marked curve is unchanged. The same comparison consequently holds with any product of descendants and insertions pulled back from Y_n . Theorem 5.5 identifies the resulting invariants with those of X_n .

For $\ell = 1$, evaluation of $f^* \mathcal{O}_{Y_n}(H)$ at the marking is surjective on every fiber and hence over the whole stack. Thus $0 \rightarrow E'_+ \rightarrow E_+ \rightarrow \text{ev}_{Y_n}^* \mathcal{O}_{Y_n}(H) \rightarrow 0$ is an exact sequence of vector bundles. It implies $e(E_+) = \text{ev}_{Y_n}^* H e(E'_+)$. Push forward (6.1), multiplied by ψ^a , along evaluation, and use $\text{ev}_{Y_n} \circ j = i \circ \text{ev}_{F_n}$. This proves (6.2). Finally pairing with b and using $\int_{Y_n} Hbu = \int_{F_n} i^*(bu)$ proves (6.3). Only this numerical pairing is asserted in general; no injectivity of i_* on the full cohomology of F_n has been used. $\square$

6.2. The mixed hypergeometric theorem. — For $\mathbf{d} \in \mathbb{N}^n$, put $D = \sum d_i$ and define

$$I^{\mu, \lambda}(\mathbf{x}, z) = e^{\sum_{i=1}^n p_i \log x_i / z} \sum_{\mathbf{d} \geq 0} \mathbf{x}^{\mathbf{d}} \frac{\prod_{r=1}^D (\mu + H + rz) \prod_{r=0}^{D-1} (\lambda - H - rz)}{\prod_{i=1}^n \prod_{r=1}^{d_i} (p_i + rz)^2}. \quad (6.4)$$

Empty products are 1. The twisting class on the moduli of maps to Y_n is

$$e_\mu(R^\bullet \pi_* f^* \mathcal{O}(H)) e_\lambda^{-1}(R^\bullet \pi_* f^* \mathcal{O}(-H)).$$

The pairing is $\int_Y ab(\mu + H)/(\lambda - H)$ over $\mathbb{Q}(\mu, \lambda)$. At positive degree its twisting factor is $e_\mu(E_+)e_\lambda(E_-)$.

6.3. The quoted mirror-theoretic inputs. — Four published mirror-theoretic results are used below in the precise form needed here. We record their relevant specializations rather than paraphrasing them, since the hypotheses, normalizations, and product conventions enter the proof. In our application the target Y_n is a smooth projective variety with trivial stabilizers, so there is a single sector and all stabilizer ages vanish. Cone points and tangent spaces always carry the functorial, dilaton-shifted formal-neighborhood meaning of [40, Appendix B], over a complete Hausdorff linearly topologized algebra R over the Novikov ring; $R\{z\}$ denotes series $\sum a_k z^k$ with $a_k \rightarrow 0$, and R_{nilp} the topologically nilpotent elements.

Theorem 6.2 (Coates–Corti–Iritani–Tseng, [41, Corollary 32, with the data of Theorem 31])

Let $\Sigma = (N, \Sigma, \rho)$ be a stacky fan defining a smooth toric Deligne–Mumford stack $\mathcal{X}$ with projective coarse moduli space, and let S be a finite set with a map to N_Σ . Then the S -extended nonequivariant I -function is a $\Lambda_{\text{nov}}[[x, t]]$ -valued point of the nonequivariant genus-zero Givental cone of $\mathcal{X}$, at $-z$ in the cone convention.

Projectivity implies the semi-projectivity condition of the source, and completion is in curve degree and input variables. In our application $\mathcal{X} = Y_n$ has trivial stabilizers and $S = \emptyset$, so the twisted-sector and extended-degree data of [41, Definition 23 and equation (19)] are absent and the general I -function reduces to

$$ze^{\sum_{i=1}^n p_i t_i / z} \sum_{\mathbf{d} \geq 0} \mathbf{q}^{\mathbf{d}} e^{\sum_{i=1}^n d_i t_i} \prod_{i=1}^n \prod_{r=1}^{d_i} (p_i + rz)^{-2}, \quad (6.5)$$

each $\mathbb{P}^1$ factor contributing two equal divisor classes p_i and effective degrees being nonnegative tuples. Its positive part is $z\mathbf{1} + \sum_{i=1}^n p_i t_i$, so uniqueness of the cone graph section makes it the small J -function. This conclusion requires no positivity assumption on T_{F_n} .

For the next two statements let $B = \bigoplus_{j=1}^r B_j$ be a sum of line bundles on Y_n , $\rho_j = c_1(B_j)$, and $c = \exp(\sum_{k \geq 0} s_k \text{ch}_k)$ with $s_{-1} = 0$; the valuation of s_k is $k + 1$, and $B_b(x)$ denotes the Bernoulli polynomial.

Theorem 6.3 (Coates–Corti–Iritani–Tseng, [40, Theorem 4.1 and Remark 4.2])

The operator

$$\Delta = \prod_{j=1}^r \exp \left(\sum_{a,b \geq 0} s_{a+b-1} \frac{B_b(0)}{b!} \frac{\rho_j^a}{a!} z^{b-1} \right)$$

is a symplectic isomorphism from the untwisted to the c -twisted loop space, and $\mathcal{L}^{c,B} = \Delta \mathcal{L}^{\text{un}}$.

The pairings vary with c , as in Remark 4.2 of the source; identifying them by $\sqrt{c(IB)}$ instead would change the displayed operator. The general statement carries a stabilizer age $f_{ij} \in [0, 1) \cap \mathbb{Q}$ in place of the argument 0 of the Bernoulli

polynomial, and is sectorwise over $I\mathcal{X}$; for Y_n there is one sector and all ages vanish. The independent characteristic classes used below are handled by commuting exponents and a proved induction, not by attributing a two-class statement to a theorem stated for one class.

For a topological type $\mathbf{t} = (0, d, (i_1, \dots, i_m))$ let $J_{\mathbf{t}}$ be the corresponding part of the untwisted J -function, the last marking being the output. With $N_{\mathbf{t},j} = \rho_j \cdot d \in \mathbb{Z}$ and $s(w) = \sum_{k \geq 0} s_k w^k / k!$, set

$$M_{\mathbf{t}}(z) = \prod_{j=1}^r \frac{\prod_{a \leq N_{\mathbf{t},j}} \exp s(\rho_j + az)}{\prod_{a \leq 0} \exp s(\rho_j + az)}, \quad I^{c,B}(\tau, z) = \sum_{\mathbf{t}} M_{\mathbf{t}}(z) J_{\mathbf{t}}(\tau, z),$$

the quotients cancelling tails to give finite products or inverse products.

Theorem 6.4 (Coates–Corti–Iritani–Tseng, [40, Theorem 4.6])

The family $\tau \mapsto I^{c,B}(\tau, -z)$ is a point of $\mathcal{L}^{c,B}$ over $\Lambda[[s_0, s_1, \dots]][[\tau]]$.

No convexity of the individual B_j is assumed. The bundles used here are actual line bundles on the smooth projective variety Y_n , and the proof below supplies the extension to two independent characteristic classes.

Proposition 6.5 (Coates–Corti–Iritani–Tseng, [40, Lemma B.1, Propositions B.2–B.3, Corollary B.7])

- (i) *If $I(t) \in \mathcal{L}^{\text{tw}}(R[[t]])$ then $I'(t) \in T_{I(t)}\mathcal{L}^{\text{tw}}(R[[t]])$.*
- (ii) *For the untwisted cone, $f \in \mathcal{L}(R)$ and $x \in R_{\text{nilp}}$ imply $(1+x)f \in \mathcal{L}(R)$.*
- (iii) *$T_f\mathcal{L}(R)$ is an $R\{z\}$ -submodule of the ambient tangent space.*
- (iv) *If $I(t) \in \mathcal{L}(R[[t]])$ and $\xi(t)$ is tangent at $I(t)$, then $z\xi'(t)$ is tangent at $I(t)$.*

Item (ii), applied over dual numbers, implies that f itself is tangent. Item (iv) does not permit arbitrary division by z ; the proof below checks precisely the potentially negative power. Finite-parameter quotients and their inverse limits use coefficient rings of the kind just specified.

The small toric J -function of Y_n is therefore the untwisted denominator series appearing in (6.4), multiplied by z , and hypergeometric modification of a J -family by an invertible multiplicative characteristic class of a sum of line bundles lies on the corresponding twisted genus-zero cone. The proof of Theorem 6.6 supplies the additional independent-characteristic-class step; it is not hidden inside the statement of any theorem quoted above.

6.4. The mixed hypergeometric theorem, continued. — Define

$$g(\mathbf{x}) = \sum_{\mathbf{d} \neq 0} \frac{(-1)^D D! (D-1)!}{\prod_{i=1}^n (d_i!)^2} \mathbf{x}^{\mathbf{d}}, \quad Q_i = x_i e^{g(\mathbf{x})}, \quad \tau = \sum_{i=1}^n p_i \log Q_i - \lambda g(\mathbf{x}).$$

Theorem 6.6 (Mixed mirror theorem). — *The series $-zI^{\mu,\lambda}(\mathbf{x}, -z)$ lies on the mixed-twisted Givental cone. In the convention $J = z\mathbf{1} + \tau + O(z^{-1})$,*

$$zI^{\mu,\lambda} = J_{Y_n}^{\mu,\lambda}(\tau, z).$$

Here Novikov and divisor variables are combined by the divisor equation, as specified in the proof. Consequently the normalized zero-input series is

$$B^{\mu,\lambda} = e^{-\sum_{i=1}^n p_i \log Q_i / z} e^{\lambda g / z} I^{\mu,\lambda} = 1 + (\lambda - H) \sum_{\beta > 0, a \geq 0} \mathbf{Q}^\beta z^{-a-2} \mathbf{E}_{a,\beta}^{\mu,\lambda}. \quad (6.6)$$

Proof. — We separate the toric input, the formal product modification, and the specialization and normalization steps.

1. Toric input, variables, and formal topology. The ambient variety $Y_n = (\mathbb{P}^1)^n$ is smooth and projective. Its fan is complete and nonsingular, its stabilizers are trivial, and its anticanonical class is $2H$. Thus Theorem 6.2 applies with no extended variables and with the ambient torus parameters set to zero. It puts the untwisted toric series on the untwisted cone. Its positive-degree terms have order at most z^{-1} after multiplication by z , since their denominators have z -degree $2D \geq 2$. The unique cone section with expansion $z\mathbf{1} + \sum_i p_i t_i + O(z^{-1})$ is therefore the small J -function, which is

$$J_{Y_n}(\mathbf{t}, z) = z e^{\sum_i p_i t_i / z} \sum_{\mathbf{d} \geq 0} \mathbf{q}^{\mathbf{d}} e^{\sum_i d_i t_i} \prod_{i=1}^n \prod_{r=1}^{d_i} (p_i + r z)^{-2}.$$

Here $\mathbf{q}$ denotes independent Novikov variables. The substitution $x_i = q_i e^{t_i}$ is a relabeling of divisor-weighted Novikov monomials; in this notation the customary divisor exponential is recorded using formal logarithms of x_i . Equivalently all identities below can be kept in $(\mathbf{q}, \mathbf{t})$ until the last step, so no evaluation of a Novikov series at $q_i = 1$ is required.

Use the full primary input $\mathbf{t}$ on $H^*(Y_n)$ during the cone argument; its restriction to the displayed divisor coordinates will be taken afterwards. We use the cone convention in which $J_{Y_n}(\mathbf{t}, -z)$ lies on $\mathcal{L}_{Y_n}$. Formal points and tangent vectors are those of Section 6.3; positive powers of z are permitted when their coefficients tend to zero in the coefficient-ring topology. Initially retain finitely many characteristic parameters and work modulo a power of their maximal ideal and of the Novikov/input ideal. The differential operators below are then finite-order operators at each characteristic-parameter degree. All constructions are compatible with these quotients and with enlarging the finite set of parameters. This specifies the inverse-limit interpretation of the subsequent series.

2. Two independent characteristic classes. Put $\rho_+ = H$, $\rho_- = -H$, and $d_+ = D$, $d_- = -D$, the cohomology classes on Y_n . Introduce parameters $s_{j,k}$ for $j \in \{+, -\}$ and $k \geq 0$, and set

$$s_j(w) = \sum_{k \geq 0} s_{j,k} \frac{w^k}{k!}, \quad G_j(w, z) = \sum_{a, b \geq 0} s_{j, a+b-1} \frac{B_b(0)}{b!} \frac{w^a}{a!} z^{b-1}, \quad s_{j,-1} = 0.$$

The twist is $\exp\left(\sum_{j,k} s_{j,k} \operatorname{ch}_k(R\pi_* f^* \mathcal{B}_j)\right)$, where $\mathcal{B}_\pm = \mathcal{O}_{Y_n}(\pm H)$. The genus-zero quantum Riemann–Roch formula of Theorem 6.3 gives

$$\mathcal{L}_{s_+, s_-} = \Delta \mathcal{L}_{Y_n}, \quad \Delta = \exp G_+(\rho_+, z) \exp G_-(\rho_-, z).$$

For independent classes, the exponents in the Riemann–Roch calculation add up, since differentiating the twisting class with respect to $s_{j,k}$ inserts the Chern character of the corresponding index bundle. The Riemann–Roch expression for that insertion does not change when the other twist is present. The two multiplication operators commute. This proves the product form of Δ . This fact alone does not prove hypergeometric modification; that is the separate argument in the next two steps.

The Bernoulli generating series gives $G_j(w+z, z) - G_j(w, z) = s_j(w)$. Consequently the degree- $\mathbf{d}$ modification at $-z$ is

$$M_{\mathbf{d}}(-z) = \prod_{j=+,-} \exp\{G_j(\rho_j, z) - G_j(\rho_j - d_j z, z)\}.$$

For positive d_j this telescopes through d_j terms; for negative d_j the same identity gives the inverse product with the complementary endpoints. In particular the formula does not require either $\mathcal{B}_j$ to be nef. The standard hypergeometric theorem with a single characteristic class is Theorem 6.4; we now supply its independent-class extension explicitly.

3. Reduction to differential operators on the untwisted cone. Let ∇_j be the constant primary direction ρ_j , so on the divisor slice $\nabla_+ = \sum_{i=1}^n \partial_{t_i}$ and $\nabla_- = -\nabla_+$. The divisor equation, applied to the degree- $\mathbf{d}$ part of the full J -family, reads

$$-z \nabla_j J_{Y_n, \mathbf{d}}(\mathbf{t}, -z) = (\rho_j - d_j z) J_{Y_n, \mathbf{d}}(\mathbf{t}, -z).$$

The descendant term at the output marking supplies the cup-product term ρ_j ; all other inputs are primary. This also follows directly on the divisor slice from $e^{-\sum p_i t_i / z} e^{\sum d_i t_i}$. The sign here is fixed by using $J(\mathbf{t}, -z)$. After applying Δ^{-1} , the modified series is therefore

$$\mathfrak{F}_s = \exp\{-G_+(-z \nabla_+, z) - G_-(-z \nabla_-, z)\} J_{Y_n}(\mathbf{t}, -z).$$

All operators in this exponent commute. For each retained parameter,

$$\partial_{s_{j,k}} \mathfrak{F}_s = D_{j,k} \mathfrak{F}_s, \quad D_{j,k} = - \sum_{b=0}^{k+1} \frac{B_b(0)}{b!(k+1-b)!} (-z \nabla_j)^{k+1-b} z^{b-1}.$$

We will prove that $\mathfrak{F}_s$ is a point of the untwisted cone.

4. Tangency and the complete induction. We use the four statements of Proposition 6.5: item (i) for derivatives of cone-valued families; item (ii) for conical scaling; item (iii) for multiplication of tangent vectors by nonnegative powers of z ; and item (iv) for the operation $\xi \mapsto z \partial_t \xi$ on a tangent family along a cone-valued

family. These statements hold over the formal coefficient rings specified above. For a cone-valued family $f(\mathbf{t})$, every term in $D_{j,k}f$ is tangent to the cone along f . The potentially negative power of z can be checked by setting $b = 0$, in which case the operator reduces to a scalar multiple of $\nabla_j(z\nabla_j)^k$. Start with the tangent vector $\nabla_j f$ and apply Corollary B.7 exactly k times. If $b \geq 1$ and $a = k + 1 - b \geq 1$, the operator is, up to sign, $z^b(z\nabla_j)^{a-1}\nabla_j$; the same argument and Proposition B.3 apply. If $a = 0$, it is a multiple of $z^k f$, which is tangent by Propositions B.2 and B.3. Thus we never divide an arbitrary tangent vector by z .

The induction is on ordinary total degree in a fixed finite set of characteristic parameters, not on their indices. In the formal polarization write the graph equations of $\mathcal{L}_{Y_n}$ as $\mathcal{E}(f) = 0$, meaning that the negative component is the differential of the genus-zero descendant potential evaluated at the positive component after the dilaton shift. At $s = 0$ the equations hold. Assume $\mathcal{E}(\mathfrak{F}_s) \in (s)^r$ for $r \geq 1$. Take $\tilde{\mathfrak{F}}_s$ to be the graph point with the same positive component as $\mathfrak{F}_s$. Then $\mathfrak{F}_s - \tilde{\mathfrak{F}}_s \in (s)^r$. All $D_{j,k}$ differentiate only the primary input and hence preserve $(s)^r$. The chain rule and the preceding tangency calculation give

$$\partial_{s_{j,k}} \mathcal{E}(\mathfrak{F}_s) = d\mathcal{E}_{\mathfrak{F}_s}(D_{j,k}\mathfrak{F}_s) = d\mathcal{E}_{\tilde{\mathfrak{F}}_s}(D_{j,k}\tilde{\mathfrak{F}}_s) + O((s)^r) = O((s)^r).$$

Each homogeneous monomial of degree r has a nonzero derivative of degree $r - 1$ with respect to some parameter. Characteristic zero now implies that the homogeneous part of degree r in every graph equation vanishes. Thus $\mathcal{E}(\mathfrak{F}_s) \in (s)^{r+1}$, proving the induction. It includes mixed monomials in the two parameter sets. Passing to the compatible completions proves cone membership. Applying Δ proves the two-class hypergeometric assertion and restricting the full primary input to divisors gives the required family.

5. Euler and inverse-Euler specialization. Specialize to $c_+(w) = \mu + w$ and $c_-(w) = (\lambda + w)^{-1}$:

$$\begin{aligned} s_{+,0} &= \log \mu, & s_{+,k} &= (-1)^{k-1}(k-1)!\mu^{-k} & (k \geq 1), \\ s_{-,0} &= -\log \lambda, & s_{-,k} &= (-1)^k(k-1)!\lambda^{-k} & (k \geq 1). \end{aligned}$$

The formal meaning is the equivariant characteristic-class specialization described in [40, Section 2.3]. Its validity in this argument is justified as follows. The rank contributions are first written as powers of $e^{s_{+,0}}$ and $e^{s_{-,0}}$, then specialized to powers of μ and λ^{-1} . The remaining operations are expansions in inverse weights. At fixed curve degree and fixed insertions the non-equivariant Chern classes have bounded degree on the proper moduli stack. Thus the resulting correlators are rational functions of the weights; their expansions determine them injectively. The same is true of each finite hypergeometric ratio. The assertion is therefore a rational-function identity, not an analytic substitution of logarithms into an arbitrary characteristic-parameter power series. Equivalently, the cone graph

equations can be specialized coefficientwise. At fixed curve degree, the positive-degree input has positive Novikov order, so only finitely many such inputs contribute. Once the remaining input degree is fixed, dimension bounds ensure that only finitely many Chern and descendant operations are nonzero. This justifies specializing the formal identity obtained in Step 4.

At z the two finite ratios are

$$\frac{\prod_{r=-\infty}^D (\mu + H + rz)}{\prod_{r=-\infty}^0 (\mu + H + rz)} = \prod_{r=1}^D (\mu + H + rz),$$

$$\frac{\prod_{r=-\infty}^{-D} (\lambda - H + rz)^{-1}}{\prod_{r=-\infty}^0 (\lambda - H + rz)^{-1}} = \prod_{r=0}^{D-1} (\lambda - H - rz).$$

The infinite-product notation only cancels identical tails. The inverse Euler twist of the negative line contributes the second numerator, since its index bundle is $-E_-$ in positive degree. In degree zero that index bundle is $\mathcal{B}_-$ instead, yielding the reciprocal in the pairing. The two factors produce exactly (6.4), proving the asserted cone membership over $\mathbb{Q}(\mu, \lambda)$.

6. The normalized J -section and the evaluation series. For $D > 0$ the numerator has z -degree $2D - 1$, while the denominator has degree $2D$. The factor at $r = 0$ in the negative product is $\lambda - H$; the leading coefficient of the summand is

$$\frac{(\lambda - H)(-1)^{D-1} D! (D - 1)!}{z \prod_{i=1}^n (d_i!)^2}.$$

It is independent of μ . Consequently $I^{\mu, \lambda} = 1 + \tau/z + O(z^{-2})$ with the τ in the statement. Uniqueness of the graph section with positive part $z\mathbf{1} + \tau$ gives $zI^{\mu, \lambda} = J_{Y_n}^{\mu, \lambda}(\tau, z)$ in the positive- z convention. The string equation, with the twist-compatible derivation in Appendix B, removes the unit input $-\lambda g$, and the divisor equation removes the divisor exponential, giving the first expression in (6.6).

To verify the second expression without taking a degenerate limit, define the ordinary evaluation pushforward

$$\mathbf{V}_{a, \beta}^{\mu, \lambda} = (\mathrm{ev}_{Y_n})_* (\psi^a e_\mu(E_+) e_\lambda(E_-) [M_{Y_n}]^{\mathrm{vir}}).$$

Setting $w = (\mu + H)/(\lambda - H)$, contraction with the inverse twisted pairing yields $w^{-1} \mathbf{V}_{a, \beta}^{\mu, \lambda}$. Indeed, pairing this class with any b gives $\int_{Y_n} b \mathbf{V}_{a, \beta}^{\mu, \lambda}$, as required. By the marked exact sequence, $\mathbf{V}_{a, \beta}^{\mu, \lambda} = (\mu + H) \mathbf{E}_{a, \beta}^{\mu, \lambda}$. Thus $w^{-1} \mathbf{V} = (\lambda - H) \mathbf{E}$. Finally the expansion $1/(z(z - \psi)) = \sum_{a \geq 0} \psi^a z^{-a-2}$ gives precisely (6.6). Every coefficient of $\mathbf{E}$ is polynomial in μ, λ , so this identity proves regularity at $\mu = 0$ before specialization. The comparison with F_n then follows from Proposition 6.1; it initially gives a comparison against ambient classes. Theorem 15.1 identifies the evaluation classes as ambient, and nondegeneracy then gives equality of the full one-point cohomology classes, including in even dimension. $\square$

Remark 6.7. — This is a theorem about a cohomological Givental cone, not a quantum- K cone. The terminology “quantized symplectic cone” is not needed. Equation (6.6) proves regularity at $\mu = 0$ before taking a limit of a degenerate pairing. Combining it with Proposition 6.1 proves the local comparison *against all ambient insertions*. The additional geometric Theorem 15.1 proves ambientness of the full one-point evaluation series also for odd n , and therefore upgrades this comparison to the full one-point cohomology-valued series. Its proof does not use the mirror theorem.

6.5. Monic division and all numerical descendants. — Write $I_{X_n}^{\lambda, \text{amb}} = I^{0, \lambda}$ and $B^\lambda = B^{0, \lambda}$. After removing the displayed divisor logarithms, all coefficients lie in $\mathcal{R}[\lambda][[\mathbf{x}, z^{-1}]]$; completion means that polynomial degree in λ may grow with the series indices.

Lemma 6.8. — *The quotient $U^\lambda = (B^\lambda - 1)/(\lambda - H)$ is uniquely defined in this completed ring. For every ambient b ,*

$$\int_F i^* b i^* U^\lambda = \sum_{\beta > 0, a \geq 0} \mathbf{Q}^\beta z^{-a-2} \int_F i^* b C_{a, \beta}^\lambda.$$

Proof. — Substitution $\lambda = H$ annihilates every positive-degree term of $I_{X_n}^{\lambda, \text{amb}}$. The remaining exponentials cancel because $\log Q_i = \log x_i + g$ and $\sum p_i = H$. Thus $B^H = 1$. The remainder theorem for the monic polynomial $\lambda - H$ applies coefficientwise even though H is nilpotent. Multiplication by this monic polynomial is injective in each polynomial coefficient ring, which also proves uniqueness after completion. Equation (6.6) identifies the quotient with the generating series of $\mathbb{E}_{a, \beta}^{0, \lambda}$ on Y_n . Use (6.3) term by term. $\square$

On the diagonal set $x_i = y$, $Q_i = Q$, and define

$$c_D = \sum_{|\mathbf{d}|=D} \binom{D}{d_1, \dots, d_n}^2, \quad g(y) = \sum_{D \geq 1} \frac{(-1)^D c_D}{D} y^D, \quad R = \log y + g(y),$$

$$Q = y e^{g(y)}, \quad A = \theta_y R. \quad (6.7)$$

Symmetry is applied only after summing over every ordered degree tuple $\mathbf{d}$ of a fixed total degree; an individual summand need not be symmetric. The symmetric-ring calculation uniquely defines $K_j(y)$ by

$$e^{-HR/z} I_{X_n}^{0, \text{amb}}(y, z) = 1 + \sum_{j=2}^n \frac{H^j}{z^j} K_j(y). \quad (6.8)$$

There are no omitted higher cohomological powers on Y_n .

Theorem 6.9 (Uniform descendant extraction). — *The mirror coordinate is invertible as a germ at zero. For every $0 \leq a \leq n - 2$,*

$$\mathcal{D}_a^\Delta(Q) = -\frac{n!}{(n-2-a)!} K_{a+2}(y(Q)). \quad (6.9)$$

These are convergent germs with rational coefficients, and the equality holds without any assumption about primitive middle cohomology.

Proof. — Each positive-degree term of the untwisted-in- λ local numerator contains $-H$. Its degree-zero part is zero, and its degree-one part is $H(-1)^D \binom{D}{\mathbf{d}}^2 / D$. This proves the formula for R and the vanishing of the z^{-1} term in (6.8). Symmetry and homogeneity give (6.8), with $K_j(0) = 0$. Specializing $(\lambda - H)U^\lambda = B^\lambda - 1$ at $\lambda = 0$ gives

$$-HU^0 = \sum_{j=2}^n H^j z^{-j} K_j.$$

In $\mathcal{R}^{S_n}$ the kernel of multiplication by H is $\mathbb{Q}H^n$. Therefore $U^0 = -\sum_{j=2}^n H^{j-1} z^{-j} K_j$ modulo that kernel. Its restriction to F_n kills H^n , so pairing the quotient with δ_a gives (6.9), using $\int_F H^{n-1} = n!$. Only the ambient pairing assertion of Lemma 6.8 was used.

For convergence, $c_D \leq n^{2D}$ because the sum of multinomial coefficients is n^D . At fixed cohomological degree, expanding the finite products and their inverses introduces at most a fixed power of $1 + \log(D+1)$ into the bound, since their coefficients are polynomials in harmonic sums of the form $\sum_{r \leq D} r^{-k}$. The same applies to the local series and its exponential normalization. Thus g and every K_j converge for $|y| < n^{-2}$. Since $Q = y + O(y^2)$, the inverse-function theorem gives $y(Q)$ and the convergent germs in (6.9). All operations used rational coefficients. $\square$

Remark 6.10 (Why the ambient lift is retained)

For $a = n - 2$ the required function is K_n . Its coefficient $H^n K_n$ is zero after restriction to F_n . Division before restriction recovers $-H^{n-1} K_n$, which can be integrated on F_n . Thus discarding the ambient top term before division loses a genuine numerical descendant. No cancellation of a nilpotent class is being performed on F_n .

7. The quantum-Serre coefficient identity

Define a signed ambient series, initially purely algebraic,

$$\mathcal{I}^\sharp(y, z) = e^{H \log y/z} \sum_{\mathbf{d} \geq 0} (-1)^D y^D \frac{\prod_{r=1}^D (H + rz)^2}{\prod_i \prod_{r=1}^{d_i} (p_i + rz)^2}. \quad (7.1)$$

It is useful to distinguish its coefficients from the normalized ones:

$$\mathcal{I}^\sharp(y, 1) = \sum_{j=0}^n H^j W_j(y), \quad W_j = A v_j \quad (0 \leq j \leq n-1), \quad v_0 = 1.$$

Each W_j is a polynomial in $\log y$ with convergent power-series coefficients. The top coefficient W_n is not needed in the physical integral over F_n .

Theorem 7.1. — *Coefficientwise in the completed ambient ring,*

$$z\theta_y I_{X_n}^{0,\text{amb}} = H\mathcal{I}^\sharp, \quad A = \sum_{D \geq 0} (-1)^D c_D y^D = \theta_y R.$$

For $0 \leq j \leq n-1$,

$$v_j = \frac{R^j}{j!} + \sum_{k=2}^j \frac{R^{j-k}}{(j-k)!} K_k + \sum_{k=2}^{j+1} \frac{R^{j-k+1}}{(j-k+1)!} \theta_Q K_k, \quad (7.2)$$

where K_k is read as a function of Q , and empty sums are zero.

Proof. — In degree D , $z\theta_y$ multiplies by $H + Dz$. For $D > 0$,

$$\prod_{r=0}^{D-1} (-H - rz) = (-1)^D H \prod_{r=1}^{D-1} (H + rz).$$

Multiplying by $H + Dz$ completes the second positive product, giving H times the degree- D summand of (7.1). In degree zero both sides are $He^{H \log y/z}$. This proves the first identity. Its coefficient of H gives $A = \theta_y R$, while the degree-zero cohomology part of (7.1) gives its displayed series. Now differentiate (6.8) at $z = 1$ and use $\theta_y = A\theta_Q$. Comparing the coefficient of H^{j+1} for $0 \leq j \leq n-1$ gives (7.2). $\square$

Corollary 7.2. — *The first logarithmic period satisfies*

$$\frac{W_1}{A} = R + \theta_Q K_2 = R - \frac{(n-2)!}{n!} \theta_Q \mathcal{D}_0^\Delta.$$

More explicitly, with $\mathfrak{h}_D = \sum_{r=1}^D 1/r$ and $\mathfrak{h}_0 = 0$,

$$W_1 = A \log y + \sum_{D \geq 1} (-1)^D y^D \sum_{|\mathbf{d}|=D} \binom{D}{\mathbf{d}}^2 \left(2\mathfrak{h}_D - \frac{2}{n} \sum_i \mathfrak{h}_{d_i} \right).$$

Proof. — Take $j = 1$ in (7.2) and use Theorem 6.9. For the explicit expansion, differentiate the finite products to first cohomological degree; their linear term is $2\mathfrak{h}_D H - 2 \sum_i \mathfrak{h}_{d_i} p_i$. Summing over all degree tuples replaces the summed expression involving the second sum by $\frac{H}{n} \sum_i \mathfrak{h}_{d_i}$, by permutation symmetry. $\square$

8. An all-loop differential operator and full-series certificate

This section derives the sunset differential operators. The derivation is independent of the geometric mirror theorem and of any claimed numerical match with a Feynman integral. The fact that the differential operator is irreducible, and therefore the minimal operator annihilating A , follows by two Fourier transforms, with their extensions checked, from irreducibility of symmetric powers of the Kloosterman connection. The latter uses the differential Galois group of the Bessel connection [42–44]; the full argument is recorded in Proposition 15.2, and is used nowhere in the arguments below.

There have been many prior derivations of these operators in the mathematical [45] and physics literature [19, 22–24, 46], so the result is not new, but our derivation provides a uniform construction for all n , together with the all-degree certificate (8.2), which is what allows the operator to be used without a finite coefficient fit.

Throughout, $\theta_y := y\partial_y$, and all differential operators are written with powers of y on the left, so $\theta_y y = y(\theta_y + 1)$.

Definition 8.1 (Finite operator algorithm). — Set $T_0 = 1$, $T_1 = \theta_y/n$, and recursively define

$$T_{k+1} = \frac{\theta_y T_k + k y T_{k-1}}{n - k} \quad (1 \leq k < n), \quad \mathcal{S}_n = n! (\theta_y T_n + n y T_{n-1}).$$

Expand uniquely

$$\mathcal{S}_n = \sum_{j \geq 0} y^j U_j(\theta_y), \quad U_0(\theta_y) = \theta_y^{n+1}.$$

Only finitely many U_j are nonzero. Define

$$P_{n-1}(y, \theta_y) = \theta_y^{n-1} + \sum_{j \geq 1} y^j U_j(\theta_y) \prod_{r=1}^{j-1} (\theta_y + r)^2. \quad (8.1)$$

This is a finite construction over $\mathbb{Q}$, for each integer $n \geq 4$.

Theorem 8.2 (Full ambient certificate). — *The operator P_{n-1} has order $n - 1$, and*

$$P_{n-1} \mathcal{I}^\sharp(y, 1) = H^{n-1} y^H = H^{n-1} + H^n \log y \quad \text{in } \mathcal{R}[[y]][\log y]. \quad (8.2)$$

Its coefficient of θ_y^{n-1} is

$$\prod_{0 \leq j < n/2} (1 + (n - 2j)^2 y). \quad (8.3)$$

Thus it has no finite singular point on $y > 0$. The functions $W_0, \dots, W_{n-2}$ form a local basis of homogeneous solutions, $P_{n-1} W_{n-1} = 1$, and $P_{n-1} W_n = \log y$. Here W_n is the ambient top coefficient already defined in Section 7; it must be retained before restriction to F_n .

Proof. — We first explain $\mathcal{S}_n$. If $f_1, \dots, f_n$ solve $\theta_y^2 f_i = -y f_i$ over a commutative coefficient algebra, let V_k be the average over all subsets of size k of the product in which those k factors are replaced by $\theta_y f_i$. The product rule gives

$$\theta_y V_k = (n - k) V_{k+1} - k y V_{k-1}, \quad \theta_y V_0 = n V_1, \quad \theta_y V_n = -n y V_{n-1}.$$

Eliminating $V_1, \dots, V_n$ yields $\mathcal{S}_n V_0 = 0$, with exactly Definition 8.1. Assign weights 1 to θ_y and 2 to y . The commutator lowers weight, so induction gives $\deg U_j \leq n + 1 - 2j$ and $U_j = 0$ for $2j > n + 1$. The leading coefficient of $\mathcal{S}_n$ is 1.

Now work in the actual square-free algebra $\mathcal{R}$ and set

$$f_i(y) = \sum_{d \geq 0} \frac{(-1)^d y^{d+p_i}}{\Gamma(d+1+p_i)^2}.$$

Gamma functions of nilpotent variables mean their finite Taylor expansions, and $y^{p_i} = 1 + p_i \log y$ with $y > 0$, so no branch is chosen anywhere. The recurrence of Γ gives

$$(\theta_y^2 + y) f_i = \frac{y^{p_i}}{\Gamma(p_i)^2} = 0,$$

because $\Gamma(w)^{-2}$ has a double zero at $w = 0$ and $p_i^2 = 0$. This is why the calculation uses the square-free algebra, not a substitution $p_i = H/n$ that would destroy its relations. Hence $h = \prod_i f_i$ satisfies $\mathcal{S}_n h = 0$.

On y^{D+H} define $\Xi = \Gamma(\theta_y + 1)^2$ by multiplication by $\Gamma(D + H + 1)^2$. Each multiplier is invertible. Then

$$\Xi y^j \Xi^{-1} = y^j \prod_{r=1}^j (\theta_y + r)^2.$$

Multiplication of Ξh by the constant invertible class $\prod_i \Gamma(1+p_i)^2 / \Gamma(1+H)^2$ yields exactly $\mathcal{I}^\#(y, 1)$ of (7.1), alternating signs included. It follows that

$$\left[\theta_y^{n+1} + \sum_{j \geq 1} y^j U_j(\theta_y) \prod_{r=1}^j (\theta_y + r)^2 \right] \mathcal{I}^\# = 0.$$

Since $\theta_y^2 y^j = y^j (\theta_y + j)^2$, the operator in brackets is $\theta_y^2 P_{n-1}$, so $\theta_y^2 P_{n-1} \mathcal{I}^\# = 0$.

Write $P_{n-1} \mathcal{I}^\# = \sum_{D \geq 0} y^{D+H} R_D$. For $D > 0$, $(D + H)^2 R_D = 0$ implies $R_D = 0$ because $D + H$ is invertible. For $D = 0$, only the constant-degree part θ_y^{n-1} acts, giving $R_0 = H^{n-1}$. This proves (8.2) including its nonzero source; division by a nilpotent element has not occurred. The degree bound for U_j proves the order is at most $n - 1$, and the constant-degree coefficient proves it is exactly $n - 1$.

For completeness, the highest weighted symbol of $\mathcal{S}_n$ is the characteristic polynomial of the displayed $(n+1)$ -dimensional system with commuting variables. Over $\mathbb{Q}(\sqrt{-y})$ its eigenvalues are $(n - 2j)\sqrt{-y}$ for $0 \leq j \leq n$ (the symmetric-power representation of the two eigenvalues $\pm\sqrt{-y}$). Pairing opposite eigenvalues, with an additional zero eigenvalue when n is even, shows that the highest coefficients of

the U_j are the coefficients of the corresponding product of $\theta_y^2 + (n - 2j)^2 y$. The Gamma multiplication and division by θ_y^2 in (8.1) convert this product to (8.3).

Finally $W_j = (\log y)^j / j! + O(y(1 + |\log y|)^j)$. Comparing coefficients of H^j in (8.2) gives $P_{n-1}W_j = 0$ for $j < n - 1$ and $P_{n-1}W_{n-1} = 1$. The different leading logarithmic powers prove linear independence of the $n - 1$ homogeneous solutions. They form a basis because the operator has order $n - 1$. $\square$

Remark 8.3 (The unsigned variable). — We use $y = -t$ throughout. This removes the alternating signs generated by $t^j = (-1)^j y^j$ and avoids expressions such as $(-y)^H$, whose interpretation would require a branch choice. The Euler operator is unchanged, $t\partial_t = y\partial_y$.

Corollary 8.4 (Frobenius normalization). — *The origin is a regular singular point with indicial polynomial ρ^{n-1} and a single maximal unipotent logarithmic block. For any constant b , all local solutions of $P_{n-1}f = b$ have the form*

$$f = bW_{n-1} + \sum_{j=0}^{n-2} b_j W_j.$$

Their boundary polynomial in $\log y$ determines the b_j uniquely. In particular a differential equation alone does not specify the physical solution.

Proof. — The coefficient of θ^{n-1} is nonzero at zero, and all other coefficient polynomials are holomorphic there; the Euler-operator form gives regular singularity and the indicial polynomial. Theorem 8.2 gives the logarithmic basis and a particular solution. Continuation $\log y \mapsto \log y + 2\pi i$ on this basis has a single Jordan block, as is already seen on its leading polynomials. The boundary normalization then determines each constant. $\square$

Example 8.5 (The two initial specializations). — As an illustration we specialize to the cases of $n = 4$ and $n = 5$ in the symbolic implementation of the finite construction given in Appendix A.

For $n = 4$, the algorithm gives

$$P_3 = \theta_y^3 + y(20\theta_y^3 + 30\theta_y^2 + 18\theta_y + 4) + 64y^2(\theta_y + 1)^3. \quad (8.4)$$

For $n = 5$, it gives

$$\begin{aligned} P_4 &= \theta_y^4 + yQ_1(\theta_y) + y^2Q_2(\theta_y) + y^3Q_3(\theta_y), \\ Q_1 &= 35\theta_y^4 + 70\theta_y^3 + 63\theta_y^2 + 28\theta_y + 5, \\ Q_2 &= 259\theta_y^4 + 1036\theta_y^3 + 1580\theta_y^2 + 1088\theta_y + 285, \\ Q_3 &= 225\theta_y^4 + 1350\theta_y^3 + 2925\theta_y^2 + 2700\theta_y + 900. \end{aligned} \quad (8.5)$$

9. The physical integral and its signed Gamma formula

The large-momentum behaviour of the sunset family and its connection to the periods of the associated Calabi–Yau geometry have been investigated using several

approaches. These include GKZ systems and relative Calabi–Yau periods in [22, 23], the all-order structure of the multiloop sunset in [28], and the two-loop case in [17]. We take the asymptotic result of [26] as our starting point.

9.1. Convergence and the positive Laplace transform. — Define the two-dimensional equal-mass sunset normalization by

$$I_{\ominus}^{(n-1)}(s) = \int_{\mathbb{R}_+^n} e^{-\sum_{i=1}^n x_i + sW(x)} W(x) \prod_{i=1}^n \frac{dx_i}{x_i}, \quad W(x) = \left(\sum_{i=1}^n x_i^{-1} \right)^{-1}. \quad (9.1)$$

Let define the coordinates $u_1, \dots, u_n$ $u_i = x_i/(x_1 + \dots + x_n)$ so that the domain of integration becomes $\Delta_n = \{u_i > 0 : \sum_{i=1}^n u_i = 1\}$, and put

$$\phi(u) = \sum_{i=1}^n u_i^{-1}, \quad \Omega = \frac{du_1 \cdots du_{n-1}}{u_1 \cdots u_n}.$$

Proposition 9.1. — *The Schwinger integral is absolutely convergent for $\operatorname{Re} s \leq n^2$ and holomorphic for $\operatorname{Re} s < n^2$. It is not absolutely convergent for $\operatorname{Re} s > n^2$. Its analytic continuation to $\mathbb{C} \setminus [n^2, \infty)$ is*

$$I_{\ominus}^{(n-1)}(s) = \int_{\Delta_n} \frac{\Omega}{\phi(u) - s}. \quad (9.2)$$

For real $s < n^2$ it also equals

$$\int_0^\infty e^{rs} \left(\int_{\Delta_n} e^{-r\phi(u)} \Omega \right) dr. \quad (9.3)$$

Proof. — Set $x_i = tu_i$. Then $\prod dx_i/x_i = (dt/t)\Omega$ and $W(x) = t/\phi(u)$. The radial integral gives $\int_0^\infty e^{-t(1-s/\phi)} dt/\phi = 1/(\phi - s)$. Applied to the absolute value of the integrand, the same computation gives $1/(\phi - \operatorname{Re} s)$ where $\operatorname{Re} s < \phi$ and $+\infty$ where $\operatorname{Re} s \geq \phi$. By Tonelli’s theorem, the Schwinger integral therefore converges absolutely if and only if $\int_{\Delta_n} \Omega/(\phi - \operatorname{Re} s)$ is finite, and when it does, Fubini’s theorem allows the radial integration to be performed first, which gives (9.2). Cauchy–Schwarz gives $\phi \geq n^2$, with equality only at $u_i = 1/n$. The required boundary integrability follows from the estimate $\phi^{-1} \leq \min_i u_i$. On the sector $u_1 \leq \dots \leq u_n$, we have $u_n \geq 1/n$, which bounds the density by $n/(u_2 \cdots u_{n-1})$. Successively integrating $0 < u_1 < \dots < u_{n-1} < 1$ is finite (each integration cancels the next denominator). There are finitely many sectors. For $\operatorname{Re} s < n^2$ the density is bounded by a constant times Ω/ϕ , locally uniformly in s . At $s = n^2$, the minimum of ϕ is nondegenerate on the simplex, so the only new interior singularity is bounded by a constant times $\|u - u_*\|^{-2}$ in real dimension $n - 1$. It is integrable because $n \geq 4$. This endpoint is one of the essential uses of the standing hypothesis $n \geq 4$, alongside the integral curve-lattice identification. For $\operatorname{Re} s > n^2$, an open set near u_* gives radial exponential growth, so absolute convergence fails. For each compact subset of the slit plane, $|\phi - s|^{-1} \leq C/\phi$, which proves holomorphic

continuation under the integral. Finally the integrand in (9.3) is nonnegative and its radial integral is (9.2). Hence Tonelli's theorem permits us to interchange the radial and simplex integrals. $\square$

The simplex representation also gives the Taylor expansion at $s = 0$ and its behaviour on the circle of convergence. Write $u_* = (1/n, \dots, 1/n)$ for the barycentre of Δ_n .

Proposition 9.2. — *For $|s| < n^2$,*

$$I_{\ominus}^{(n-1)}(s) = \sum_{m \geq 0} a_m s^m, \quad a_m = \int_{\Delta_n} \phi(u)^{-m-1} \Omega > 0, \quad (9.4)$$

and, as $m \rightarrow \infty$,

$$a_m = \frac{\pi^{\frac{n-1}{2}} n^{n/2-2-2m}}{m^{\frac{n-1}{2}}} (1 + O(m^{-1})). \quad (9.5)$$

Consequently the Taylor series of $I_{\ominus}^{(n-1)}$ at $s = 0$ has radius of convergence exactly n^2 , and it converges absolutely and uniformly on $|s| \leq n^2$. For an integer $j \geq 0$, the j th derivative has a finite limit as $s \uparrow n^2$ along the real axis if and only if $j < (n-3)/2$. In particular, for $n = 4$ and $n = 5$ the integral is continuous, but not C^1 , at the thresholds $s = 16$ and $s = 25$.

Proof. — Since $\phi \geq n^2 > |s|$ on Δ_n , the expansion $(\phi - s)^{-1} = \sum_{m \geq 0} s^m \phi^{-m-1}$ is dominated by $(\phi - |s|)^{-1} \leq n^2(n^2 - |s|)^{-1} \phi^{-1}$, and $\int_{\Delta_n} \Omega/\phi = I_{\ominus}^{(n-1)}(0)$ is finite by Proposition 9.1. Integrating (9.2) term by term gives (9.4).

For (9.5), write $u = u_* + v$ with $\sum_i v_i = 0$ and use $v' = (v_1, \dots, v_{n-1})$ as coordinates. Since $\sum_i v_i = 0$,

$$\phi(u_* + v) = \sum_{i=1}^n \frac{n}{1 + nv_i} = n^2 + n^3 \sum_{i=1}^n v_i^2 + O(|v|^3), \quad \sum_{i=1}^n v_i^2 = v'^T (I + J) v',$$

where J is the $(n-1) \times (n-1)$ matrix with all entries 1, so that $\det(I + J) = n$. The density of Ω at u_* is n^n . By the Cauchy–Schwarz argument in the proof of Proposition 9.1, u_* is the only point of Δ_n at which $\phi = n^2$. Hence for every neighbourhood N of u_* there is $\delta > 0$ with $\phi \geq n^2(1 + \delta)$ on $\Delta_n \setminus N$, and the contribution of $\Delta_n \setminus N$ to a_m is at most $(n^2(1 + \delta))^{-m} I_{\ominus}^{(n-1)}(0)$. On N write $\phi^{-m-1} = n^{-2m-2} \exp(-(m+1)\psi)$ with $\psi = \log(\phi/n^2) = n \sum_i v_i^2 + O(|v|^3)$. The multidimensional Laplace method, for a smooth phase with a nondegenerate minimum and a smooth amplitude, gives

$$a_m = n^{n-2m-2} \left(\frac{\pi}{nm} \right)^{\frac{n-1}{2}} \det(I + J)^{-1/2} (1 + O(m^{-1})),$$

which is (9.5). The root test now gives the radius n^2 . On $|s| \leq n^2$ the terms are bounded by $a_m n^{2m}$, which is of exact order $m^{-(n-1)/2}$ and hence summable since $n \geq 4$. For the j th derivative, the terms $m(m-1) \cdots (m-j+1) a_m s^{m-j}$ are nonnegative and nondecreasing in $s \in [0, n^2]$, so by monotone convergence the

limit as $s \uparrow n^2$ equals $\sum_m m(m-1) \cdots (m-j+1) a_m n^{2(m-j)}$, whose terms are of exact order $m^{j-(n-1)/2}$. This is finite exactly when $j < (n-3)/2$. $\square$

Remark 9.3 (First correction). — The remainder in (9.5) can be made explicit to the next order:

$$a_m = \frac{\pi^{\frac{n-1}{2}} n^{n/2-2-2m}}{m^{\frac{n-1}{2}}} \left(1 - \frac{3(n-1)}{8m} + O(m^{-2}) \right). \quad (9.6)$$

Unlike Proposition 9.2, the argument uses the Bessel representation (17.1) of Section 17; this remark is not used in any proof. Replacing $J_0(\sqrt{T}r)$ in (17.1) by $I_0(\sqrt{s}r)$ gives a function of s that is holomorphic on $|s| < n^2$, because $|I_0(\sqrt{s}r)| \leq e^{\sqrt{|s|}r}$ and $K_0(r)^n = O(r^{-n/2}e^{-nr})$ as $r \rightarrow \infty$. Since $J_0(iz) = I_0(z)$, it agrees with $I_{\ominus}^{(n-1)}$ for $s = -T \leq 0$, and hence on the whole disk. Expanding I_0 termwise gives

$$a_m = \frac{2^{n-1}}{4^m(m!)^2} \int_0^\infty r^{2m+1} K_0(r)^n dr.$$

By [47, §10.40], $K_0(r)^n = (\pi/(2r))^{n/2} e^{-nr} (1 - n/(8r) + O(r^{-2}))$ uniformly for $r \geq 1$, and the integrals over $[0, 1]$ are negligible compared with the main term. Hence

$$\int_0^\infty r^{2m+1} K_0(r)^n dr = \left(\frac{\pi}{2}\right)^{n/2} \frac{\Gamma(2m+2-n/2)}{n^{2m+2-n/2}} \left(1 - \frac{n^2}{8(2m+1-n/2)} + O(m^{-2}) \right).$$

Finally $\binom{2m}{m} = 4^m (\pi m)^{-1/2} (1 - \frac{1}{8m} + O(m^{-2}))$ and $\Gamma(2m+2-n/2)/\Gamma(2m+1) = (2m)^{1-n/2} (1 + \frac{(2-n)(4-n)}{16m} + O(m^{-2}))$. The leading factors reproduce (9.5), and the three first-order terms add up to

$$\frac{1}{8m} \left(-1 + \frac{(2-n)(4-n)}{2} - \frac{n^2}{2} \right) = -\frac{3(n-1)}{8m}.$$

Remark 9.4. — By (8.3), the operator of Theorem 8.2 is also singular at the pseudothresholds $s = (n-2j)^2$, $1 \leq j < n/2$, which lie inside the disk of Proposition 9.2. These points are genuine singularities of the operator and have the same local exponents as the threshold. For $n = 4$, the local exponents at both $y = -1/16$ and $y = -1/4$ are $\{0, \frac{1}{2}, 1\}$. For $n = 5$, the local exponents at $y = -1/25$, $y = -1/9$, and $y = -1$ are $\{0, 1, 1, 2\}$. Nevertheless, Proposition 9.1 shows that the physical integral is holomorphic at the pseudothresholds. The vanishing there concerns only its singular component, not the singularity of the operator itself. The exponent $(n-3)/2$ in Proposition 9.2 is the familiar threshold exponent of n -particle phase space in two dimensions. The disk of Proposition 9.2 lives in the physical variable s and should not be confused with the Q -Taylor disk of Theorem 11.4.

9.2. The analytic literature input and the degree sign. — Define

$$I_{F_n}(r\mathbf{1}, z) = e^{H \log r/z} \sum_{\mathbf{d} \geq 0} r^D \frac{\prod_{k=1}^D (H + kz)}{\prod_{i=1}^n \prod_{k=1}^{d_i} (p_i + kz)^2}.$$

We quote the result of Iritani [26, Section 2, equation (10)], who studies the large complex structure asymptotics of the same family of amplitudes.

Theorem 9.5 (Iritani, [26, Section 2, equation (10)])

For this smooth Fano hypersurface, the positive oscillatory integral is

$$\int_{\Delta_n} e^{-r\phi(u)} \Omega = \int_{F_n} I_{F_n}(r\mathbf{1}, -1) \widehat{\Gamma}_{F_n} \quad (r > 0).$$

This is the specialization to $q_1 = \dots = q_n = r$ of [26, Section 2, equation (10)], which states, for a smooth Fano $F_n \subset (\mathbb{P}^1)^n$ of degree $(1, \dots, 1)$ and $q_1, \dots, q_n > 0$,

$$\begin{aligned} \int_{\mathbb{R}_+^{n-1}} e^{-\phi_q(w)} \prod_{i=1}^{n-1} \frac{dw_i}{w_i} &= \int_{F_n} I_{F_n}(q, -1) \widehat{\Gamma}_{F_n}, \\ \phi_q(w) &= \left(\sum_{i=1}^{n-1} q_i w_i + q_n \right) \left(\sum_{i=1}^{n-1} w_i^{-1} + 1 \right), \end{aligned}$$

with ordinary cohomological Chern roots in the Gamma class. We use this identity only as an analytic input. More precisely, equation (10) of [26] gives the positive-real oscillatory integral in the $(n-1)$ independent variables w_i , paired with the $\widehat{\Gamma}$ -class of the degree- $(1, \dots, 1)$ Fano hypersurface. Specializing $q_1 = \dots = q_n = r$ and making the change of variables

$$u_i = \frac{w_i}{1 + \sum_{j=1}^{n-1} w_j}, \quad u_n = \frac{1}{1 + \sum_{j=1}^{n-1} w_j},$$

identifies its positive-real integral with

$$\int_{\Delta_n} e^{-r\phi(u)} \Omega.$$

Thus the only external analytic identity used here is the corresponding positive-real $\widehat{\Gamma}$ -pairing. The passage from this identity to the physical Schwinger integral is not quoted from [26]: it is the Laplace-transform argument of Theorem 9.7, including the factor $(-1)^D$ derived in (9.11).

This is the only analytic input in the paper. Every other external result quoted here is a foundational statement about virtual classes, including the comparison of algebraic and symplectic virtual classes (Theorem 14.12), about toric mirror symmetry and its twisted versions (Section 6.3), about quantum K -theory (Theorem 14.8), or about the symplectic integrality of Gromov–Witten invariants (Theorem 14.11). The entire passage from the Feynman integral to Gromov–Witten theory goes through this one identity. It is the specialization to the present family

of the $\widehat{\Gamma}$ -integral structure of a Fano manifold, in which the oscillatory integral of the Landau–Ginzburg mirror over its positive real form is paired against $\widehat{\Gamma}_{F_n}$. We apply the cited result under its standing hypotheses and with the convention $\widehat{\Gamma}(E) = \prod_j \Gamma(1 + \delta_j)$, for ordinary cohomological Chern roots δ_j , with no inserted factors of $2\pi i$. Let ϵ be the cohomological involution $p_i \mapsto -p_i$; it acts trivially on scalar logarithms and on the variable y .

Proposition 9.6 (Uniform Gamma polynomial). — *On F_n ,*

$$\widehat{\Gamma}_{F_n} \Gamma(1 - H) = \widehat{\Gamma}_{X_n}|_{F_n} = \sum_{k=0}^{n-1} \gamma_k H^k, \quad \sum_{k \geq 0} \gamma_k t^k = \exp \left(\sum_{r \geq 1} \frac{2\zeta(2r+1)}{2r+1} t^{2r+1} \right). \quad (9.7)$$

Thus $\gamma_0 = 1$, $\gamma_1 = \gamma_2 = 0$, $\gamma_3 = 2\zeta(3)/3$, $\gamma_5 = 2\zeta(5)/5$, and $\gamma_6 = 2\zeta(3)^2/9$.

Proof. — The tangent K -class is $[T_{X_n}|_{F_n}] = [T_{Y_n}|_{F_n}] - [\mathcal{O}(H)] + [\mathcal{O}(-H)]$. Therefore the Gamma class equals $\prod_i \Gamma(1 + 2p_i) \Gamma(1 - H)/\Gamma(1 + H)$. In the Taylor expansion $\log \Gamma(1 + w) = -\gamma_E w + \sum_{k \geq 2} (-1)^k \zeta(k) w^k/k$, the factor $p_i^2 = 0$ makes the first logarithm sum to $-2\gamma_E H$. The Gamma ratio cancels this linear term and all even powers, leaving $\sum_{r \geq 1} 2\zeta(2r+1) H^{2r+1}/(2r+1)$. Truncate above degree $n-1$ and exponentiate. This also explains why products of odd zeta values first occur when $n \geq 7$. $\square$

We need the following theorem about the Laplace transform which is not in Iritani’s paper [26]. For the relation between the conventions in that work and the present work we refer to remark 9.8.

Theorem 9.7 (Signed Laplace/Gamma formula). — *Let $s = -T < 0$ and $y = T^{-1} = -1/s > 0$, with the real logarithm. For T sufficiently large (in particular for $T > n^2$),*

$$I_{\ominus}^{(n-1)}(-1/y) = y \int_{F_n} \epsilon(\mathcal{I}^{\sharp}(y, 1)) \widehat{\Gamma}_{X_n}|_{F_n}. \quad (9.8)$$

It is an identity of convergent logarithmic germs. In particular

$$P_{n-1} \left(\frac{I_{\ominus}^{(n-1)}(-1/y)}{y} \right) = (-1)^{n-1} n!. \quad (9.9)$$

Its leading boundary polynomial is

$$I_{\ominus}^{(n-1)}(-1/y) = n! y \sum_{k=0}^{n-1} \frac{(-1)^{n-1-k} \gamma_k}{(n-1-k)!} (\log y)^{n-1-k} + O(y^2(1 + |\log y|)^{n-1}). \quad (9.10)$$

Proof. — Insert Theorem 9.5 into (9.3). The degree- D coefficient has r^{D-H} , so its Laplace integral is

$$\int_0^{\infty} e^{-Tr} r^{D-H} dr = T^{-1-D+H} \Gamma(1 + D - H).$$

This identity for a nilpotent H means finitely many derivatives of the elementary Gamma integral. Crucially,

$$\Gamma(1 + D - H) = (-1)^D \Gamma(1 - H) \prod_{k=1}^D (H - k). \quad (9.11)$$

Multiplying the existing Fano numerator $\prod (H - k)$ gives the square *and* the factor $(-1)^D$. Thus the result is

$$y \int_{F_n} y^{-H} \sum_{\mathbf{d} \geq 0} (-1)^D y^D \frac{\prod_{k=1}^D (H - k)^2}{\prod_i \prod_{k=1}^{d_i} (p_i - k)^2} \widehat{\Gamma}_{F_n} \Gamma(1 - H),$$

which is precisely (9.8).

To justify termwise integration, expand only to the finitely many cohomological degrees that can contribute. The Fano scalar coefficients are $D! / \prod d_i!^2$; integrating r^D multiplies by $D! T^{-D-1}$. Logarithmic derivatives introduce only fixed powers of $1 + \log(D + 1) + |\log T|$; the same holds for finite product expansions. The sum of the resulting absolute integrals is bounded by a constant times $T^{-1} \sum_D (n^2/T)^D$ times such a fixed logarithmic power. It converges for $T > n^2$, giving the required interchange.

Apply P_{n-1} to (9.8) and use Theorem 8.2. On F_n , the top-degree term is $\epsilon(H^{n-1}y^H) = (-1)^{n-1}H^{n-1}$; the Gamma factor contributes only its constant term. This gives (9.9). For (9.10) retain the degree-zero Novikov term y^{-H} ; all other terms satisfy the stated convergent-germ bound. $\square$

Remark 9.8 (The analytic source and the sign conventions)

Three separate conventions must be matched before any displayed formula is transcribed between the present work and the one of Iritani [26].

First, the denominator in (9.2) is $\phi - s$, whereas the amplitude in [26] uses $s - \phi$; this is an overall sign, independent of the curve degree. Second, the period normalizations $\Pi_{n-1} = (-1)^{n-1}yA$ and $\pi = -yA$ differ with loop parity, as detailed in Section 16. Third, and least visible, is the degree factor itself. The degree factor can be incorporated in either of two equivalent ways. Here it appears explicitly in the series (7.1), which contains $(-1)^D$, with $y > 0$ and the real logarithm understood. Alternatively one may evaluate an unsigned squared-numerator series at a negative argument, since $(-x)^{\mathbf{d}} = (-1)^D x^{\mathbf{d}}$ when $|\mathbf{d}| = D$; this produces the same factor automatically, but then leaves the branch of x^H at negative x to be fixed (or its cohomological phase to be removed separately). The displayed formula in Iritani's Theorem 8 therefore does not exhibit an additional explicit $(-1)^D$ in that convention. In the convention of the present paper the termwise Laplace transform of equation (10) produces the factor $(-1)^D$ in (9.11); this is the convention used in Theorem 9.7 and in the subsequent numerical checks.

10. The uniform descendant assembly

This section gives the all-loop analytic formula (10.2) for the sunset integral, and its zeta-free bracket (10.3) with rational coefficients. The proof uses the uniform Gamma polynomial of Proposition 9.6 and the uniform differential operator of Theorem 8.2, together with the top-degree extraction of Theorem 6.9 and the triangular relation of Theorem 7.1. This expansion near the large-momentum limit, where the monodromy is maximally unipotent, is the main result of the paper. It is the uniform all-loop form of the two-loop theorem of [17]. Its analytic shape is that of [28, Theorem 4.1], proved there by a Mellin–Barnes residue computation; what is established here is the enumerative meaning of every coefficient, as set out in the introduction.

Define the signed period prefactor

$$\Pi_{n-1}(s) = (-1)^{n-1} y A(y), \quad y = -1/s.$$

Thus $dR/ds = yA(y) = (-1)^{n-1} \Pi_{n-1}$. In particular, $\Pi_3 = -yA$ and $\Pi_4 = yA$. For $0 \leq j \leq n-1$ define

$$\begin{aligned} \mathcal{T}_j(R, Q) &= \frac{n!}{j!} R^j - \sum_{a=0}^{j-2} \frac{(n-2-a)!}{(j-a-2)!} R^{j-a-2} \mathcal{D}_a^\Delta(Q) \\ &\quad - \sum_{a=0}^{j-1} \frac{(n-2-a)!}{(j-a-1)!} R^{j-a-1} \theta_Q \mathcal{D}_a^\Delta(Q). \end{aligned} \quad (10.1)$$

An empty sum is zero, so $\mathcal{T}_0 = n!$.

Theorem 10.1 (All-loop analytic–enumerative identity)

As convergent logarithmic germs at $s = -\infty$,

$$I_\ominus^{(n-1)}(s) = \Pi_{n-1}(s) \sum_{k=0}^{n-1} (-1)^k \gamma_k \mathcal{T}_{n-1-k}(R(s), Q(s)). \quad (10.2)$$

Its zeta-free bracket is

$$\mathcal{T}_{n-1} = n R^{n-1} + \sum_{j=0}^{n-2} h_j(Q) R^j, \quad (10.3)$$

where

$$h_{n-2} = -\theta_Q \mathcal{D}_0^\Delta, \quad h_j = -(j+1) \mathcal{D}_{n-3-j}^\Delta - \theta_Q \mathcal{D}_{n-2-j}^\Delta \quad (0 \leq j \leq n-3). \quad (10.4)$$

Here $\theta_Q = Q \partial_Q$ and “zeta-free” denotes the $k = 0$ contribution in the explicit Gamma expansion, and does not assume any unproved algebraic independence of zeta values. Each h_j has rational Taylor coefficients and zero constant term.

Proof. — The involution in (9.8) acts by $(-1)^j$ on H^j . Top-degree extraction therefore gives

$$I_{\ominus}^{(n-1)}(s) = n!yA \sum_{k=0}^{n-1} (-1)^{n-1-k} \gamma_k v_{n-1-k} = \Pi_{n-1} \sum_{k=0}^{n-1} (-1)^k \gamma_k (n!v_{n-1-k}).$$

Substitute (6.9) into (7.2). The resulting expression for $n!v_j$ is exactly (10.1). For $j = n - 1$ the first coefficient is $n!/(n - 1)! = n$; the two factorial ratios simplify respectively to $n - 2 - a$ and 1, giving (10.4). Rationality and convergence follow from Theorem 6.9. All signs and the factor of the curve degree in θ_Q are retained. $\square$

Corollary 10.2 (Specialization to three loops). — For $n = 4$ put $\Pi_3 = -yA$ and $\delta_0 = H^2/2$. Then

$$I_3 = \Pi_3 [4R^3 + f_2 R^2 + f_1 R + f_0 - 16\zeta(3)],$$

with

$$f_2 = -\theta_Q \mathcal{D}_0^\Delta, \quad f_1 = -2\mathcal{D}_0^\Delta - \theta_Q \mathcal{D}_1^\Delta, \quad f_0 = -\mathcal{D}_1^\Delta - \theta_Q \mathcal{D}_2^\Delta,$$

and $\mathcal{D}_0^\Delta = -12K_2$, $\mathcal{D}_1^\Delta = -24K_3$, $\mathcal{D}_2^\Delta = -24K_4$.

Proof. — Use $n! = 24$, $\gamma_3 = 2\zeta(3)/3$ and $\mathcal{T}_0 = 24$ in (10.2); then apply (10.4) and (6.9). $\square$

Corollary 10.3 (Specialization to four loops). — For $n = 5$ put $\Pi_4 = yA$. Then

$$I_4 = \Pi_4 \left[5R^4 + \sum_{j=0}^3 h_j R^j - 80\zeta(3) \left(R - \frac{1}{20} \theta_Q \mathcal{D}_0^\Delta \right) \right],$$

where

$$\begin{aligned} h_3 &= -\theta_Q \mathcal{D}_0^\Delta, & h_2 &= -3\mathcal{D}_0^\Delta - \theta_Q \mathcal{D}_1^\Delta, \\ h_1 &= -2\mathcal{D}_1^\Delta - \theta_Q \mathcal{D}_2^\Delta, & h_0 &= -\mathcal{D}_2^\Delta - \theta_Q \mathcal{D}_3^\Delta, \end{aligned}$$

and $(\mathcal{D}_0^\Delta, \mathcal{D}_1^\Delta, \mathcal{D}_2^\Delta, \mathcal{D}_3^\Delta) = (-20K_2, -60K_3, -120K_4, -120K_5)$.

Proof. — Now $n! = 120$ and $\mathcal{T}_1 = 120R - 6\theta_Q \mathcal{D}_0^\Delta$. Only $k = 0, 3$ contribute in (10.2). $\square$

Remark 10.4 (The two initial specializations). — Corollaries 10.2 and 10.3 are the statements used in the companion paper [27], that gives an independent analytic normalization and a separate primary coefficient bridge for the $n = 4$ identity from the Picard–Fuchs equation and a Mellin transform, an explicit quantum- K formula and fourfold integrality statement at $n = 4$, the numerical descendant potentials, and the comparison of the $n = 3, 4, 5$ normalizations with the local–relative Euler factors.

Remark 10.5 (What starts at higher loop order)

At $n = 6$ the $\zeta(5)$ term first appears, in addition to the $\zeta(3)$ term multiplying $\mathcal{T}_2$. At $n = 7$ the product $\zeta(3)^2$ appears. Thus it would be incorrect to extrapolate

the special $n = 4, 5$ pattern to a claim that only a single $\zeta(3)$ boundary term occurs for every loop number. Formula (10.2) fixes both the zeta polynomial and the descendant correction multiplying each of its terms.

11. A global real mirror coordinate and continuation

Let ν be normalized Haar measure on $(S^1)^n$ and put $\mathbf{q}(\mathbf{w}) = |w_1 + \cdots + w_n|^2$, so $0 \leq \mathbf{q} \leq n^2$. The symbol $\mathbf{q}(\mathbf{w})$ in this section is a real function on the torus, not a Novikov variable Q .

Theorem 11.1 (Positive torus formula). — *The power series in (6.7) extend to $\mathbb{C} \setminus (-\infty, -n^{-2}]$ by*

$$A(y) = \int \frac{d\nu}{1 + y\mathbf{q}(\mathbf{w})}, \quad g(y) = - \int \log(1 + y\mathbf{q}(\mathbf{w})) d\nu,$$

using the principal logarithm. Their Taylor radii at zero are exactly n^{-2} . For $y > 0$, $A(y) > 0$ and $Q(y) = ye^{g(y)}$ is strictly increasing. It maps $(0, \infty)$ bijectively onto $(0, Q_\infty)$, where

$$Q_\infty = \exp \left(- \int \log |w_1 + \cdots + w_n|^2 d\nu \right), \quad n^{-2} \leq Q_\infty \leq 1. \quad (11.1)$$

The inverse $y(Q)$ is real analytic on that open interval.

Proof. — The constant term of $(w_1 + \cdots + w_n)^D (w_1^{-1} + \cdots + w_n^{-1})^D$ is c_D , so $c_D = \int \mathbf{q}^D d\nu$. Summing the geometric and logarithmic series for $|y| < n^{-2}$ gives the formulas. On compact subsets of $\mathbb{C} \setminus (-\infty, -n^{-2}]$ their denominators and logarithm branches are uniformly controlled for $0 \leq \mathbf{q} \leq n^2$; hence they are holomorphic there. If $M_D = \binom{D+n-1}{n-1}$ is the number of compositions, then Cauchy–Schwarz and the multinomial theorem give

$$\frac{n^{2D}}{M_D} \leq c_D \leq n^{2D}.$$

Thus $\lim c_D^{1/D} = n^2$, proving the exact radii for A and g . For positive y , $yQ'(y)/Q(y) = 1 + yg'(y) = A(y) > 0$. Also

$$\log Q(y) = - \int \log(\mathbf{q}(\mathbf{w}) + y^{-1}) d\nu.$$

The limiting logarithmic integral is finite. Indeed, conditioning on $w_1, \dots, w_{n-1}$ and writing $b = \sum_{i < n} w_i$, the elementary one-variable Jensen formula gives $\int_{S^1} \log |b + w_n|^2 d\nu_n = 2 \log \max(1, |b|)$. It is between 0 and $2 \log(n-1)$. For completeness, the circle identity follows by expanding $\log(1 + w/b)$ when $|b| > 1$, and $\log(1 + b/w)$ when $|b| < 1$; all nonconstant Fourier coefficients have mean zero. At $|b| = 1$ the remaining singularity is $2 \log |2 \sin(t/2)|$, which is integrable, and gives the same limit. The positive part is bounded uniformly by $2 \log n$. Since the circle mean is nonnegative, the mean of the negative part is also at

most $2 \log n$, uniformly for $|b| \leq n - 1$. Tonelli's theorem, applied to the two nonnegative parts, now proves absolute integrability on the full torus. Fubini's theorem then computes $\int \log \mathbf{q} \, d\nu$ by first taking the circle mean in w_n and then integrating over $w_1, \dots, w_{n-1}$; since that circle mean is nonnegative, so is the integral. Monotone convergence applied after subtracting an upper bound to $\log(\mathbf{q} + y^{-1})$ proves the limit. Since also $\mathbf{q} \leq n^2$, the integral of $\log \mathbf{q}$ lies between 0 and $2 \log n$. This proves (11.1) and the endpoint behavior. Strict monotonicity and the real inverse-function theorem complete the proof. $\square$

Corollary 11.2 (All-negative-axis continuation). — *The germs $\mathcal{D}_a^\Delta$ admit a specified real analytic continuation along $0 < Q < Q_\infty$. Using these continued functions, (10.2) holds for every real $s < 0$. This assertion does not say that the original Taylor series in Q converges on the entire interval.*

Proof. — The leading coefficient (8.3) never vanishes for $y > 0$. Thus all the W_j used in Theorem 7.1 continue along the positive real axis as solutions of their homogeneous or inhomogeneous ordinary differential equations. Write $I_{X_n}^{0,\text{amb}}(y, 1) = \sum_{j=0}^n H^j J_j(y)$. The identity $\theta_y I_{X_n}^{0,\text{amb}} = H\mathcal{I}^\sharp$ says $\theta_y J_{j+1} = W_j$ for $0 \leq j \leq n - 1$. It therefore continues the J_j by integration, and hence the K_j by the exponential normalization in (6.8). Since $A > 0$, $Q'(y) > 0$; Theorem 11.1 and (6.9) give the asserted continuation of the descendants. The physical integral is analytic on $s < 0$ by Proposition 9.1. Both sides of (10.2) continue from the same germ, so the identity persists along this axis. No passage through a singularity of the differential operator is involved. $\square$

11.1. Global univalence and the Q -Taylor disk. — Corollary 11.2 leaves open where the Taylor series in Q converge. We now show that the mirror map is injective on all of $\mathbb{C} \setminus (-\infty, -n^{-2}]$ and bound the Q -Taylor radius above and below by explicit constants. Write $\mathbb{E}$ for integration against ν . Integrating the cross terms gives $\mathbb{E}\mathbf{q} = n$ and $\mathbb{E}\mathbf{q}^2 = 2n^2 - n$. By Theorem 11.1,

$$Q(y) = y \exp(-\mathbb{E} \operatorname{Log}(1 + y\mathbf{q})) \quad (y \in \mathbb{C} \setminus (-\infty, -n^{-2}]), \quad (11.2)$$

with $Q(0) = 0$ and $Q'(0) = 1$. It is convenient to use the coordinate $x = 1/y$ and to argue for a general random variable. Let $\mathbf{p}$ be a random variable with values in $[0, \varkappa]$, $\varkappa > 0$, and put

$$R_{\mathbf{p}}(x) = -\mathbb{E} \operatorname{Log}(x + \mathbf{p}), \quad \mathcal{Q}_{\mathbf{p}}(x) = e^{R_{\mathbf{p}}(x)} \quad (x \in \mathbb{C} \setminus (-\infty, 0]),$$

and

$$r_{\mathbf{p}}(u) = \exp(-\mathbb{E} \log |\mathbf{p} - u|) \in (0, \infty] \quad (0 \leq u \leq \varkappa),$$

which is positive because $\mathbb{E} \log |\mathbf{p} - u| \leq \log \varkappa$. For $\mathbf{p} = \mathbf{q}$ we have $\varkappa = n^2$, and $\mathcal{Q}_{\mathbf{q}}(1/y) = Q(y)$ on $\mathbb{C} \setminus (-\infty, -n^{-2}]$ once $\mathcal{Q}_{\mathbf{q}}$ is extended as in Theorem 11.3: both sides are holomorphic on $\mathbb{C} \setminus (-\infty, -n^{-2}]$ and agree for $y > 0$. We then drop the subscript and write $r(u) = r_{\mathbf{q}}(u)$.

Theorem 11.3 (Global univalence). — *The map $\mathcal{Q}_{\mathfrak{p}}$ extends holomorphically to $\widehat{\mathbb{C}} \setminus [-\varkappa, 0]$, with $\mathcal{Q}_{\mathfrak{p}}(\infty) = 0$, and is injective there. Its image contains the disk of radius*

$$r_{\mathfrak{p}} = \inf_{0 \leq u \leq \varkappa} r_{\mathfrak{p}}(u) \geq \frac{1}{\max\{\mathbb{E}\mathfrak{p}, \varkappa - \mathbb{E}\mathfrak{p}\}}.$$

In particular $\mathcal{Q}$ is injective on $\mathbb{C} \setminus (-\infty, -n^{-2}]$, and its image $\Omega_n = \mathcal{Q}(\mathbb{C} \setminus (-\infty, -n^{-2}])$ contains the disk $|Q| < r_n$, where

$$r_n = \inf_{0 \leq u \leq n^2} \exp(-\mathbb{E} \log |\mathfrak{q} - u|) \geq \frac{1}{n(n-1)}. \quad (11.3)$$

Proof. — Write $R = R_{\mathfrak{p}}$ and $\mathcal{Q} = \mathcal{Q}_{\mathfrak{p}}$. On the upper half-plane, $-\pi < \operatorname{Im} R(x) < 0$ and

$$\operatorname{Im} R'(x) = \operatorname{Im} x \mathbb{E}|x + \mathfrak{p}|^{-2} > 0.$$

For distinct x_1, x_2 there, $(R(x_2) - R(x_1))/(x_2 - x_1) = \int_0^1 R'(x_1 + t(x_2 - x_1)) dt$ has positive imaginary part, so R is injective; this is the Noshiro–Warschawski criterion [48, Theorem 2.16] applied to $-iR$. The exponential is injective on the strip $-\pi < \operatorname{Im} R < 0$, so $\mathcal{Q}$ is injective on the upper half-plane, with values in the open lower half-plane. Since $\mathcal{Q}(\bar{x}) = \overline{\mathcal{Q}(x)}$, the same holds on the lower half-plane, with values in the open upper half-plane. For $x > 0$, $\mathcal{Q}(x) > 0$ and $\mathcal{Q}'(x) = -\mathcal{Q}(x) \mathbb{E}(x + \mathfrak{p})^{-1} < 0$. For $x < -\varkappa$ one has $x + \mathfrak{p} < 0$, so the boundary values of R from the two half-planes differ by exactly $2\pi i$; hence $\mathcal{Q}$ extends holomorphically across $(-\infty, -\varkappa)$, where

$$\mathcal{Q}(x) = -\exp(-\mathbb{E} \log |x + \mathfrak{p}|) < 0, \quad \mathcal{Q}'(x) = -\mathcal{Q}(x) \mathbb{E}(x + \mathfrak{p})^{-1} < 0.$$

Finally $\mathcal{Q}(x) = x^{-1} + O(x^{-2})$ as $x \rightarrow \infty$, so $\mathcal{Q}$ extends holomorphically to ∞ , with $\mathcal{Q}(\infty) = 0$ and nonzero derivative in $1/x$. The images of the two half-planes, of $(0, \infty)$, of $(-\infty, -\varkappa)$ and of ∞ are pairwise disjoint, and $\mathcal{Q}$ is injective on each of these pieces; hence it is injective on $\widehat{\mathbb{C}} \setminus [-\varkappa, 0]$.

For the image, let $x_k \rightarrow -u$ with $u \in [0, \varkappa]$ and $x_k \notin [-\varkappa, 0]$. The functions $\log |x_k + \mathfrak{p}|$ are bounded above by $\log(\varkappa + \sup_k |x_k|)$ and converge pointwise to $\log |\mathfrak{p} - u| \in [-\infty, \infty)$. The reverse Fatou lemma gives $\limsup_k \mathbb{E} \log |x_k + \mathfrak{p}| \leq \mathbb{E} \log |\mathfrak{p} - u|$, that is,

$$\liminf_{k \rightarrow \infty} |\mathcal{Q}(x_k)| \geq r_{\mathfrak{p}}(u) \geq r_{\mathfrak{p}}. \quad (11.4)$$

Let w be a boundary point of the image with $|w| < r_{\mathfrak{p}}$, and choose x_k with $\mathcal{Q}(x_k) \rightarrow w$. A subsequence of (x_k) converges in $\widehat{\mathbb{C}}$. Its limit cannot lie in $\widehat{\mathbb{C}} \setminus [-\varkappa, 0]$, since then w would lie in the image, which is open; so the limit is a point $-u$ of the slit, contradicting (11.4). Thus the image, which is open, connected and contains 0, has no boundary point in the disk $|w| < r_{\mathfrak{p}}$, and therefore contains that disk (a segment from 0 to a missing point of the disk would meet the boundary). Finally, Jensen’s inequality and the convexity of $u \mapsto \mathbb{E}|\mathfrak{p} - u|$ give $\mathbb{E} \log |\mathfrak{p} - u| \leq \log \mathbb{E}|\mathfrak{p} - u| \leq \log \max\{\mathbb{E}\mathfrak{p}, \varkappa - \mathbb{E}\mathfrak{p}\}$. For $\mathfrak{p} = \mathfrak{q}$ one has $\varkappa = n^2$ and $\mathbb{E}\mathfrak{q} = n$. $\square$

The constant r_n is attained at the right endpoint if and only if the following inequality holds; we use it only as a hypothesis:

$$\mathbb{E} \log |\mathfrak{q} - u| \leq \mathbb{E} \log(n^2 - \mathfrak{q}) \quad (0 \leq u \leq n^2). \quad (11.5)$$

Theorem 11.4 (The Q -Taylor radius of the mirror map)

Let $n \geq 4$. Then $\mathbb{E}(n^2 - \mathfrak{q})^{-1} < \infty$, and the constant

$$\rho_n = \exp(-\mathbb{E} \log(n^2 - \mathfrak{q})) = r(n^2) \quad (11.6)$$

satisfies $Q(y) \rightarrow -\rho_n$ as $y \downarrow -n^{-2}$; the map Q sends $(-n^{-2}, 0)$ increasingly onto $(-\rho_n, 0)$. The Taylor radius R_y of the inverse mirror map $y(Q)$ at $Q = 0$ satisfies

$$\frac{1}{n(n-1)} \leq r_n \leq R_y \leq \rho_n \leq \frac{2}{n(n-1)} < \frac{1}{n} \leq Q_\infty. \quad (11.7)$$

If (11.5) holds, then $r_n = R_y = \rho_n$.

Proof. — Lagrange's identity gives $n^2 - \mathfrak{q} = \sum_{j < k} |w_j - w_k|^2$. Keeping only the pairs with $j = 1$ and writing $w_k = w_1 e^{i\theta_k}$ with $\theta_k \in (-\pi, \pi]$,

$$n^2 - \mathfrak{q} \geq \sum_{k=2}^n 4 \sin^2(\theta_k/2) \geq \frac{4}{\pi^2} \sum_{k=2}^n \theta_k^2.$$

The θ_k are independent and uniform, and $|\theta|^{-2}$ is integrable on $(-\pi, \pi]^{n-1}$ because $n-1 \geq 3$. Hence $\mathbb{E}(n^2 - \mathfrak{q})^{-1} < \infty$; in particular $\mathbb{E} \log(n^2 - \mathfrak{q})$ is finite.

For $-n^{-2} < y < 0$ one has $0 < 1 + y\mathfrak{q} \leq 1$, so $Q(y) < 0$ and, as in the proof of Theorem 11.1, $yQ'(y)/Q(y) = A(y) > 0$; thus $Q' > 0$. As $y \downarrow -n^{-2}$, $1 + y\mathfrak{q}$ decreases to $1 - \mathfrak{q}/n^2$, and monotone convergence gives $Q(y) \rightarrow -\rho_n$ and $A(y) \uparrow n^2 \mathbb{E}(n^2 - \mathfrak{q})^{-1}$. Hence

$$Q'(y) \longrightarrow Q'_c := n^4 \rho_n \mathbb{E}(n^2 - \mathfrak{q})^{-1} \in (0, \infty) \quad (y \downarrow -n^{-2}). \quad (11.8)$$

By Theorem 11.3 the inverse $y(Q)$ is holomorphic on $\Omega_n \supset \{|Q| < r_n\}$, so $R_y \geq r_n \geq 1/(n(n-1))$; and $r_n \leq r(n^2) = \rho_n$. The arithmetic-geometric mean inequality applied to Lagrange's identity gives

$$\mathbb{E} \log(n^2 - \mathfrak{q}) \geq \log \binom{n}{2} + \binom{n}{2}^{-1} \sum_{j < k} \mathbb{E} \log |w_j - w_k|^2 = \log \binom{n}{2},$$

because $w_j \bar{w}_k$ is uniformly distributed on the circle and $\int_{S^1} \log |1 - w| d\nu_1 = 0$. This is $\rho_n \leq 2/(n(n-1))$, which is less than $1/n$ since $n \geq 4$. Jensen's inequality gives $\mathbb{E} \log \mathfrak{q} \leq \log n$, that is, $Q_\infty \geq 1/n$.

Suppose now that $R_y > \rho_n$, and let

$$\mathcal{Y}(Q) := \sum_{m \geq 0} c_m Q^m$$

be the analytic sum of the Taylor series of $y(Q)$ on $|Q| < R_y$. It agrees with $y(Q)$ on the connected component of $\{|Q| < R_y\} \cap \Omega_n$ that contains 0, and this component contains $(-\rho_n, 0]$. By (11.8),

$$\mathcal{Y}(-\rho_n) = -n^{-2}, \quad \mathcal{Y}'(-\rho_n) = \frac{1}{Q'_c} \neq 0.$$

Hence $\mathcal{Y}$ maps a neighbourhood N of $-\rho_n$, with $0 \notin N$, biholomorphically onto a disk W centred at $-n^{-2}$. Let $G: W \rightarrow N$ be the inverse. For $y \in W \cap (-n^{-2}, 0)$ sufficiently close to $-n^{-2}$, one has $Q(y) \in N$ and

$$\mathcal{Y}(Q(y)) = y,$$

so $G(y) = Q(y)$. Since $W \setminus (-\infty, -n^{-2}]$ is connected, the identity theorem gives $G = Q$ there. Thus Q extends holomorphically to W and has no zeros there, contradicting the singularity at $y = -n^{-2}$. Pick $y_0 \in W$ with $y_0 < -n^{-2}$, put $u = -1/y_0 \in (0, n^2)$, and let $x_- = 1/(y_0 + i\varepsilon)$ and $x_+ = \overline{x_-}$. As $\varepsilon \downarrow 0$, $x_{\pm} \rightarrow -u$ from the lower and upper half-plane. The function $\mathbf{q}$ is a nonconstant real-analytic function on the connected torus, so it has no atoms, and dominated convergence ($|\operatorname{Arg}| \leq \pi$) gives $\operatorname{Im} R(x_-) \rightarrow \pi \nu(\mathbf{q} < u)$. Since $R(\bar{x}) = \overline{R(x)}$,

$$\frac{Q(y_0 + i\varepsilon)}{Q(y_0 - i\varepsilon)} = \exp(2i \operatorname{Im} R(x_-)) \rightarrow e^{2\pi i \nu(\mathbf{q} < u)},$$

whereas holomorphy of Q at y_0 , with $Q(y_0) \neq 0$, forces the limit to be 1. Since $0 < \nu(\mathbf{q} < u) < 1$, and $\mathbf{q}$ is continuous and takes the values 0 and n^2 on the connected torus, both $\{\mathbf{q} < u\}$ and $\{\mathbf{q} > u\}$ are nonempty open sets. This contradiction proves $R_y \leq \rho_n$. Under (11.5), $r(u) \geq r(n^2) = \rho_n$ for all $u \in [0, n^2]$, so $r_n = \rho_n$, and (11.7) forces $R_y = \rho_n$. $\square$

Corollary 11.5. — *Let $n \geq 4$. For $2 \leq k \leq n$ the normalized ambient germs $K_k(y(Q))$, with K_k as in (6.8), and for $0 \leq a \leq n-2$ the diagonal descendant germs $\mathcal{D}_a^\Delta(Q)$ have Taylor radius at least r_n , hence at least $1/(n(n-1))$, and at least ρ_n if (11.5) holds. In particular, if $\mathcal{D}_a^\Delta(Q) = \sum_m d_{a,m} Q^m$, then $\limsup_m |d_{a,m}|^{1/m} \leq 1/r_n \leq n(n-1)$.*

Proof. — By (8.3) and Theorem 8.2, the finite nonzero singular points of the ambient equation are among $y = -1/(n-2j)^2$, $0 \leq j < n/2$, all on the removed ray $(-\infty, -n^{-2}]$. The functions $K_k(y)$ are convergent holomorphic germs at $y = 0$, so their continuations to $\mathbb{C}^* \setminus (-\infty, -n^{-2}]$ have trivial monodromy around 0, which generates $\pi_1(\mathbb{C}^* \setminus (-\infty, -n^{-2}])$; hence they are single-valued and holomorphic on $\mathbb{C} \setminus (-\infty, -n^{-2}]$. (Alternatively, the logarithmic monodromy cancels directly in the normalized ambient series.) Composing with the holomorphic inverse $y(Q): \Omega_n \rightarrow \mathbb{C} \setminus (-\infty, -n^{-2}]$ of Theorem 11.3 shows that $K_k(y(Q))$ is holomorphic on $|Q| < r_n$, and Theorem 6.9 transfers this to $\mathcal{D}_a^\Delta(Q)$. The last assertion is the Cauchy–Hadamard formula. $\square$

Remark 11.6. — Condition (11.5) says that the logarithmic potential of the law of $\mathbf{q}$ on $[0, n^2]$ is maximal at the right endpoint; it is not used anywhere else in the paper. We have checked it numerically for $4 \leq n \leq 9$ by conditioning on $w_1, \dots, w_{n-1}$ and averaging over the last angle using the circle formula from the proof of Theorem 11.1. In each case, the maximum over $[0, n^2]$ is attained only at $u = n^2$, and the potential is increasing on $[n^2 - 1, n^2]$. For $n = 4, 5$, ratio tests on the Q -expansions up to order about 120 place the dominant singularity of $y(Q)$ and of the $\mathcal{D}_a^\Delta(Q)$ at $Q = -\rho_n$, the image of the conifold point $y = -n^{-2}$; accordingly the coefficients eventually alternate in sign. The table lists the constants in (11.7) together with the Euclidean momenta T_* defined by $Q(1/T_*) = 1/(n(n-1))$ and by $Q(1/T_*) = \rho_n$.

n	$1/(n(n-1))$	ρ_n	T_* for $1/(n(n-1))$	T_* for ρ_n
4	0.083333	0.089553	8.42	7.61
5	0.050000	0.051919	15.43	14.70
6	0.033333	0.034107	24.43	23.76
7	0.023810	0.024180	35.44	34.80

Since $Q(1/T)$ decreases in T , Corollary 11.5 shows that the Q -series of the descendants converge at $s = -T$ for every T larger than the value in the fourth column, whereas Theorem 9.7 is stated for $T > n^2$; by Theorem 11.4 the series of $y(Q)$ diverges for T smaller than the value in the last column. Together with $\rho_n < Q_\infty$ this makes the caveat in Corollary 11.2 precise. Finally, $\text{Var } \mathbf{q} = n(n-1)$, and expanding $\mathbb{E} \log(n^2 - \mathbf{q})$ about the mean suggests $n(n-1)\rho_n = 1 + \frac{1}{2n(n-1)} + O(n^{-3})$. The observed values $n(n-1)\rho_n = 1.0746, 1.0384, 1.0232, 1.0155$ for $n = 4, \dots, 7$ are consistent with this, so the lower bound in (11.7) appears to be asymptotically sharp.

12. Degree-one geometry

Theorem 12.1. — For each coordinate curve class e_i and all $n \geq 4$,

$$N_{0,e_i} = (n-1)(n-2), \quad N_{1,e_i} = -2(n-1)(n-2), \quad N_{a,e_i} = 0 \quad (2 \leq a \leq n-2).$$

Therefore the coefficient of Q in the diagonal descendant tuple is

$$n(n-1)(n-2)(1, -2, 0, \dots, 0).$$

Proof. — Fix i , put $Y_i = (\mathbb{P}^1)^{n-1}$, and let $h = \sum_{j \neq i} p_j$. The equation of F_n is $x_0 a_i + x_1 b_i = 0$ over Y_i . A coordinate fiber is contained in F_n exactly on $S_i = \{a_i = b_i = 0\} \subset Y_i$. At a point of S_i , if da_i, db_i were dependent, a nonzero combination $x_0 da_i + x_1 db_i$ would vanish. At that point of $\mathbb{P}^1 \times Y_i$ both the fiber and base derivatives of the equation would vanish, contradicting smoothness of F_n . Thus S_i is a smooth regular codimension-two locus of class h^2 .

The ambient one-marked degree-one moduli is $\mathbb{P}^1 \times Y_i$; the hypersurface moduli is its zero locus $\mathbb{P}^1 \times S_i$ for the induced section. The degree-one nonconstant component maps isomorphically to its fiber. A contracted tree attached to it cannot be stable with only one marking: for a nonempty subtree with $v \geq 1$ vertices and at most one marking, the total number of special points is at most $2(v-1) + 1 + 1 = 2v < 3v$. There are consequently no such boundary trees and no residual map automorphisms. This description also holds for families, since a fiberwise degree-one map to $\mathbb{P}^1$ identifies the universal domain with the trivial $\mathbb{P}^1$ -bundle, with its marking specifying a point.

The induced rank-two positive-bundle section on the smooth ambient moduli has regular zero locus $\mathbb{P}^1 \times S_i$; its virtual class is therefore fundamental. The local obstruction bundle has rank $m-1=0$, so the local virtual class is fundamental as well. On this moduli space let p be the class of the marked point in the fiber. The universal cotangent line has $\psi = -2p$, and $\text{ev}_{F_n}^* H = p + h$, with $p^2 = 0$. Since $\int_{S_i} h^{n-3} = \int_{Y_i} h^{n-1} = (n-1)!$, the primary invariant is $(n-1)!/(n-3)! = (n-1)(n-2)$, the first descendant is $-2(n-1)!/(n-3)!$, and the remaining ones vanish by $p^2 = 0$. There are n coordinate classes, giving the final assertion. $\square$

13. Degree two and degree-dependent descendant bounds

The primary calculations in this section use only the ambient virtual comparison, and do not depend on the ambientness theorem or Iritani's results in [26]. Use δ_a for the insertions of Proposition 4.1; the scalar Gamma coefficients remain γ_k , and the assembly polynomials are $\mathcal{T}_j$.

Theorem 13.1 (Degree-two primaries). — *For all $n \geq 4$ and distinct i, j ,*

$$N_{0,2e_i} = \frac{(n-1)(n-2)}{4}, \quad N_{0,e_i+e_j} = -2(n-1)(n-2).$$

Consequently, writing $N_m = \sum_{|\beta|=m} N_{0,\beta}$,

$$N_2 = -\frac{n(n-1)(n-2)(4n-5)}{4}.$$

The multiple-cover transform defined by $N_{0,\beta} = \sum_{k\alpha=\beta} b_\alpha/k^2$ gives

$$b_{2e_i} = 0, \quad b_{e_i+e_j} = -2(n-1)(n-2), \quad n_2 := \sum_{|\beta|=2} b_\beta = -n(n-2)(n-1)^2.$$

These explicit integrality statements require no general integrality theorem.

Proof. — On the ambient moduli use the bundles $E_+ = \pi_* f^* \mathcal{O}_{Y_n}(H)$ and $E_- = R^1 \pi_* f^* \mathcal{O}_{Y_n}(-H)$ of Lemma 5.3. The virtual comparison computes the invariant by integrating $e(E_+)e(E_-)\text{ev}^* \delta_0$. We compute the two possible degree types.

For $2e_i$, separate the passive factor $(\mathbb{P}^1)^{n-1}$ and write $\bar{p} = \sum_{j \neq i} p_j$. On the active unmarked moduli $M = \overline{\mathcal{M}}_{0,0}(\mathbb{P}^1, 2)$ put

$$E_+^{\text{act}} = \pi_* f^* \mathcal{O}_{\mathbb{P}^1}(1), \quad E_-^{\text{act}} = R^1 \pi_* f^* \mathcal{O}_{\mathbb{P}^1}(-1), \quad c = c_1(E_-^{\text{act}}).$$

These have ranks three and one. The line bundles on the target are $\mathcal{O}(1)$ and $\mathcal{O}(-1)$, whose pullbacks have degrees 2 and -2 . Let B be the trace-free line of the double cover recalled below (unrelated to the bundles $\mathcal{B}_{\pm}$ on Y_n), so that $f_* \mathcal{O}_C = \mathcal{O} \oplus (\mathcal{O}(-1) \otimes B)$. The projection formula gives

$$E_+^{\text{act}} = H^0(\mathbb{P}^1, \mathcal{O}(1)) \otimes \mathcal{O} \oplus B, \quad E_-^{\text{act}} = H^1(\mathbb{P}^1, \mathcal{O}(-2)) \otimes B \cong B,$$

so $E_+^{\text{act}} \cong \mathcal{O}^2 \oplus E_-^{\text{act}}$ and $c_1(E_+^{\text{act}}) = c$. After passive twisting,

$$e(E_+)e(E_-) = \bar{p}^2(\bar{p} + c)(c - \bar{p}) = c^2 \bar{p}^2 - \bar{p}^4.$$

Here and below these bundles pull back under forgetting the marking; contraction of an unstable degree-zero component preserves their cohomology.

We give the stack integral used in this computation. An unmarked degree-two map to $\mathbb{P}^1$ has no contracted components, since a contracted subtree attached to the at most two positive components cannot have three special points per vertex. It is a finite flat double cover, including the reducible boundary consisting of two degree-one components. Conversely every nonzero binary quadratic defines such a cover; a double root gives the nodal reducible cover. Trace splitting of the rank-two algebra gives $\mathcal{O} \oplus \mathcal{O}(-1) \otimes B$ over a base, with multiplication specified by a section of $\mathcal{O}(2) \otimes B^{-2}$. This description is compatible with base change (the finite morphism is flat of degree two, and the trace-free line has relative degree -1). It identifies

$$M \cong [(H^0(\mathbb{P}^1, \mathcal{O}(2)) \setminus \{0\})/\mathbb{G}_m] = \mathbb{P}(2, 2, 2).$$

The branch map to $\mathbb{P}^2$ has generic stabilizer μ_2 . If h is the pullback of its hyperplane class, the character line E_-^{act} has $c = \pm h/2$, as also follows by tensoring the trace splitting with $\mathcal{O}(-1)$ and taking H^1 . Thus $\int_M c^2 = \frac{1}{2} \frac{1}{4} = \frac{1}{8}$. The divisor equation (Appendix B) for the marking gives $\int_{\overline{\mathcal{M}}_{0,1}(\mathbb{P}^1, 2)} \text{ev}^* p_i c^2 = 2/8$. Only the term $p_i \bar{p}^{n-3}/(n-3)!$ of δ_0 contributes, since the other terms have excessive passive degree. Because $\int_{(\mathbb{P}^1)^{n-1}} \bar{p}^{n-1} = (n-1)!$, the answer is $(n-1)(n-2)/4$.

For $e_i + e_j$, the active unmarked moduli is $\mathbb{P}^3$, the complete linear system of $(1, 1)$ divisors in $Y_A = \mathbb{P}^1 \times \mathbb{P}^1$. Every geometric map has at most two positive components. A contracted unmarked subtree would need at least three external attachments by stability, so none occurs. The map is the inclusion of a smooth graph or the transverse nodal union of one fiber of each ruling. Every automorphism of such an inclusion is the identity.

For a family over S , the map $C \rightarrow Y_A \times S$ is proper and fiberwise quasi-finite, hence finite. The cokernel of $\mathcal{O}_{Y_A \times S} \rightarrow f_* \mathcal{O}_C$ vanishes on fibers; Nakayama, applied to this finite module locally, makes it zero. The map is therefore a closed immersion. Its flat finitely presented image is fiberwise an effective Cartier divisor,

and the fiberwise Cartier criterion makes it a relative effective Cartier divisor. Its equation determines a line in the fixed four-dimensional space $H^0(Y_A, \mathcal{O}(1, 1))$, giving the projective-space functor. These constructions commute with base change; the finite-presentation data can first be descended to a Noetherian base if required. This proves the stack isomorphism, including all finite stabilizers, which here are trivial.

The one-marked moduli is the universal divisor $U \subset \mathbb{P}^3 \times Y_A$ by the stable-map universal-curve theorem [32, Proposition 4.5 and Corollary 4.6]. A point at the node represents the map obtained by inserting a contracted $\mathbb{P}^1$ between the two branches and placing the marking on it. The two nodes and the marking make that component stable. Forgetting the marking contracts it. A section through a node before this insertion is not an admissible marked prestable curve. Pullback target bundles retain their cohomology under the contraction, because the contracted component has $H^0(\mathcal{O}) = \mathbb{C}$ and $H^1(\mathcal{O}) = 0$ and values match at its two nodes. Let $h = c_1(\mathcal{O}_{\mathbb{P}^3}(1))$ and $\bar{p} = \sum_{k \neq i, j} p_k$, a sum of $n - 2$ passive classes. The universal divisor sequences give

$$c(E_+^{\text{act}}) = (1 - h)^{-1}, \quad E_-^{\text{act}} = \mathcal{O}_{\mathbb{P}^3}(-1).$$

Indeed, for the positive bundle the sequence has first term $\mathcal{O}_{\mathbb{P}^3}(-1) \boxtimes \mathcal{O}$ and middle term $\mathcal{O} \boxtimes \mathcal{O}(1, 1)$; for the negative bundle the first term is $\mathcal{O}(-h) \boxtimes \mathcal{O}(-2, -2)$ and the middle term is $\mathcal{O} \boxtimes \mathcal{O}(-1, -1)$. Their direct images give precisely the two assertions. Hence

$$e(E_+)e(E_-) = -(\bar{p}^4 + 2h\bar{p}^3 + 2h^2\bar{p}^2 + 2h^3\bar{p}), \quad h^4 = 0.$$

Integrate this against $(\bar{p} + p_i + p_j)^{n-2}/(n-2)!$ on U times the passive factors, using $[U] = h + p_i + p_j$. Only two terms survive. The term $-2h^3\bar{p}$ combines with $p_i + p_j$ from $[U]$ and the linear active term of the insertion, giving $-4(n-2)$. The term $-2h^2\bar{p}^2$ combines with h from $[U]$ and the quadratic active term, giving $-2(n-2)(n-3)$. Their sum is $-2(n-1)(n-2)$, as asserted. There are n classes of the first type and $\binom{n}{2}$ of the second. Sum them, and subtract $N_{0, e_i}/4$ in class $2e_i$ to obtain the multiple-cover claims. $\square$

Theorem 13.2 (Support bound for descendants)

Let $\beta = (d_1, \dots, d_n)$ have total degree $m > 0$ and let $k = \#\{i : d_i > 0\}$. For the insertions defined here,

$$N_{a, \beta} = 0 \quad \text{if } a > 2m + k - 2.$$

In particular $N_{a, \beta} = 0$ if $a > 3m - 2$. At degree two this gives $N_{a, \beta} = 0$ for $a \geq 5$; these assertions are understood within $0 \leq a \leq n - 2$.

Proof. — Every passive coordinate is constant on the connected genus-zero domain. This is functorial over arbitrary bases since $\pi_*\mathcal{O}_C = \mathcal{O}_B$. At any geometric base point, choose an affine chart containing the constant fiber image. Properness of the universal curve makes the image of the complement closed in the base.

Removing it puts the whole family over that neighborhood in the chart, where $\pi_*\mathcal{O}_C = \mathcal{O}_B$ gives the factorization. The local factorizations glue uniquely. Here $R^1\pi_*\mathcal{O}_C = 0$ and the isomorphism $\mathcal{O}_B \rightarrow \pi_*\mathcal{O}_C$ follow by the same split-complex argument as Lemma 5.3, since each connected genus-zero fiber has $h^0 = 1$, $h^1 = 0$, and the constant section generates H^0 . Thus the ambient one-marked moduli is

$$\overline{\mathcal{M}}_{0,1}((\mathbb{P}^1)^k, \beta_{\text{act}}) \times (\mathbb{P}^1)^{n-k}.$$

Stability and the cotangent line depend only on the active map. The active moduli is a smooth proper Deligne–Mumford stack of dimension $k-3+2m+1 = 2m+k-2$, since the target is convex and the genus-zero obstruction $H^1(C, f^*T)$ vanishes. Its rational Chow groups vanish above this dimension, so $\psi^a = 0$ there for the stated range. Pull back and use the ambient virtual comparison; multiplying by its Euler classes and the insertion cannot change this zero. Finally $k \leq m$. $\square$

Proposition 13.3 (Sharp diagonal bound). — *For $m \geq 1$ and $n \geq 3m$,*

$$[Q^m]\mathcal{D}_{3m-2}^\Delta(Q) = -\frac{2^m n!}{m!(n-3m)!} \neq 0.$$

Thus $3m-2$ cannot be replaced by $2m$ as a uniform descendant cutoff. For example, $n = 9$, $m = 3$ gives $[Q^3]\mathcal{D}_7^\Delta = -483840$, although $7 > 2m$.

Proof. — Work with the local ambient I -series at $z = 1$ and remove its bare exponential $e^{H \log y}$. For a degree tuple of total degree m and support k , the local numerator has degree at most $2m$ in H . Because $p_i^2 = 0$, expanding each active denominator contributes at most one power of p_i , hence at most k additional cohomological powers. The total degree is at most $2m + k \leq 3m$. A coefficient of y^r in $e^{-Hg(y)}$ has degree at most r in H . For $r > 0$, its product with a term of degree $m - r$ has degree at most $3(m - r) + r < 3m$. Thus normalization cannot change the top coefficient. Neither can substituting $y(Q) = Q + O(Q^2)$, since lower degrees have cohomological degree strictly less than $3m$.

The top degree is possible only for tuples with m distinct entries one. For each such tuple the leading numerator is $(-1)^m H^{2m}$, and the top denominator term is $(-2)^m \prod_{i:d_i=1} p_i$. Summing over these subsets gives $2^m H^{2m} e_m(p) = 2^m H^{3m}/m!$. Apply Theorem 6.9 with $a = 3m - 2$. $\square$

14. Compactification and primary arithmetic

14.1. The compact target and its curve lattice. — Use the lines convention

$$Z = \mathbb{P}_{\text{lines}}(\mathcal{O}_{F_n} \oplus K_{F_n}), \quad p: Z \rightarrow F_n, \quad \xi = c_1(\mathcal{O}_Z(1)).$$

Let D_0 be the section from the first summand and D_∞ that from K_{F_n} ; write $i_0: F_n = D_0 \hookrightarrow Z$ and let ℓ_f be the fiber curve class.

Theorem 14.1 (Compactification and class comparison)

One has

$$Z \setminus D_\infty = X_n, \quad [D_\infty] = \xi, \quad i_0^* \xi = 0, \quad \mathcal{O}_Z(D_\infty)|_{D_\infty} = K_{F_n}^{-1}, \quad c_1(TZ) = 2\xi.$$

The class ξ is nef. For every positive β and every $\ell \geq 0$,

$$\overline{\mathcal{M}}_{0,\ell}(X_n, \beta) \cong \overline{\mathcal{M}}_{0,\ell}(Z, i_{0*}\beta)$$

with their standard obstruction theories, virtual classes, and virtual structure sheaves. Moreover

$$H_2(Z, \mathbb{Z}) = i_{0*}H_2(F_n, \mathbb{Z}) \oplus \mathbb{Z}\ell_f, \quad \xi \cdot \ell_f = 1,$$

and $k\mathfrak{b} = i_{0*}\beta$ if and only if $\mathfrak{b} = i_{0*}\alpha$ and $k\alpha = \beta$. For such classes, $\mathfrak{b}$ is effective if and only if α is effective.

Proof. — A line with nonzero first coordinate has a unique representative $(1, v)$, identifying the open complement with K_{F_n} . Projection of the tautological subline to $\mathcal{O}_{F_n}$ cuts out D_∞ , proving its divisor class and its restriction to the zero section. On the infinity section the tautological subline is K_{F_n} , giving the displayed positive normal bundle. The relative Euler sequence gives $c_1(T_{Z/F_n}) = 2\xi + p^*c_1(K_{F_n})$; adding $p^*c_1(T_{F_n})$ leaves 2ξ . The dual bundle $\mathcal{O}_{F_n} \oplus K_{F_n}^{-1}$ is globally generated, and surjects onto $\mathcal{O}_Z(1)$, so ξ is nef.

A curve not contained in D_∞ has nonnegative intersection with that effective divisor, and a nonconstant curve contained in it has strictly positive intersection because $K_{F_n}^{-1}$ is ample. In a stable map of class $i_{0*}\beta$ the total ξ -degree is zero. Every nonconstant component must therefore avoid D_∞ . A contracted component in D_∞ would connect, through contracted components, to a nonconstant component meeting it, which is impossible. Thus every geometric map lands in X_n . For families, maps with image in an open target form an open substack: the bad locus is the proper image from the universal curve of the inverse image of D_∞ . It has no geometric points by the preceding argument and hence is empty, over arbitrary bases. The open immersion of moduli stacks is therefore an isomorphism including nilpotents. Since $T_Z|_{X_n} = T_{X_n}$, the actual obstruction complexes and their maps agree; the virtual classes and the virtual structure sheaves agree too.

The projective-bundle formula and the section give the direct-sum lattice decomposition. Intersecting $k\mathfrak{b} = i_{0*}\beta$ with ξ annihilates the fiber coefficient, and projection to F_n gives $k\alpha = \beta$. For effectivity, one implication uses the zero section. Conversely push an effective cycle of class $i_{0*}\alpha$ to F_n ; its pushforward is an effective cycle of class α . This is a statement about effective cycles, not an assertion that all such classes have nonempty genus-zero stable-map moduli. $\square$

Remark 14.2 (Positivity and the scope of the arithmetic theorems)

Algebraically, $c_1(TZ) = 2\xi$ is nef. The symplectic hypothesis in Theorem 14.11 below is different, because in complex dimension n there must be no spherical

class $\mathfrak{a}$ with positive symplectic area and $3 - n \leq c_1(\mathfrak{a}) < 0$. For $n = 4$ the only forbidden value is -1 , impossible since $c_1(TZ)$ is even. For $n \geq 5$, the classes

$$\mathfrak{a}_k = i_{0*}(ke_i) - \ell_f$$

have $c_1(\mathfrak{a}_k) = -2$ and positive area for all sufficiently large k , for every fixed Kähler form on Z . They are spherical, since coordinate curves in F_n exist by Theorem 12.1, as their parameter scheme has nonzero class h^2 , and the fiber is a sphere. The image of $\pi_2(Z) \rightarrow H_2(Z, \mathbb{Z})$ is a subgroup, so their indicated difference is spherical. Thus Z fails this symplectic condition for $n \geq 5$. The existence of these topological classes does not say that they are represented by holomorphic curves for the algebraic complex structure.

14.2. A compact target with trivial canonical bundle. —

Theorem 14.3 (Calabi–Yau double cover and isolation of the local sector)

There is a smooth connected projective n -fold $\tilde{Z}$, a finite flat double cover $r : \tilde{Z} \rightarrow Z$, and disjoint divisors $F_{n,+}, F_{n,-} \cong F_n$ over D_0 , such that

$$K_{\tilde{Z}} \cong \mathcal{O}_{\tilde{Z}}, \quad N_{F_{n,\pm}/\tilde{Z}} \cong K_{F_n}.$$

Write $q = p \circ r$ and $j_{\pm} : F_n \hookrightarrow \tilde{Z}$. For every $\beta > 0$ and $\ell \geq 0$,

$$\overline{\mathcal{M}}_{0,\ell}(\tilde{Z}, j_{+*}\beta) \cong \overline{\mathcal{M}}_{0,\ell}(F_n, \beta) \cong \overline{\mathcal{M}}_{0,\ell}(X_n, \beta),$$

with the standard obstruction theories on the first and last stacks identified. The virtual classes and virtual structure sheaves agree. Every effective integral covering root of $j_{+}\beta$ is $j_{+*}\alpha$ with $k\alpha = \beta$; conversely each effective such α gives a covering root. The same assertion holds modulo torsion.*

Proof. — Write $\mathcal{O}_{F_n}(H) = K_{F_n}^{-1}$, which is very ample, and put $M = \mathcal{O}_Z(2\xi) = -K_Z$. Indeed the relative tangent determinant is $\det T_{Z/F_n} = \mathcal{O}_Z(2\xi) \otimes p^*K_{F_n}$; multiplication by $p^*\det T_{F_n}$ cancels the K_{F_n} factor. Thus $K_Z = \mathcal{O}_Z(-2\xi)$ as an actual line bundle, not merely as a rational first Chern class. In the lines convention, the two tautological projections give sections

$$u \in H^0(Z, \mathcal{O}_Z(\xi)), \quad v \in H^0(Z, \mathcal{O}_Z(\xi) \otimes p^*K_{F_n}).$$

They never vanish simultaneously; $D_{\infty} = (u = 0)$ and $D_0 = (v = 0)$.

Choose $a \in H^0(F_n, \mathcal{O}_{F_n}(4H))$ with smooth zero divisor. Existence can be seen directly, without invoking a smoothness assertion for an arbitrary semiample system. Embed F_n by the very ample $\mathcal{O}_{F_n}(4H)$ in $\mathbb{P}^N$. If the hyperplane corresponding to a section is singular at a point x , it contains the projective tangent $(n-1)$ -plane $T_x F$. The incidence variety of such pairs has dimension $n-1+(N-n) = N-1$ and is projective over the dual $\mathbb{P}^N$. Its image is therefore a proper closed subset. Hence a hyperplane outside this image has smooth section. The section is nonempty since an ample divisor on the positive-dimensional projective variety F_n has positive intersection with H^{n-2} .

The section

$$b = u^4 + p^* a v^4 \in H^0(Z, \mathcal{O}_Z(4\xi)) = H^0(Z, M^2)$$

has smooth nonempty zero divisor B disjoint from D_0 . Indeed, on $u \neq 0$ its equation is $1 + aw^4 = 0$, where $w = v/u$ is the fiber coordinate of K_{F_n} . On its zero set $aw \neq 0$, so the fiber derivative $4aw^3$ does not vanish. On $v \neq 0$ the equation in a local trivialization is $t^4 + a = 0$. If $t \neq 0$ its fiber derivative is nonzero; if $t = 0$, smoothness of $(a = 0)$ gives a nonzero base derivative. This proves smoothness everywhere, including along infinity.

Define

$$\tilde{Z} = \operatorname{Spec}_Z(\mathcal{O}_Z \oplus M^{-1}), \quad M^{-1} \otimes M^{-1} \xrightarrow{b} \mathcal{O}_Z.$$

Locally the equation is $\tau^2 = b$; hence the morphism is finite flat of degree two and the total space is projective. Away from B it is étale. On B , its equation has nonzero base differential db , so $\tilde{Z}$ is smooth. The section b has a zero of order one along a branch component, so it cannot be a square in the function field after a rational trivialization of M . The generic cover is a field extension. The algebra is locally free over the integral Z , and embeds in that field, proving that $\tilde{Z}$ is integral and thus connected.

Viewed as the hypersurface $\tau^2 - b = 0$ in $\operatorname{Tot}(M)$, its canonical bundle is

$$K_{\tilde{Z}} = r^*(K_Z \otimes M^{-1} \otimes M^2) = r^*(K_Z \otimes M) = \mathcal{O}_{\tilde{Z}}.$$

This follows from the tangent determinant sequence for the smooth hypersurface, just as in Proposition 4.1. Equivalently, if $R_b = (\tau = 0)$ is the ramification divisor, the local equation $b = \tau^2$ gives $db = 2\tau d\tau$ and hence $K_{\tilde{Z}} = r^*K_Z \otimes \mathcal{O}(R_b)$. The tautological section τ of r^*M has divisor R_b , so this again gives the trivial canonical line bundle. This computation proves an isomorphism of line bundles, not just vanishing of a rational Chern class. Along D_0 , u trivializes $\mathcal{O}_Z(\xi)$ and $b = u^4$. Consequently $r^{-1}(D_0)$ is the disjoint union $\tau = +u^2$ and $\tau = -u^2$, each isomorphic to F_n . Étaleness along both components identifies their normal bundles with $N_{D_0/Z} = K_{F_n}$. In particular

$$F_{n,+} \cdot j_{+*}\alpha = -H \cdot \alpha, \quad F_{n,+} \cdot j_{-*}\alpha = 0, \quad q_* j_{\pm*} = \operatorname{id}. \quad (14.1)$$

Now let a family of stable maps have class $j_{+*}\beta$. Composition with the finite morphism r preserves stability: exactly the same components are contracted, and a nonconstant map still has finite automorphism group. Its class on Z is $i_{0*}\beta$. Theorem 14.1, followed by zero-section factorization, shows that this composed map factors scheme-theoretically through D_0 . The original map therefore factors through $r^{-1}(D_0) = F_{n,+} \amalg F_{n,-}$ over the entire base. For a geometric fiber the connected domain lies in one component. The intersection numbers in eq. (14.1) and $H \cdot \beta > 0$ exclude $F_{n,-}$. For a family, the two alternatives define open-and-closed subsets of the base. Equivalently, the two idempotents pull back through

$\pi_* \mathcal{O}_C = \mathcal{O}_S$. Since no geometric fiber lies in $F_{n,-}$, the entire family lies in $F_{n,+}$. This proves the asserted stack isomorphism over arbitrary bases.

Along $F_{n,+}$ the differential dr is an isomorphism $j_+^* T_{\tilde{Z}} \cong i_0^* T_Z = T_{F_n} \oplus K_{F_n}$. It identifies the universal stable-map obstruction morphisms, because they are induced by the same composed universal differential. For an explicit comparison, let $\mathfrak{M}$ be the common stable-map stack, let $\mathfrak{B}$ be the same smooth prestable-curve stack used above, and let $\pi : \mathcal{C} \rightarrow \mathfrak{M}$ be the universal curve. Composition with r does not stabilize or alter this curve. The obstruction morphisms fit into the commuting diagram

$$\begin{array}{ccc} (R\pi_* f_Z^* T_Z)^\vee & \xrightarrow{\sim} & (R\pi_* f_{\tilde{Z}}^* T_{\tilde{Z}})^\vee \\ \phi_Z \downarrow & & \downarrow \phi_{\tilde{Z}} \\ \mathbb{L}_{\mathfrak{M}/\mathfrak{B}} & \xrightarrow{\text{id}} & \mathbb{L}_{\mathfrak{M}/\mathfrak{B}}. \end{array} \quad (14.2)$$

The chain rule for the universal differential, followed by the same relative-duality and trace construction on $\mathcal{C}$, proves commutativity. Global resolutions are supplied by [33, Proposition 5]. Restriction through the open immersion $X_n \subset Z$ makes the same comparison with the theory of X_n . Thus the theories agree with those of X_n , including their maps to the cotangent complex, not merely their ranks or K -classes. Consequently, the induced isomorphism of stable-map stacks identifies the perfect obstruction theories and hence the associated virtual classes and virtual structure sheaves. The standard theory of F_n , by contrast, has the excess factor described in Theorem 5.5 and Remark 5.6.

Finally suppose $k\gamma = j_{+*}\beta$ and γ is effective. Set $\alpha = q_*\gamma$; then $k\alpha = \beta$. The torsion-free splitting of $H_2(Z)$ in Theorem 14.1 gives $r_*\gamma = i_{0*}\alpha$. Represent γ by an effective cycle. Nefness of $r^*\xi$ and zero total degree force every component downstairs into D_0 , by the curve argument in that theorem. More explicitly, a complete irreducible curve of zero ξ -degree avoids D_∞ and hence lies in X_n . On its normalization, a nonconstant projection to F_n pulls K_{F_n} back to a negative-degree line bundle, whose section is zero. A curve projecting to a point cannot be complete and nonconstant inside an affine-line fiber. Thus every such curve lies in D_0 ; this argument applies to effective cycles of any geometric genus. Thus the cycle splits between $F_{n,+}$ and $F_{n,-}$; write its projected effective classes as α_+ and α_- . Equation (14.1) gives $H \cdot \alpha_+ = H \cdot \beta/k = H \cdot \alpha$. Since $\alpha = \alpha_+ + \alpha_-$, ampleness forces $\alpha_- = 0$. The cycle lies in $F_{n,+}$, so $\gamma = j_{+*}\alpha$. Conversely the inclusion preserves effectivity. The proof still works if the initial equality is modulo torsion: intersection numbers kill torsion, $H_2(Z)$ is free, and $q_*j_{+*} = \text{id}$ with free $H_2(F_n)$ gives the same conclusion. In particular there are no additional effective torsion lifts contributing to the invariants in this sector. $\square$

Lemma 14.4 (Formal realization of the local neighborhood)

The morphism r induces an isomorphism

$$\widehat{\widetilde{Z}}_{F_n,+} \cong \widehat{X}_{F_n}$$

of formal completions along the indicated divisors.

Proof. — On $u \neq 0$ write $w = v/u$ and $\eta = \tau/u^2$. The product aw^4 is a global regular function, although w is locally the coordinate of K_{F_n} . The equation of the cover is $\eta^2 = 1 + aw^4$. In the completion along the zero section the unique root with constant term $+1$ is

$$\eta = (1 + aw^4)^{1/2} = \sum_{j \geq 0} \binom{1/2}{j} (aw^4)^j.$$

This converges in the ideal-adic topology and is compatible on every infinitesimal neighborhood. Uniqueness also follows because the sum of two roots with constant term 1 is a unit. Since aw^4 is global, the expressions glue and supply the inverse to the completed morphism r . $\square$

Remark 14.5 (A compact sector, not an asserted global open embedding)

The scheme Z is an open-embedding compactification of X_n . The variety $\widetilde{Z}$ is a compact target realizing the local stable-map sector, with the formal-neighborhood identification just proved. No isomorphism of an entire Zariski-open neighborhood of $F_{n,+}$ with all of X_n is needed or asserted. The direct family argument in Theorem 14.3 already establishes the sector over arbitrary bases; the formal calculation is an additional description.

Integral roots and the Novikov convention. The local transforms below use actual integral classes in the free lattice $H_2(F_n, \mathbb{Z})$. When an external formula on $\widetilde{Z}$ is written using $H_2(\widetilde{Z}, \mathbb{Z})/\text{tors}$, its coefficient is understood as the sum of the invariants of the integral lifts. For a class in the sector and for each of its roots, Theorem 14.3 proves that there is at most one effective lift, namely $j_{+*}\alpha$; all other lifts have zero stable-map invariants. The retraction $q_*j_{+*} = \text{id}$ also shows that the divisibility index of $j_{+*}\beta$ agrees with that of β , both integrally and modulo torsion. Thus no root is gained or lost on passing to the external convention. This does not require the entire group $H_2(\widetilde{Z}, \mathbb{Z})$ to be torsion-free.

14.3. The rational primary transform. —

Theorem 14.6 (Primary transform). — *There are unique rational numbers b_β , with the zero convention for noneffective classes, such that*

$$N_{0,\beta} = \sum_{k\alpha=\beta} \frac{b_\alpha}{k^2}, \quad b_\beta = \sum_{k\alpha=\beta} \frac{\mu(k)}{k^2} N_{0,\alpha}.$$

Here k is a positive integer, α is an integral curve class, and μ is the arithmetic Mobius function. With $N_m = \sum_{|\beta|=m} N_{0,\beta}$ and $n_m = \sum_{|\beta|=m} b_\beta$, one has the formal identities

$$\mathcal{D}_0^\Delta(Q) = \sum_{m \geq 1} n_m \operatorname{Li}_2(Q^m), \quad (14.3)$$

$$N_m = \sum_{k|m} \frac{n_{m/k}}{k^2}, \quad n_m = \sum_{k|m} \frac{\mu(k)}{k^2} N_{m/k} \in \mathbb{Q}. \quad (14.4)$$

These statements alone make no integrality or enumerative assertion.

Proof. — In a nonzero nonnegative degree tuple, any integral root has order dividing its total degree. There are therefore finitely many roots. The term $k = 1$ defines b_β inductively. Substitution of the proposed inverse gives the coefficient $r^{-2} \sum_{k|r} \mu(k)$ in front of $N_{0,\beta/r}$. It is one for $r = 1$ and zero otherwise. If $r > 1$, expanding $\prod_{p|r} (1 - 1)$ proves the last assertion. This proves the inverse without an external inversion theorem. A multiple of an effective class is effective, so the zero convention is consistent. The number of tuples at a fixed total degree is finite. Summation over them and the defining series $\operatorname{Li}_2(q) = \sum_{k \geq 1} q^k/k^2$ give (14.3)–(14.4). $\square$

The two-loop case with $n = 3$ is not covered by the hypotheses used here; its relation to local mirror symmetry is the subject of Bloch–Kerr–Vanhove [17]. The comparison between the unmarked r^{-3} prepotential convention of that setting and the one-primary r^{-2} convention used throughout this paper, and the separation of the cover exponent from the genuine lower-loop restrictions on curve lattices and threshold convergence, are carried out in the companion paper [27].

14.4. An exact integral K -class and the quantum- K identity. — Put $\mathcal{P}_i = p^* \mathcal{O}_{F_n}(p_i)$ and

$$E_\delta = \sum_{|S|=n-2} \prod_{i \in S} (1 - [\mathcal{P}_i^{-1}]) \in K_{\text{alg}}^0(Z). \quad (14.5)$$

This is an integral virtual vector-bundle class, and its pullback $\tilde{E}_\delta = r^* E_\delta$ is an integral algebraic K -class on $\tilde{Z}$.

Lemma 14.7. — *Its full Chern character, not merely its leading component, is $\operatorname{ch}(E_\delta) = p^*(H^{n-2}/(n-2)!)$.*

Proof. — The square of each $p^* p_i$ is zero, so $\operatorname{ch}(1 - [\mathcal{P}_i^{-1}]) = 1 - e^{-p^* p_i} = p^* p_i$ exactly. Expansion of the products gives the assertion. $\square$

In particular $\operatorname{ch}(\tilde{E}_\delta) = q^* \delta_0$ exactly in every degree. No assertion that an arbitrary integral cohomology class has an integral inverse Chern character is used. The construction is for the distinguished insertion, not a general integral-lifting principle.

Define

$$\kappa_\alpha = \chi(\overline{\mathcal{M}}_{0,1}(Z, i_{0*}\alpha), \text{ev}^* E_\delta \otimes \mathcal{O}^{\text{vir}}) \in \mathbb{Z}.$$

The virtual structure sheaf and the requisite perfect obstruction theory are specified in [38, Section 2.3 and Proposition 1]. This is an Euler characteristic of a coherent K -class on a proper Deligne–Mumford stack, not an integral on its coarse space. In characteristic zero, its coarse-space pushforward is exact. Étale-locally, this is the invariants functor for a finite group, and the averaging idempotent splits the invariants. Coherent cohomology on the proper coarse space is finite and bounded. Thus the Euler characteristic is an integer. Equivalently, it is the true, rather than the fake, quantum- K invariant. Its definition on the smooth projective target Z does not require semipositivity. Semipositivity is used for the external identity below, which is applied on $\tilde{Z}$. Theorem 14.3 identifies that target’s invariant with κ_α , including the insertion and virtual structure sheaf.

The standing target of [49] is a smooth complex projective variety M , assumed in its Section 3 to be semipositive of complex dimension $d \geq 3$; that paper identifies its Theorem 1.1 with the Ionel–Parker result.

Theorem 14.8 (Chou, [49, Proposition 3.4]). — *In that setting let $\beta \neq 0$ be a positive integral curve class, $\gamma \in H^{2(d-K_M\beta-2)}(M)$ homogeneous, and $\Gamma = \text{ch}^{-1}(\gamma) \in K^0(M)_\mathbb{Q}$. Then*

$$\text{GV}_\beta(\gamma) = \text{QK}_\beta(\Gamma) + \delta_{K_M\beta,0} \sum_{\substack{r|\text{ind}(\beta) \\ r>1}} \mu(r)r \text{QK}_{\beta/r}(\Gamma),$$

where $\text{ind}(\beta) = \max\{r \geq 1 : \beta/r \in H_2(M, \mathbb{Z})\}$, QK is the true one-point genus-zero quantum- K invariant, and GV uses primary weight r^{-2} in the zero-Chern sector and equals GW in the positive-Chern sector.

We apply this to $\tilde{Z}$, whose canonical bundle is trivial, so that both the symplectic condition and nonnegative Chern degree on algebraic curves hold automatically; no extension to a nef-anticanonical target outside the stated semipositive scope is invoked. The source uses rational topological K -theory, and its statement alone furnishes no integral lift. Here E_δ is an integral algebraic K -class with exactly homogeneous full Chern character, the lattices of F_n and Z are free, and for $\tilde{Z}$ we use the torsion-free quotient with the sector identification of Theorem 14.3, so every contributing root is unambiguous; for lattices with torsion a root-index convention would need additional care. Theorem 14.9 applies the statement to $\tilde{Z}$ and not to Z , with exact homogeneous lift r^*E_δ .

Theorem 14.9 (Quantum- K expression). — *For every $n \geq 4$,*

$$b_\beta = \sum_{k\alpha=\beta} \mu(k)k \kappa_\alpha, \quad \kappa_\beta = \sum_{k\alpha=\beta} k b_\alpha. \quad (14.6)$$

Proof. — On $\tilde{Z}$ use the integral algebraic class r^*E_δ . Its full character is $q^*\delta_0$, with complex degree $n - 2$. Since $\tilde{Z}$ is a smooth projective variety, its underlying

space has the homotopy type of a finite CW complex, and the Chern character $\text{ch} : K_{\text{top}}^0(\tilde{Z}) \otimes \mathbb{Q} \rightarrow H^{\text{even}}(\tilde{Z}, \mathbb{Q})$ is an isomorphism. Thus the equality of characters identifies the image of r^*E_δ with the rational topological class $\text{ch}^{-1}(q^*\delta_0)$ required by Chou's Proposition 3.4. The target is smooth projective, dimension $n \geq 4$, and $c_1 = 0$; both symplectic and algebraic positivity conditions hold automatically. Thus Theorem 14.8 applies without a nef-scope extension. Theorem 14.3 identifies the moduli and actual obstruction theories for all effective roots with those on Z and X_n . Consequently their virtual structure sheaves and the insertions agree, so their true quantum- K invariants are precisely the κ_α defined above.

The retraction q_* to $H_2(F_n, \mathbb{Z}) = \mathbb{Z}^n$ shows that the divisibility index of $j_{+*}\beta$ is exactly that of β . The effective-root proof of Theorem 14.3 shows that passing to this quotient introduces no extra effective class or stable-map component in the sector at hand. Thus every root in the source formula is exactly accounted for. The GV coefficients in the cited proposition are defined by the same one-primary k^{-2} transform as Theorem 14.6. Uniqueness of that rational transform identifies them with b_β . The first equality in (14.6) is therefore the cited proposition. The convolution of $k \mapsto k$ and $k \mapsto \mu(k)k$ at r is $r \sum_{k|r} \mu(k)$, which gives the inverse equality. This also proves integrality directly from true K -theoretic Euler characteristics, independently of the symplectic integrality route. $\square$

Corollary 14.10 (Primary integrality in every loop order)

For every $n \geq 4$, $b_\beta \in \mathbb{Z}$ and $n_m \in \mathbb{Z}$.

Algebraic proof. — Theorem 14.9 writes b_β as the finite sum $\sum_{k\alpha=\beta} \mu(k)k \kappa_\alpha$ of integers, because the κ_α are true coherent Euler characteristics on proper Deligne–Mumford stacks. Finite summation at fixed H -degree gives $n_m \in \mathbb{Z}$. This proof uses the compact sector, the exact integral K -lift, and Chou's one-primary identity, but neither a mirror theorem nor an algebraic–symplectic comparison. $\square$

Theorem 14.11 (Ionel–Parker, [50, Theorem 9.2 and equations (9.9)–(9.10)])

Let (M, ω) be a closed semipositive symplectic manifold of real dimension $2d \geq 6$ and let $\gamma_1, \dots, \gamma_k \in H^{\text{even}}(M, \mathbb{Z})$ be integral primary insertions of even degree. Define $n_{\mathbf{a},0}$ in the zero-Chern sector by

$$\sum_{\mathbf{a} \neq 0, c_1(\mathbf{a})=0} \text{GW}_{\mathbf{a},0}(\gamma_1, \dots, \gamma_k) q^{\mathbf{a}} = \sum_{\mathbf{a} \neq 0, c_1(\mathbf{a})=0} n_{\mathbf{a},0}(\gamma_1, \dots, \gamma_k) \sum_{r \geq 1} r^{k-3} q^{r\mathbf{a}},$$

and put $n_{\mathbf{a},0} = \text{GW}_{\mathbf{a},0}$ when $c_1(\mathbf{a}) > 0$. Then all these $n_{\mathbf{a},0}$ are integers.

Semipositivity here excludes spherical classes $\mathbf{a}$ with $\omega(\mathbf{a}) > 0$ and $3 - d \leq c_1(\mathbf{a}) < 0$. It is a condition on a closed smooth manifold and its symplectic form, not only on effective algebraic curves, and the theorem concerns primary genus-zero invariants. Its hypothesis holds for Z when $n = 4$ and fails for $n \geq 5$, by Remark 14.2. The all-dimensional application below is instead to $\tilde{Z}$, which has $c_1(T\tilde{Z}) = 0$, real dimension $2n \geq 8$, and integral insertion $q^*\delta_0$ of complex

codimension $n - 2$, equal to the one-marked virtual dimension; these check the target, degree, and insertion hypotheses for every $n \geq 4$.

Theorem 14.12 (Siebert, [51, Theorem 0.1]). — *Let M be a complex projective manifold, $R \in H_2(M, \mathbb{Z})$, and $g, k \geq 0$. On the moduli space $C_{R,g,k}(M)$ of genus- g , k -marked stable maps in class R , the homology class associated to Behrend’s algebraic virtual fundamental class and Siebert’s symplectic virtual fundamental class coincide. In particular their numerical invariants with matching insertions coincide.*

The comparison target for the all-dimensional primary-integrality argument is the smooth projective $\tilde{Z}$, whose canonical bundle is trivial; the local-to-compact moduli and actual obstruction-theory identification is proved in Theorem 14.3, before this use. No comparison theorem for a noncompact target is presumed.

The semipositive numerical convention. The symplectic proof of Corollary 14.10 uses the semipositive perturbed-map interpretation permitted by [50, Remark 1.1]. Theorem 14.12 compares Behrend’s algebraic class with Siebert’s symplectic construction, and [52, introduction] states agreement of Siebert’s invariants with the Ruan–Tian invariants on semipositive targets. The numerical route used here is therefore

$$\mathrm{GW}^{\mathrm{Behrend}} = \mathrm{GW}^{\mathrm{Siebert}} = \mathrm{GW}^{\mathrm{Ruan-Tian}},$$

with Ionel–Parker read in its permitted semipositive convention. This specifies the comparison being used; it does not assert that every virtual-cycle framework is identified by Theorem 0.1 of [51]. The algebraic proof through Theorem 14.9 is independent of it.

The one-marked case. The construction of [52] includes the convention that the unstable moduli of marked domains is a point. Alternatively the one-primary numerical comparison reduces to three markings. For a nonzero curve class A and an ample divisor D on the compact target, the divisor equation gives $\langle \eta, D, D \rangle_{0,3,A} = (D \cdot A)^2 \langle \eta \rangle_{0,1,A}$, and the factor is nonzero for effective nonzero A , so comparison of the three-point rational invariants implies comparison of the one-point ones. Dividing by that factor compares invariants; it does not prove one-point BPS integrality, which uses the one-primary conclusion of Theorem 14.11 or the algebraic argument already given.

Symplectic proof with the semipositive convention. — Theorem 14.3 gives the smooth projective n -fold $\tilde{Z}$ with trivial canonical bundle. With any Kähler form, it is a closed symplectic $2n$ -manifold and $c_1 = 0$; hence it is semipositive in the exact sense of Theorem 14.11, in every dimension under consideration. The insertion $q^* \delta_0$ is integral and has complex codimension $n - 2$, the one-marked virtual dimension. Algebraic and symplectic invariants agree through Theorem 14.12 and the semipositive comparison convention recorded above; this is a specified numerical comparison, not an assertion equating every virtual-cycle construction.

Apply the one-primary formula of Theorem 14.11 to $j_{+*}\beta$. Theorem 14.3 identifies its invariants, insertions, and all effective covering roots with the local ones. Uniqueness of the rational transform identifies its integral coefficients with b_β . Finite summation gives n_m . This proof does not use the quantum- K identity. $\square$

Remark 14.13 (True and fake invariants, and boundary automorphisms)

Let $\mathfrak{M}_\beta = \overline{\mathcal{M}}_{0,1}(\tilde{Z}, j_{+*}\beta)$ and use the insertion $\tilde{E}_\delta$. The fake invariant is the identity-sector Todd integral

$$\kappa_\beta^{\text{fake}} := \int_{[\mathfrak{M}_\beta]^{\text{vir}}} \text{ch}(\text{ev}^* \tilde{E}_\delta) \text{Td}(T_{\mathfrak{M}_\beta}^{\text{vir}}) = N_{0,\beta}.$$

The equality follows because the full Chern character is homogeneous of exactly the virtual dimension, so only the constant Todd term contributes. True quantum K -theory instead takes coherent cohomology on the stack. The nonidentity inertia terms in the virtual Kawasaki–Riemann–Roch calculation behind Theorem 14.8 must be retained. Theorems 14.6 and 14.9 give

$$\kappa_\beta - \kappa_\beta^{\text{fake}} = \sum_{r\alpha=\beta} (r - r^{-2}) b_\alpha.$$

Primitivity of the total curve class alone does not exclude a boundary map with a multiply-covered component and nontrivial automorphisms. For a primitive class the displayed identity asserts that the total nonidentity correction vanishes, not that every nonidentity stratum is empty. Trivial stabilizers in the specific classes e_i and $e_i + e_j$ follow from the separate geometric descriptions in Sections 12–13.

Remark 14.14 (The integral ambient lattice for other insertions)

The compact-sector comparison transfers any insertion $\eta \in H^{2n-4}(F_n, \mathbb{Z})$ by $q^*\eta$. Rational ambientness, however, does not imply that every such integral class is an integral combination of the square-free monomials used in E_δ . Here is a precise lattice calculation. Poincaré duality, the integral weak-Lefschetz identification, and the universal coefficient theorem identify $H^{2n-4}(F_n, \mathbb{Z})$ with the integral dual of $H^2(F_n, \mathbb{Z})$, with coordinates $\eta \mapsto (\int_{F_n} \eta p_j)_{j=1}^n$. For $|S| = n - 2$,

$$\int_{F_n} p_S p_j = \int_{Y_n} H p_S p_j = \begin{cases} 1, & j \notin S, \\ 0, & j \in S. \end{cases}$$

Thus a monomial with complement $\{a, b\}$ gives $e_a + e_b$. These vectors generate the index-two sublattice

$$\{(v_1, \dots, v_n) \in \mathbb{Z}^n : \sum_i v_i \text{ is even}\}.$$

Indeed their sums have even parity, whereas differences $e_i - e_j$ and the vectors $2e_i$ are integer combinations of pair sums; these generate every vector of even total sum. The distinguished δ_0 is itself the sum of these monomials, so its lift is unaffected. This calculation neither rules out other integral K -lifts nor weakens the primary theorem. It only prevents extending the displayed exact algebraic

lift to arbitrary integral insertions without a further argument. The symplectic primary theorem concerns integral insertions more generally; an exact algebraic quantum- K expression for another insertion must check its own Chern-character lift. Corollary 18.2 below nevertheless proves integral coefficients of the full primary cover transform by pairing with integral duals of coordinate curves.

Corollary 14.15 (The highest logarithmic coefficient)

Write $a_{n-2}(m) = [Q^m]h_{n-2}(Q)$. Unconditionally,

$$a_{n-2}(m) = -mN_m = -m \sum_{k|m} \frac{n_{m/k}}{k^2}, \quad h_{n-2}(Q) = \sum_{m \geq 1} mn_m \log(1 - Q^m),$$

and

$$\exp(h_{n-2}(Q)) = \prod_{m \geq 1} (1 - Q^m)^{mn_m}.$$

Here rational powers mean their formal binomial expansions. The n_m are integral for every $n \geq 4$ by Corollary 14.10. The symbol a_{n-2} is distinct from the coefficient d_{n-2} in the alternative period normalization.

Proof. — Use $h_{n-2} = -\theta_Q \mathcal{D}_0^\Delta$ and $\theta_Q \text{Li}_2(Q^m) = -m \log(1 - Q^m)$. At any fixed degree only finitely many factors contribute. $\square$

Remark 14.16 (Interpretation for a Calabi–Yau fivefold)

For $n = 5$, $\dim_{\mathbb{C}} X_n = 5$ and the primary insertion is $H^3/6$. The exponent k^{-2} comes from one primary marking, not from the dimension of X_n . Formula (14.3) is a definitionally invertible transformation of rational invariants. Integrality adds a separate arithmetic assertion; it does not, by itself, identify the coefficients with counts of embedded curves or with contributions of individual covering loci. In particular no weight for the descendants $\psi^a \delta_a$, $a > 0$, follows from this primary transform. No genus-one or higher-genus covering formula is asserted here.

Primary integers and higher-dimensional interpretation. The b_β are integral genus-zero primary GV-type invariants for the specified insertion δ_0 . The general primary convention with ℓ insertions has cover weight $r^{\ell-3}$, so its one-marked specialization is r^{-2} ; the complex codimension of an insertion is not its marking number. The comparison with r^{-3} on the same threefold is derived in the companion paper [27].

Equation (0.2) of [30], in its introductory setting of a smooth projective complex Calabi–Yau d -fold, defines primary numbers by

$$\sum_{\beta \neq 0} N_{0,\beta}(\gamma_1, \dots, \gamma_k) q^\beta = \sum_{\beta \neq 0} n_{0,\beta}(\gamma_1, \dots, \gamma_k) \sum_{r \geq 1} \frac{q^{r\beta}}{r^{3-k}}, \quad \gamma_i \in H^*(X_n, \mathbb{Z}).$$

This is a definition and integrality prediction in that source, not an unconditional theorem about all covering loci. The fivefold genus-one calculations in that paper require its stated ideal-geometry framework; none of those calculations is an input

here. The weight for one primary insertion is r^{-2} . This convention is distinct from any assertion about descendant or genus-one covering laws.

The targets have complex dimension four at $n = 4$, five at $n = 5$, and at least six at $n \geq 6$. The ideal-curve and moving-cover constructions discussed in [30, Sections 0.5 and 1.1] have further restrictions in higher dimensions. Those restrictions are not hypotheses of the algebraic primary proof above. In particular the proof does not assume that every simple curve in the zero section is unobstructed, and does not identify the integers with unsigned embedded-curve counts, a specified sheaf invariant, or physical BPS-state multiplicities. The rational descendant description of the full sunset integral is proved independently; only its highest nonconstant logarithmic coefficient is singled out here by the primary integer transform. These restrictions are stated once in Section 1 and under the standing caveats of Section 3; they are not repeated at each occurrence.

15. Ambient evaluation classes and minimality

The following results have proofs independent of the analytic descendant assembly. The ambientness argument appears in Appendix C; the extension and Fourier-transform arguments used for minimality appear in Appendix D.

Theorem 15.1 (Ambient one-point theory in Chow groups)

For every $n \geq 4$, every smooth $F_n \in |\mathcal{O}_{Y_n}(H)|$, every positive curve class β , and every $a \geq 0$, the class defined by (5.2) satisfies

$$C_{a,\beta}^\lambda \in \text{im}(i^* : \text{CH}^*(Y_n)_\mathbb{Q} \longrightarrow \text{CH}^*(F_n)_\mathbb{Q})[\lambda].$$

Consequently every pairing with the Poincaré-orthogonal nonambient cohomology vanishes. The ambient comparison (6.3) identifies the full one-point evaluation classes in cohomology, and hence their normalized one-point generating series in cohomology.

Proof. — The universal hypersurface is a projective bundle over Y_n . Chow localization and its projective-bundle formula imply that every cycle defined over the entire smooth hypersurface family restricts to an ambient cycle on each fiber. Appendix C constructs the universal virtual evaluation class and proves its specialization by refined Gysin maps on the actual stable-map stacks. This verifies the required universal-cycle hypothesis, including fixed curve classes and nonreduced moduli spaces. Nondegeneracy of the ambient pairing in Proposition 4.2 then gives the full cohomological comparison. $\square$

Proposition 15.2 (Irreducibility and the minimal period equation)

The operator P_{n-1} of Definition 8.1 is irreducible over $\mathbb{C}(y)$ and has order $n - 1$. Consequently every nonzero scalar differential operator over $\mathbb{C}(y)$ annihilating $A(y) = \sum_{D \geq 0} (-1)^D c_D y^D$ has order at least $n - 1$, so P_{n-1} is the minimal operator of the period.

Proof. — Write $S_n = \mathcal{S}_n = \theta^{n+1} + \sum_{j \geq 1} y^j U_j(\theta)$ as in Definition 8.1. Its cyclic module on $\mathbb{G}_m$ is the n th symmetric power of the rank-two connection $\theta^2 + y = 0$. The monomial basis satisfies $\theta v_k = (n - k)v_{k+1} - kyv_{k-1}$, which gives exactly the defining recursion. The recursion makes v_0 cyclic; the leading coefficient of S_n in ordinary derivatives is y^{n+1} , a unit on $\mathbb{G}_m$, so the induced surjection from its cyclic module to the rank- $(n + 1)$ symmetric power is an isomorphism. To identify this symmetric power with the connection considered in [44], first take the rank-two Kloosterman connection Kl_2 of [44, Proposition 2.4(1)]. In their coordinate z , it is the connection associated with $(z\partial_z)^2 - z$. After the change of coordinate $z = -y$, one has $z\partial_z = y\partial_y = \theta$, so this operator becomes $\theta^2 + y$. Thus the rank-two connection occurring here is precisely the pullback of Kl_2 under $z = -y$. Consequently, the rank- $(n + 1)$ connection defined by S_n is the n -th symmetric power $\text{Sym}^n \text{Kl}_2$, after the same coordinate change. Proposition 2.7 of [44] states that $\text{Sym}^k \text{Kl}_2$ is irreducible for every $k \geq 0$. It follows that the cyclic module of S_n on $\mathbb{G}_m$ is irreducible.

For an operator $L = \theta^r + \sum_{j \geq 1} y^j p_j(\theta)$, Lemma D.1 identifies $\mathcal{D}_{\mathbb{A}^1}/\mathcal{D}_{\mathbb{A}^1}(y^{-1}L)$ as a simple extension whenever the module on $\mathbb{G}_m$ is simple. Lemma D.3 identifies its Fourier transform, after restriction, inversion of the coordinate, and the gauge change $f = yg$, with

$$\mathfrak{T}(L) = \theta^{r-1} + \sum_{j \geq 1} y^j p_j(\theta) \prod_{a=1}^{j-1} (\theta + a).$$

A positive-order output is again simple. Apply this twice to S_n . The first output has order n , constant coefficient θ^n , and nonconstant coefficients $U_j(\theta) \prod_{a=1}^{j-1} (\theta + a)$, by the bound $\deg U_j \leq n + 1 - 2j$. The second output is exactly P_{n-1} and has order $n - 1$. It is therefore irreducible over $\mathbb{C}(y)$.

Theorem 8.2 gives $P_{n-1}A = 0$, and $A(0) = 1$. The minimal annihilator of A is a positive-order right divisor of P_{n-1} , so irreducibility forces its order to be $n - 1$. $\square$

This statement is logically independent of everything else in this paper, and nothing in Sections 4–17 depends on it. It is recorded only so that the order- $(n - 1)$ equation certified in Theorem 8.2 is known to be the minimal one for the period; for minimal Picard–Fuchs operators of Feynman integrals in general see [22–24, 45].

For any fixed proposed exponent, a cover transform of rational descendant data is formally invertible over $\mathbb{Q}$. That fact alone is not an integrality theorem or an enumerative interpretation. A proposed descendant law must specify an additional arithmetic or geometric assertion, with its insertion normalization and any boundary corrections; a finite check can address only that specified assertion.

16. Normalization dictionary and analytic asymptotics

16.1. The local mirror map and the physical operator. —

Remark 16.1 (Which coefficient defines the mirror map)

On the diagonal, in our convention $I = 1 + \tau/z + O(z^{-2})$,

$$I_{X_n}^{0,\lambda} = 1 + \frac{HR - \lambda g(y)}{z} + O(z^{-2}).$$

Thus R is the divisor flat coordinate; its unit companion is $-\lambda g(y)$, not $-\lambda R$. Equivalently these occur in the z^0 coefficient of zI , not of I . The identity $A = \theta_y R$ identifies the derivative of this local mirror map with the fundamental period of the quantum-Serre family. No identity of $\theta_Q K_2$ with R is asserted.

Proposition 16.2 (Change to the physical variable)

Let $\theta_s = s\partial_s$ and let M_s denote multiplication by s . Define the order- $n-1$ scalar operator

$$\mathcal{L}_{n-1} = (-1)^{n-1} P_{n-1}(-1/s, -\theta_s) \circ M_s.$$

Then $\mathcal{L}_{n-1} I_{\odot}^{(n-1)} = -n!$. Its coefficients in powers of ∂_s are polynomials in s , and its leading coefficient is

$$s^n \prod_{0 \leq j < n/2} \left(1 - \frac{(n-2j)^2}{s} \right).$$

The period $\pi(s) = -yA(y)$, $y = -1/s$, is homogeneous for this operator.

Proof. — For any f the operator is equivalently

$$(\mathcal{L}_{n-1} f)(s) = (-1)^n \left[P_{n-1}(y, \theta_y) \frac{f(-1/y)}{y} \right]_{y=-1/s}.$$

Use (9.9) and $P_{n-1}A = 0$ for the source and period. The leading coefficient follows from (8.3) and the leading term of $\theta_s^{n-1} = s^{n-1}\partial_s^{n-1} + \dots$. For polynomiality, write each nonconstant y -term in the algorithm as $y^j Q_j(\theta_y)$, where Q_j is divisible by $\prod_{r=1}^{j-1} (\theta_y + r)^2$. Composing with M_s gives a constant multiple of $s^{1-j} Q_j(-\theta_s - 1)$. The latter polynomial in θ_s vanishes at $0, \dots, j-2$. Its expansion in the falling-factorial basis $\theta_s(\theta_s - 1) \cdots (\theta_s - r + 1) = s^r \partial_s^r$ therefore starts at $r = j-1$. All powers s^{r+1-j} are nonnegative. The $j = 0$ term is already polynomial. $\square$

For a second useful period normalization put $\pi = -yA$ and define $d_j(m) = (-1)^n [Q^m] h_j(Q)$. Then

$$I_{n-1}^{\text{rat}} = \pi(s) \left[-n(-R)^{n-1} + \sum_{m \geq 1} \sum_{j=0}^{n-2} d_j(m) Q^m R^j \right], \quad d_{n-2}(m) = (-1)^{n-1} m N_m,$$

where I_{n-1}^{rat} means the γ_0 contribution, not a claim of algebraic independence of zeta values. Direct differentiation gives $dR/ds = -\pi$. This fixes the dictionary without a logarithmic branch ambiguity.

16.2. Fixing the homogeneous ambiguity in the zeta terms. — For $0 \leq r \leq n-2$ define a fully normalized homogeneous basis

$$\mathfrak{f}_r(s) = (-1)^{r+1} y W_r(y), \quad y = -1/s.$$

This definition specifies all power-series coefficients, not just a leading logarithmic term; in particular $\mathfrak{f}_0 = \pi$. From the signed Gamma formula the coefficient of $\mathfrak{f}_r$ in the Gamma correction to the rational particular solution is exactly

$$\alpha_r^{\text{can}} = -n! \gamma_{n-1-r}.$$

Indeed the corresponding physical term is $n!(-1)^r \gamma_{n-1-r} y W_r = -n! \gamma_{n-1-r} \mathfrak{f}_r$. Since $\gamma_3 = 2\zeta(3)/3$, this gives

$$\alpha_{n-4}^{\text{can}} = -\frac{2n!}{3} \zeta(3).$$

At $n = 4$ this is $-16\zeta(3)$ multiplying π ; at $n = 5$ it is $-80\zeta(3)$ multiplying $\mathfrak{f}_1$. A full basis, rather than just the leading powers of $\log y$, is needed to specify these coefficients, since lower homogeneous additions alter the answer. Equation (9.10) fixes this freedom.

16.3. The first correction at large Euclidean momentum. —

Proposition 16.3 (Uniform first correction at large Euclidean momentum)

Put $\ell_T = \log T$ and

$$\mathbf{F}_{n-1}(\ell_T) = n! [t^{n-1}] \exp(\ell_T t) \exp \left(\sum_{r \geq 1} \frac{2\zeta(2r+1)}{2r+1} t^{2r+1} \right).$$

Primes below denote derivatives in ℓ_T . Then, as $T \rightarrow +\infty$,

$$T I_{\odot}^{(n-1)}(-T) = \mathbf{F}_{n-1}(\ell_T) + \frac{\mathbf{C}_{n-1}(\ell_T)}{T} + O(T^{-2}(1 + \ell_T)^{n-1}),$$

where

$$\mathbf{C}_{n-1} = -n\mathbf{F}_{n-1} + (2n-2)\mathbf{F}'_{n-1} - (n-4)\mathbf{F}''_{n-1} - 2\mathbf{F}'''_{n-1}.$$

In particular, writing $\ell = \log T$ for these examples,

$$\mathbf{F}_3 = 4\ell^3 + 16\zeta(3),$$

$$\mathbf{C}_3 = -16\ell^3 + 72\ell^2 - 48 - 64\zeta(3),$$

$$\mathbf{F}_4 = 5\ell^4 + 80\zeta(3)\ell,$$

$$\mathbf{C}_4 = -25\ell^4 + 160\ell^3 - 60\ell^2 - 240\ell - 400\zeta(3)\ell + 640\zeta(3).$$

Proof. — The degree-one coefficient of $e^{-H \log y} \mathcal{I}^\sharp(y, 1)$ is $-(1+H)^2 \sum_i (1+p_i)^{-2} = -(1+H)^2(n-2H)$. After the involution ϵ it becomes $-n + (2n-2)H - (n-4)H^2 - 2H^3$. Extract the top coefficient against $e^{H\ell_T} \widehat{\Gamma}_{X_n}$ and use $H^j e^{H\ell_T} = \partial_{\ell_T}^j e^{H\ell_T}$. The remaining terms of the convergent logarithmic germ are bounded by a constant times $T^{-2}(1 + \ell_T)^{n-1}$ after multiplication by T . $\square$

17. Independent consistency checks

The statements below are proved in the preceding sections. The finite computations recorded here were carried out independently, in exact rational arithmetic where the objects are rational and in high-precision floating point where they are transcendental. They are verifications, not ingredients of any proof; no coefficient fit is used anywhere in this paper.

The operator and its certificate. The algorithm of Definition 8.1, implemented as in Appendix A, returns the operators of Example 8.5 for $n = 4, 5$. For $n = 6$ it returns an operator of order five whose coefficient of θ^5 is $1 + 56y + 784y^2 + 2304y^3$, in agreement with (8.3), which reads $(1 + 36y)(1 + 16y)(1 + 4y)$ for $n = 6$. The certificate (8.2) was verified in the square-free algebra $\mathcal{R}$ for $n = 4, 5, 6$. By expanding $\mathcal{I}^\sharp$ to order y^6 determines the coefficients R_D of the proof of Theorem 8.2 for $D \leq 6 - \deg_y P_{n-1}$, and each of them vanishes identically for $D > 0$, with $R_0 = H^{n-1}$.

Mirror extraction against direct geometry. Computing K_j , the multivariable mirror map, and the pairing (6.9) from the I -series alone reproduces, for $n = 4$ and $n = 5$, the invariants obtained geometrically in Sections 12–13, class by class and not only in the diagonal sum:

$$N_{0,e_i} = (n-1)(n-2), \quad N_{1,e_i} = -2(n-1)(n-2), \quad N_{a \geq 2, e_i} = 0,$$

$$N_{0,2e_i} = \frac{(n-1)(n-2)}{4}, \quad N_{0,e_i+e_j} = -2(n-1)(n-2).$$

The sharp bound of Proposition 13.3 was checked at $m = 2, n = 6$, where $[Q^2]\mathcal{D}_4^\Delta = -1440$ as predicted.

Integrality. The diagonal primaries and their transforms (14.4) were computed to $m = 8$ at $n = 4$ and to $m = 6$ at $n = 5$:

$$n_m(n=4) = 24, -72, 264, -1488, 9696, -69240, 528072, -4238880,$$

$$n_m(n=5) = 60, -240, 1400, -12480, 131220, -1529520.$$

All are integers, as Corollary 14.10 asserts, while the underlying N_m are rational numbers, with at $n = 4$ the denominators 3, 2, 25, 3, 49, 8 for $m = 3, \dots, 8$. The degree-two entries agree with $n_2 = -(n-1)^2 n(n-2)$ from Theorem 13.1.

An independent evaluation of the amplitude. The left-hand side of the identity is evaluated without any of the machinery of this paper, using the classical Bessel-moment form of the equal-mass sunset family (see [45, 53, 54] and the references there):

$$I_{\ominus}^{(n-1)}(-T) = 2^{n-1} \int_0^\infty r K_0(r)^n J_0(\sqrt{T}r) dr \quad (T \geq 0). \quad (17.1)$$

This follows from

$$K_0(r) = \frac{1}{2} \int_0^\infty e^{-x-r^2/(4x)} dx/x$$

together with the Gaussian Hankel integral

$$\int_0^\infty r J_0(kr) e^{-ar^2} dr = (2a)^{-1} e^{-k^2/(4a)}.$$

With $a = \frac{1}{4} \sum_i x_i^{-1} = 1/(4W)$ the prefactor collapses to W and the exponential to e^{-TW} , which is (9.1). The interchange of the r - and x -integrations is justified by the Fubini–Tonelli theorem: since $|J_0| \leq 1$, Tonelli’s theorem bounds the double integral of the absolute value by the finite $T = 0$ integral $I_\odot^{(n-1)}(0)$, so Fubini’s theorem applies. The truncation tail has an explicit rigorous bound. From $K_0(r) = \int_0^\infty e^{-r \cosh \tau} d\tau$ and $\cosh \tau \geq 1 + \tau^2/2$ one gets $K_0(r) \leq \sqrt{\pi/(2r)} e^{-r}$, so for $B \geq 1$ the omitted part of (17.1) over $r > B$, prefactor included, is at most

$$\frac{2^{n-1}}{n} \left(\frac{\pi}{2}\right)^{n/2} B^{1-n/2} e^{-nB}. \quad (17.2)$$

The integrand has integrable behavior $r \log^n r$ at zero, and a certified value also requires an error bound for the quadrature on $[0, B]$. Reducing the n -dimensional Schwinger integral to this one-dimensional one is what makes forty digits reachable. Apart from Remark 9.3, which is not used in any proof, nothing else in the paper uses it. The same representation was used to check (9.6) against high-precision moments for $4 \leq n \leq 7$ and $m \leq 320$. After dividing a_m by the right-hand side of (9.6) without its remainder, m^2 times the deviation from 1 is essentially constant for $20 \leq m \leq 320$ (about 1.13, 1.94, 3.00 and 4.36 for $n = 4, 5, 6, 7$), as the stated $O(m^{-2})$ remainder predicts.

The analytic identity. The representation (17.1) was evaluated at forty-digit working precision and compared numerically with the right-hand sides of (9.8) and of the assembly (10.2):

n	T	$I_\odot^{(n-1)}(-T)$	agreement
4	50	5.17515590864132484...	35 digits
4	100	4.08241320270405960...	40 digits
5	100	26.9629554067580390...	40 digits
6	200	167.462014631377349...	40 digits

The reported agreement is observed numerical agreement, not an interval-certified enclosure of the displayed digits. It uses the descendant series, the mirror map, and the $\zeta(3)$ and $\zeta(5)$ terms of (9.7) together. Deleting the degree factor $(-1)^D$ of (9.11) destroys the agreement, namely at $n = 4$, $T = 100$ the unsigned variant gives 4.1143... instead of 4.0824..., which is why Remark 9.8 treats that factor separately from the two overall signs. Finally the first correction of Proposition 16.3 was checked at $n = 4$. In particular, with $\ell = \log T$, the quantity

$TI_3(-T) - F_3(\ell)$ equals -0.6514761 at $T = 10^4$ and -0.1499680 at $T = 10^5$, against $C_3(\ell)/T = -0.6518210$ and -0.1499766 , the residuals scaling as the stated $O(T^{-2}(1 + \ell)^{n-1})$.

18. Full primary classes and integral cover transforms

For a positive curve class β , let $D_\beta = C_{0,\beta}^0 \in H^2(F_n, \mathbb{Q})$ be the proper primary evaluation class of (5.2). By Proposition 4.1, the classes $p_1, \dots, p_n$ form the natural integral coordinate basis in degree two, so write

$$D_\beta = \sum_{i=1}^n d_{\beta,i} p_i, \quad N_{0,\beta}(b) = \int_F D_\beta b, \quad b \in H^{2n-4}(F_n, \mathbb{Q}).$$

This full vector carries more information than the single scalar obtained from the distinguished insertion $\delta_0 = H^{n-2}/(n-2)!$. We use the curve-lattice coordinates of Proposition 4.1, so $\mathbf{x}^\beta := \prod_i x_i^{p_i \cdot \beta}$.

Theorem 18.1 (Dimension-uniform primary extraction)

Let $h_j = \sum_{\ell=1}^j 1/\ell$, $h_0 = 0$, and

$$g(\mathbf{x}) = \sum_{\mathbf{d} \neq 0} g_{\mathbf{d}} \mathbf{x}^{\mathbf{d}}, \quad g_{\mathbf{d}} = \frac{(-1)^{|\mathbf{d}|} |\mathbf{d}|! (|\mathbf{d}| - 1)!}{\prod_i (d_i!)^2}.$$

With $\Theta_x = \sum_i x_i \partial_{x_i}$, set

$$f_i = \frac{1}{2} g^2 + \sum_{\mathbf{d} \neq 0} g_{\mathbf{d}} \left(2h_{d_i} - 2h_{|\mathbf{d}|} + \frac{1}{|\mathbf{d}|} \right) \mathbf{x}^{\mathbf{d}}.$$

For every $n \geq 4$ and every positive curve class β ,

$$d_{\beta,i} = [\mathbf{x}^\beta] f_i e^{-|\beta|g} (1 + \Theta_x g). \quad (18.1)$$

Proof. — By Lemma 6.8 and (6.6), the z^{-2} coefficient of the divided normalized series at $\lambda = 0$ is the multivariable primary evaluation series $\sum_{\beta > 0} D_\beta \mathbf{Q}^\beta$, after restriction to F_n . It may be computed in the square-free ambient ring before that restriction. The linear part of the Euler-divided seed is $g_{\mathbf{d}} \{ (2h_{d_i} - 1/|\mathbf{d}|) H - 2 \sum_i h_{d_i} p_i \}$. Normalization by e^{-Hg} and the term $(1 - e^{-Hg})/H$ give the divisor-valued germ $\sum_i f_i p_i$.

It remains only to convert from $\mathbf{x}$ to $\mathbf{Q}$. For $Q_i = x_i e^{g(\mathbf{x})}$, multivariable coefficient extraction gives, for any scalar germ f with zero constant term,

$$[\mathbf{Q}^\beta] f(\mathbf{x}(\mathbf{Q})) = [\mathbf{x}^\beta] f(\mathbf{x}) e^{-|\beta|g(\mathbf{x})} \det \left(\frac{\partial \log Q_i}{\partial \log x_j} \right)_{i,j}.$$

The logarithmic Jacobian is a rank-one perturbation of the identity, so the matrix determinant lemma gives $\det(\partial \log Q_i / \partial \log x_j) = 1 + \Theta_x g$. Applying this componentwise to $\sum_i f_i p_i$ proves (18.1). $\square$

As a check against the direct geometry of Sections 12 and 13, formula (18.1) gives

$$D_{e_i} = H - 2p_i, \quad D_{2e_i} = \frac{H - 2p_i}{4}, \quad D_{e_i+e_j} = -2(H - p_i - p_j) \quad (i \neq j).$$

Pairing with δ_0 reproduces exactly the degree-one and degree-two primary invariants proved there. More generally, on the coordinate ray $\mathbf{x} = xe_i$ one has $g = -\log(1+x)$ and hence $Q = x/(1+x)$. The active coefficient function is

$$\frac{1}{2} \log^2(1+x) + \text{Li}_2(-x),$$

while every passive coefficient is its negative. The germ identity $\text{Li}_2(-Q/(1-Q)) = -\text{Li}_2(Q) - \frac{1}{2} \log^2(1-Q)$ therefore gives

$$D_{re_i} = \frac{H - 2p_i}{r^2} \quad (r \geq 1). \quad (18.2)$$

This all-degree ray formula is consistent with the first two displayed checks. It also gives $B_{e_i} = H - 2p_i$ and $B_{re_i} = 0$ for every $r > 1$, in agreement with the degree-two calculation.

Here and below $r \mid \beta$ means that β/r is an integral curve class. Define

$$B_\beta = \sum_{r \mid \beta} \frac{\mu(r)}{r^2} D_{\beta/r}, \quad D_\beta = \sum_{r \mid \beta} \frac{1}{r^2} B_{\beta/r}.$$

Corollary 18.2 (Integral primary vector). — *Under the hypotheses of Theorem 14.11, every coefficient of B_β in the basis $p_1, \dots, p_n$ is integral, for every $n \geq 4$. More generally, for integral even primary insertions whose total degree has the required dimension, the local genus-zero transform with ℓ markings and cover weight $r^{\ell-3}$ is integral.*

Proof. — By Theorem 12.1, for each i the class e_i is represented by a coordinate fiber $C_i \cong \mathbb{P}^1$ contained in F_n . Choose one such fiber and let $\eta_i \in H^{2n-4}(F_n, \mathbb{Z})$ be its integral Poincaré dual. Then

$$\int_F p_j \eta_i = \int_{C_i} p_j = \delta_{ij}.$$

Pull all insertions back by $q : \tilde{Z} \rightarrow F_n$. The compact target is smooth projective Calabi–Yau, and the compact-sector comparison identifies all marked virtual theories and effective covering roots. Theorem 14.11 therefore applies to the local sector. The transform with insertion η_i is precisely the i th coefficient of B_β . No exact integral inverse Chern character for η_i is assumed. $\square$

Integral ambient monomial pairings alone do not suffice, because their vectors $e_i + e_j$ generate the index-two lattice of even total coordinate sum. Integral pairwise sums allow every coefficient to be a half-integer. The integral dual curves remove

this ambiguity. This result does not prove integrality of arbitrary descendant transforms.

19. All-loop endpoint reduction

The analytic statement in this section is valid for every integer $n \geq 1$; the sunset applications in the body of the paper have $n \geq 4$. Put

$$G(H) = \exp \left(\sum_{r \geq 1} \frac{2\zeta(2r+1)}{2r+1} H^{2r+1} \right), \quad d = \left\lfloor \frac{n-1}{2} \right\rfloor. \quad (19.1)$$

The ambient primitive normalization is specified in Section 19.1.

Theorem 19.1 (All-loop endpoint reduction). — *For every integer $n \geq 1$, all ambient primitive endpoint limits $C_j(\infty)$, $0 \leq j \leq n$, exist along the positive Euclidean axis. There are uniquely determined real numbers $b_1, \dots, b_d$ such that*

$$\sum_{j=0}^n C_j(\infty) H^j = G(H) \left(1 + \sum_{j=1}^d b_j H^j \right) \pmod{H^{n+1}}. \quad (19.2)$$

The upper-half-plane contour identities used in the proof are equivalent to the vanishing of the coefficients of $H^{d+1}, \dots, H^n$ in the quotient on the right. As an affine linear system in n finite-part moments, they have rank $n - d = \lfloor n/2 \rfloor + 1$.

The rank statement is a statement about these contour equations. It does *not* assert algebraic independence, transcendence, or minimality of the actual constants b_j under all possible arithmetic relations. It does prove that adding more linear combinations of the same contour identities cannot determine the remaining parameters.

19.1. Ambient normalization and Bessel endpoint formula. — Work in the square-free algebra $\mathcal{A}_n = \mathbb{C}[p_1, \dots, p_n]/(p_1^2, \dots, p_n^2)$, with $H = \sum p_i$ and $H^{n+1} = 0$. Symmetric homogeneous elements of degree j are multiples of H^j . Set $x = 1/y$, $y > 0$, and define the germ

$$\mathcal{W}(H, y) = \sum_{j=0}^n H^j W_j(y) = e^{H \log y} \sum_{\mathbf{a} \geq 0} (-y)^{|\mathbf{a}|} \frac{\prod_{r=1}^{|\mathbf{a}|} (H+r)^2}{\prod_{i=1}^n \prod_{r=1}^{a_i} (p_i+r)^2}. \quad (19.3)$$

The primitives are normalized by

$$C_0 = 1, \quad \theta_y C_{j+1} = W_j, \quad C_{j+1}(y) - \frac{\log^{j+1} y}{(j+1)!} \longrightarrow 0 \quad (y \downarrow 0), \quad \theta_y = y \frac{d}{dy}. \quad (19.4)$$

Use the positive-axis continuation of this germ throughout. These primitives are the functions denoted J_j in the proof of Corollary 11.2, and they both satisfy the

same differential relations and the same normalization at $y = 0$. In particular $C_1 = R$, and the normalized ambient coefficients of (6.8) satisfy

$$K_k = \sum_{j=0}^k \frac{(-R)^{k-j}}{(k-j)!} C_j, \quad 2 \leq k \leq n. \quad (19.5)$$

Thus the endpoint limits below apply directly to the mirror coordinate and to the descendant coefficients used earlier in the paper.

Let γ be Euler's constant, $r_0 = 2e^{-\gamma}$, $\ell(r) = \log(r/2) + \gamma$, and

$$\mathcal{B}_j(r) = \frac{\pi^j}{j!} J_0(r)^{n-j} Y_0(r)^j, \quad M_j = \int_0^{r_0} \frac{\mathcal{B}_j(r) - (2\ell(r))^j/j!}{r} dr + \int_{r_0}^{\infty} \frac{\mathcal{B}_j(r)}{r} dr. \quad (19.6)$$

These are convergent subtracted integrals: $\mathcal{B}_j - (2\ell)^j/j! = O(r^2(1 + |\log r|)^n)$ at zero and $\mathcal{B}_j = O(r^{-n/2})$ at infinity. Changing the subtraction scale would change the constants, so it is part of the definition.

Lemma 19.2. — *With $P(H) = e^{-2\gamma H}/\Gamma(1+H)^2$, one has*

$$\mathcal{W}(H, y) = P(H)x \int_0^{\infty} r K_0(r\sqrt{x}) \sum_{j=0}^n H^j \mathcal{B}_j(r) dr, \quad (19.7)$$

$$\sum_{j=0}^n C_j(\infty) H^j = P(H) \left(1 + 2H \sum_{j=0}^{n-1} M_j H^j \right) \pmod{H^{n+1}}, \quad (19.8)$$

$$P(H) = G(H) \frac{\sin \pi H}{\pi H}. \quad (19.9)$$

In particular, $C_1(\infty) = 2M_0$ and $C_2(\infty) = 2M_1 - \zeta(2)$.

Proof. — Modulo p^2 , the Bessel defining series and its order derivative give

$$\sum_{a \geq 0} \frac{(-z)^a}{\prod_{r=1}^a (p+r)^2} = \Gamma(1+2p) z^{-p} J_{2p}(2\sqrt{z}), \quad J_{2p}(r) = J_0(r) + \pi p Y_0(r).$$

The derivative used here is DLMF 10.15.5 [47]. Represent $\Gamma(|\mathbf{a}| + 1 + H)^2$ by two Laplace integrals in u, v . The powers of yuv cancel, while $\prod_i \Gamma(1+2p_i) = e^{-2\gamma H}$. Termwise integration is legitimate for $0 < y < n^{-2}$, using a majorant with exponential part $e^{-u-v+2n\sqrt{yuv}}$ and polynomial logarithmic factors. The product of the n Bessel factors has coefficient $\mathcal{B}_j$ in front of H^j . Integrating in u at fixed uv , then setting $r = 2\sqrt{yuv}$, gives (19.7). Its exponentially convergent integral continues the germ for every $x > 0$.

For the primitive, set $w_x(r) = r\sqrt{x}K_1(r\sqrt{x})$ and let $M_j(x)$ be (19.6) with $\mathcal{B}_j$ replaced by $w_x \mathcal{B}_j$, retaining the same subtraction. Since $\theta_y w_x = xr^2 K_0(r\sqrt{x})/2$, the expression

$$P(H) \left(1 + 2H \sum_{j=0}^{n-1} M_j(x) H^j \right)$$

has the required derivative. Its additive constants follow from the Mellin identity

$$2H \int_0^\infty w_x(r) e^{2H\ell(r)} \frac{dr}{r} = e^{2\gamma H} \Gamma(1+H)^2 x^{-H}, \quad \operatorname{Re} H > 0,$$

using DLMF 10.43.19 [47]. The Laurent pole of the integral is $1/(2H)$; its finite coefficients use exactly the subtraction at r_0 . Multiplication by $P(H)$ gives y^H . The difference from the leading small- r terms is $O(y(1+|\log y|)^n)$, so (19.4) is satisfied. Finally $0 < w_x \leq 1$ and $w_x \rightarrow 1$ as $x \downarrow 0$. After subtraction, near zero a uniform dominating function is $Cr(1+|\log r|)^{n+1}$; at infinity use $|\mathcal{B}_j(r)|/r = O(r^{-1-n/2})$. Dominated convergence proves (19.8). Expansion of $\log \Gamma$ and Euler's sine product prove (19.9) and the stated first coefficients. $\square$

19.2. The contour system and its exact rank. — For $k = \lceil n/2 \rceil, \dots, n$, put

$$T_k(J, Y_n) = (J + iY_n)^k (J - iY_n)^{n-k}.$$

The finite-part contour identity needed below is

$$\operatorname{FP} \int_0^\infty \frac{T_k(J_0, Y_0) - T_k(J_0, Y_0 + 2iJ_0)}{r} dr = i \int_0^\pi T_k(1, 2i\theta/\pi) d\theta. \quad (19.10)$$

Here finite part is defined by (19.6), extended linearly; all Bessel functions on the left have argument r .

To justify (19.10), integrate over an upper half-annulus. On the negative axis $J_0(-r + i0) = J_0(r)$ and $Y_0(-r + i0) = Y_0(r) + 2iJ_0(r)$ [47]. The large-arc contribution tends to zero, because the Hankel estimates [47] bound it by $Ca^{-n/2} \exp[-(2k-n)a \sin \theta] d\theta$, where a is the radius. This includes the balanced case $2k = n$. These bounds are uniform on the upper semicircle, using continuation of the asymptotic formulas at its ends. On the clockwise small arc, replace J_0 by 1 and Y_0 by $2(\ell(\epsilon) + i\theta)/\pi$; the error tends to zero. Taking the constant coefficient in the resulting polynomial in $\ell(\epsilon)$ gives (19.10). This explains both its sign and its finite-part normalization. The numerator's Y_n^n coefficient cancels, so only $M_0, \dots, M_{n-1}$ occur.

19.2.1. The differential-functional transformation. — The rank can be computed without an n -dependent elimination. Define

$$m_j = \frac{M_j}{\pi^{j+1}}, \quad \mathfrak{f}_n(v) = \sum_{j=0}^{n-1} m_j v^j, \quad A(v) = \frac{\sin v}{v} (1 + 2v\mathfrak{f}_n(v)) \pmod{v^{n+1}}.$$

Write $A(v) = \sum_{j=0}^n a_j v^j$, with $a_0 = 1$. By Lemma 19.2, $A(v)$ is the reduced endpoint series evaluated at $H = v/\pi$; thus $b_j = \pi^j a_j$. Let $\Lambda(t^j) = j!m_j$ for $0 \leq j < n$ and $D = d/dt$. For any polynomial of degree at most $n-1$, $\Lambda(q) = [\mathfrak{f}_n(D)q](0)$.

For $p_k(t) = (1 + it)^k(1 - it)^{n-k}$, dividing (19.10) by π gives

$$\Lambda(p_k(t) - p_k(t + 2i)) = i \int_0^1 p_k(2iu) du. \quad (19.11)$$

All operator series below are finite when applied to polynomials. The definition of A gives, through the needed order,

$$\begin{aligned} \mathfrak{f}_n(D)(1 - e^{2iD}) &= \frac{DA(D)/\sin D - 1}{2D}(1 - e^{2iD}) \\ &= -ie^{iD}A(D) + \frac{e^{2iD} - 1}{2D}. \end{aligned}$$

The last term at $t = 0$ equals $i \int_0^1 p_k(2iu) du$ and cancels the right-hand side of (19.11). Consequently the entire contour system is equivalent to

$$[A(D)p_k](i) = 0, \quad [n/2] \leq k \leq n. \quad (19.12)$$

There is no division by a differential operator with a kernel here: the quotients denote their ordinary formal power series, with the removable factors canceled before evaluation.

Proof of Theorem 19.1. — At $t = i + z$,

$$p_k(i + z) = (iz)^k(2 - iz)^{n-k}.$$

It has a zero of order exactly k , and $p_k^{(k)}(i) = k! i^k 2^{n-k} \neq 0$. Starting with $k = n$, equation (19.12) forces $a_n = 0$. Descending in k forces $a_{n-1} = \dots = a_{[n/2]} = 0$. Conversely these vanishings make all equations (19.12) hold, because all their lower derivatives vanish. They impose precisely $n - d$ independent conditions; the coefficients $a_1, \dots, a_d$ are arbitrary in the solution space. The transformation from $m_0, \dots, m_{n-1}$ to $a_1, \dots, a_n$ is affine triangular with diagonal entries 2, so it preserves rank and dimension. Lemma 19.2 gives existence of the actual endpoint limits and identifies their quotient with A . This proves (19.2) and the rank statement for every n . $\square$

For the two cases used most often below, write $g_3 = 2\zeta(3)/3$ and $g_5 = 2\zeta(5)/5$. For $n = 4$, with $R_* = C_1(\infty)$,

$$(C_1, C_2, C_3, C_4)_\infty = (R_*, 0, g_3, g_3 R_*).$$

For $n = 5$, with $R_* = C_1(\infty)$ and $S_* = C_2(\infty)$,

$$(C_1, C_2, C_3, C_4, C_5)_\infty = (R_*, S_*, g_3, g_3 R_*, g_3 S_* + g_5).$$

These are direct specializations of the universal factorization (19.2), not additional assumptions on the endpoint constants.

Appendix A. A symbolic implementation of the operator algorithm

The following Python code uses only the standard library and exact rational arithmetic. This program is available from the [GitHub repository](https://github.com/pierrevanhove/Sunset-Gromov-Witten)

<https://github.com/pierrevanhove/Sunset-Gromov-Witten>.

An operator is stored as a dictionary $(j, k) \mapsto c_{jk}$ for $\sum c_{jk} y^j \theta^k$. Left multiplication by θ uses $\theta y^j = y^j(\theta + j)$; left multiplication by y increments j . The code is an implementation of a proved finite construction and is not used to guess an operator from a series.

```

1 from fractions import Fraction as Fn
2 from math import factorial, comb
3
4 # (j,k) denotes y**j theta**k, with y on the left. Signs are already in the
5 # recursion (Definition labelled def:operator), so no (-1)**j is applied at the
6 # end.
7 def add(a,b):
8     c=a.copy()
9     for key,v in b.items(): c[key]=c.get(key,Fn(0))+v
10    return {key:v for key,v in c.items() if v}
11 def scale(a,c): return {key:c*v for key,v in a.items() if c*v}
12 def theta_left(a):
13     c={}
14     for (j,k),v in a.items():
15         c=add(c,{(j,k+1):v,(j,k):j*v})
16     return c
17 def y_left(a): return {(j+1,k):v for (j,k),v in a.items()}
18 def right_theta_plus(a,r):
19     c={}
20     for (j,k),v in a.items():
21         c=add(c,{(j,k+1):v,(j,k):r*v})
22     return c
23 def sunset_operator(n):
24     T=[{(0,0):Fn(1)},{(0,1):Fn(1,n)}]
25     for k in range(1,n):
26         T.append(scale(add(theta_left(T[k]),
27                             scale(y_left(T[k-1]),k)),Fn(1,n-k)))
28     U=scale(add(theta_left(T[n]),scale(y_left(T[n-1]),n)),factorial(n))
29     # Exact normalization sanity check; the all-degree proof is in the text.
30     assert {key:v for key,v in U.items() if key[0]==0}=={(0,n+1):Fn(1)}
31     P={(0,n-1):Fn(1)}
32     for j in sorted({j for j,k in U if j}):
33         term={(j,k):v for (jj,k),v in U.items() if jj==j}
34         for r in range(1,j):
35             term=right_theta_plus(right_theta_plus(term,r),r)
36         P=add(P,term)
37     return P
38
39 print(sunset_operator(4))
40 print(sunset_operator(5))

```

Appendix B. String, dilaton and divisor identities for the twisted theory

This appendix derives the string and divisor identities used in the text, in the form needed for our twists. Nothing here is quoted; the point of the derivation is that the index twists pull back under forgetting a marking. Let V be a smooth projective variety, $\beta \neq 0$, and let the twist be an invertible multiplicative class of an index bundle pulled back under forgetting a marking, as is the case for our two index bundles. For insertions α_i and integers $a_i \geq 0$, the identities are

$$\begin{aligned} \left\langle 1, \prod_i \tau_{a_i}(\alpha_i) \right\rangle_{0,\beta}^{\text{tw}} &= \sum_{i:a_i>0} \left\langle \tau_{a_i-1}(\alpha_i) \prod_{j \neq i} \tau_{a_j}(\alpha_j) \right\rangle_{0,\beta}^{\text{tw}}, \\ \left\langle D, \prod_i \tau_{a_i}(\alpha_i) \right\rangle_{0,\beta}^{\text{tw}} &= (D \cdot \beta) \left\langle \prod_i \tau_{a_i}(\alpha_i) \right\rangle_{0,\beta}^{\text{tw}} \\ &\quad + \sum_{i:a_i>0} \left\langle \tau_{a_i-1}(D\alpha_i) \prod_{j \neq i} \tau_{a_j}(\alpha_j) \right\rangle_{0,\beta}^{\text{tw}}, \end{aligned}$$

where $D \in H^2(V, \mathbb{Q})$ is a divisor. For the forgetful universal-curve morphism p , write D_i for the boundary section where the new and i th markings lie on a contracted three-pointed twig. Then $\psi_i = p^*\psi_i + D_i$, $D_i D_j = 0$ for $i \neq j$, and $D_i^2 = -p^*\psi_i D_i$. The latter is the normal bundle of the section, the tangent rather than the cotangent line. Expansion gives $\psi_i^a = p^*\psi_i^a + D_i p^*\psi_i^{a-1}$ for $a > 0$. On D_i the two evaluation maps agree. Pushing down, $p_* D_i = 1$, $p_* 1 = 0$, and $p_* \text{ev}_{\text{new}}^* D = D \cdot \beta$. These identities give the formulas, with the virtual universal-curve pullback in place of an ordinary fundamental class. That pullback is the base-change construction of the stable-map theory of [33, 39]; the index twists pull back because contraction of a degree-zero twig preserves their derived cohomology. The calculation is therefore on the actual Deligne–Mumford stacks and includes the contracted-twig locus. The unstable degree-zero terms in J are its specified normalization $z\mathbf{1} + \tau$. Inserting arbitrarily many units exponentiates to $e^{u/z}$, and divisor insertions give the displayed exponential and Novikov change in the mirror proof.

Appendix C. Ambientness of the one-point evaluation classes

C.1. Statements and conventions. — Fix $n \geq 4$. All varieties, schemes, and stacks below are over $\mathbb{C}$. Set

$$Y_n = (\mathbb{P}^1)^n, \quad H = p_1 + \cdots + p_n, \quad F_n \in |\mathcal{O}_{Y_n}(H)| \text{ smooth}, \quad i : F_n \hookrightarrow Y_n.$$

Here p_i is the pullback of the hyperplane class from the i th factor. Adjunction gives $K_{F_n} \simeq \mathcal{O}_{F_n}(-H)$; write $X_n = \text{Tot}(K_{F_n})$. The notation $\text{CH}^*(-)_{\mathbb{Q}}$ means

Chow groups with rational coefficients, with the intersection-ring convention for smooth targets. Cohomology means rational singular cohomology unless specified otherwise.

For a positive effective curve class $\beta \in H_2(F_n, \mathbb{Z})$, let

$$m = H \cdot \beta > 0, \quad M_{F_n} = \overline{\mathcal{M}}_{0,1}(F_n, \beta).$$

This is the proper Deligne–Mumford stack of stable maps, with its usual perfect obstruction theory and virtual class. On its universal curve write π for the projection and f for the universal map. The bundle

$$\mathcal{V} = R^1\pi_* f^* \mathcal{O}_{F_n}(-H)$$

has rank $m - 1$. Let λ be the equivariant first Chern class of the weight-one fiber-scaling character. For a rank- r bundle E our convention is

$$e_\lambda(E) = \sum_{j=0}^r \lambda^{r-j} c_j(E).$$

For every integer $a \geq 0$, define the Chow-valued evaluation class

$$C_{a,\beta}^\lambda = (\mathrm{ev}_{F_n})_* (\psi^a e_\lambda(\mathcal{V}) \cap [M_{F_n}]^{\mathrm{vir}}) \in \mathrm{CH}^*(F_n)_{\mathbb{Q}}[\lambda]. \quad (\mathrm{C.1})$$

The same notation will denote its cohomological realization. All pushforwards and virtual classes are those of stacks; no coarse-space substitution is made.

Theorem C.1 (Ambientness, including a Chow strengthening)

For every $n \geq 4$, every smooth F_n as above, every positive effective β , and every $a \geq 0$,

$$C_{a,\beta}^\lambda \in \mathrm{im}(i^* : \mathrm{CH}^*(Y_n)_{\mathbb{Q}} \longrightarrow \mathrm{CH}^*(F_n)_{\mathbb{Q}})[\lambda]. \quad (\mathrm{C.2})$$

Consequently $C_{a,\beta}^\lambda \in V_{\mathrm{amb}}[\lambda]$, where $V_{\mathrm{amb}} = i^ H^*(Y_n, \mathbb{Q})$. Every pairing of this class with the Poincaré-orthogonal complement of V_{amb} vanishes coefficientwise in λ .*

The statements concern fixed curve classes and arbitrary loop number; no diagonal summation or restriction to a finite set of loop orders is imposed on the proofs.

C.2. Universal hypersurfaces and ambient Chow classes. — Let $V = H^0(Y_n, \mathcal{O}_{Y_n}(H))$, so $\dim V = 2^n$, and let $B = \mathbb{P}(V)$ parametrize one-dimensional subspaces of sections. Put $N = \dim B = 2^n - 1$. The universal hypersurface is

$$\mathcal{F} = \{(x, [s]) \in Y_n \times B : s(x) = 0\}, \quad q : \mathcal{F} \rightarrow B, \quad p : \mathcal{F} \rightarrow Y_n.$$

Let $U \subset B$ be the nonempty open set of smooth hypersurfaces and $\mathcal{F}_U = q^{-1}(U)$. Bertini’s theorem gives nonemptiness. The space $U(\mathbb{C})$ is path connected in the classical topology. Indeed the projective line through any two distinct points of U is not contained in $B \setminus U$, and meets this algebraic closed set in only finitely many points; the complement of those points in the projective line is path connected. Fundamental groups and monodromy below use this classical topology.

Lemma C.2 (Restriction of a universally defined cycle)

For every $s \in U$ and every $\alpha \in \mathrm{CH}^*(\mathcal{F}_U)_{\mathbb{Q}}$, the restriction $\alpha|_{F_{n,s}}$ belongs to $i_s^* \mathrm{CH}^*(Y_n)_{\mathbb{Q}}$.

Proof. — The evaluation map of the globally generated line bundle is surjective:

$$0 \longrightarrow \mathcal{K} \longrightarrow V \otimes \mathcal{O}_{Y_n} \longrightarrow \mathcal{O}_{Y_n}(H) \longrightarrow 0.$$

The kernel $\mathcal{K}$ is a vector bundle of rank N . The incidence variety is the bundle of lines in $\mathcal{K}$: $\mathcal{F} = \mathbb{P}_{Y_n}(\mathcal{K})$. We always use the lines convention: $\mathbb{P}_{Y_n}(\mathcal{K}) = \mathrm{Proj}_{Y_n} \mathrm{Sym}(\mathcal{K}^\vee)$. In the quotient convention used in [36, Tag 02TX], this is the projective bundle associated with $\mathcal{K}^\vee$; in both descriptions $\mathcal{O}(1)$ is the dual tautological line. In particular it is a smooth projective variety, including above the locus of singular hypersurfaces. Its relative hyperplane class is

$$\xi = c_1(\mathcal{O}_{\mathbb{P}_{Y_n}(\mathcal{K})}(1)) = q^* c_1(\mathcal{O}_B(1)).$$

The Chow localization sequence makes $\mathrm{CH}^*(\mathcal{F})_{\mathbb{Q}} \rightarrow \mathrm{CH}^*(\mathcal{F}_U)_{\mathbb{Q}}$ surjective. Choose a lift $\bar{\alpha}$ of α . By the projective-bundle formula, every such lift can be written

$$\bar{\alpha} = \sum_{j=0}^{N-1} \xi^j p^* \alpha_j, \quad \alpha_j \in \mathrm{CH}^*(Y_n)_{\mathbb{Q}}. \quad (\text{C.3})$$

These two intersection-theoretic inputs are, respectively, the usual localization sequence and the projective-bundle formula; see [55, Section 1.8 and Theorem 3.3], and [36, Tag 02TX] for the latter.

Since $q|_{F_{n,s}}$ is constant, $\xi|_{F_{n,s}} = 0$. Restricting (C.3) gives $\alpha|_{F_{n,s}} = i_s^* \alpha_0$. The restriction is a refined pullback. The fiber inclusion is a regular embedding because q is smooth over U . Thus the argument applies even when a chosen representative cycle does not meet $F_{n,s}$ properly. $\square$

Remark C.3. — The lemma concerns cycles that exist on the entire universal family over U . An arbitrary cycle on one special fiber need not have this property. This is exactly the issue that must be checked for (C.1); it is not enough to invoke permutation symmetry or to declare that a class is deformation invariant.

C.3. Constructing the universal evaluation class. — We use rational intersection theory on separated Deligne–Mumford stacks as developed by Vistoli [56], together with the normal-cone and vector-bundle constructions of Kresch [57]. In particular refined Gysin maps, localized top Chern classes, Chern class operations, and proper pushforward are stack operations. The proper evaluation morphism to a scheme need not be representable; its rational pushforward includes the stabilizer multiplicities. This is why the rational Deligne–Mumford theory, rather than just integral pushforward for representable morphisms, is specified.

C.3.1. The curve class and the bundles. — Integral weak Lefschetz for the smooth ample hypersurface [34, Corollary 7.3], with $\dim F_n = n - 1 \geq 3$, identifies

$$H_2(F_{n,s}, \mathbb{Z}) \xrightarrow{\sim} H_2(Y_n, \mathbb{Z}) = \mathbb{Z}^n.$$

Here Y_n is embedded by its Segre embedding and $F_{n,s}$ is a hyperplane section. Milnor's homology isomorphism applies in degrees less than $n - 1$, and $2 < n - 1$. Thus its hypotheses and its integral, rather than only rational, conclusion apply in the required degree. The tuple $\mathbf{d} = (d_1, \dots, d_n)$, with $d_i = p_i \cdot \beta$, therefore specifies the curve class on every fiber unambiguously. It is fixed under monodromy. Its entries are nonnegative and $\sum_i d_i = m > 0$. We work with the proper Deligne–Mumford stack

$$M_{Y_n} = \overline{\mathcal{M}}_{0,1}(Y_n, \mathbf{d}).$$

We record the required cohomology facts on connected nodal genus-zero curves. Such a curve has a tree as dual graph. A line bundle of nonnegative degree on every component has zero H^1 . To see that remove a leaf, use the normalization sequence, and use surjectivity of evaluation at the attaching point on a nonnegative-degree line bundle on $\mathbb{P}^1$. For a line bundle of nonpositive component degrees and negative total degree, every global section is zero. It vanishes on negative-degree components and then on all degree-zero components by propagation through the connected tree. Riemann–Roch on a genus-zero curve now gives constant dimensions in both applications below.

Consequently the universal curve and map over M_{Y_n} give vector bundles

$$E_+ = \pi_* f^* \mathcal{O}_{Y_n}(H), \quad \mathrm{rk} E_+ = m+1; \quad E_- = R^1 \pi_* f^* \mathcal{O}_{Y_n}(-H), \quad \mathrm{rk} E_- = m-1. \quad (\text{C.4})$$

These assertions hold over the entire moduli stack, not only on its reduced locus. To see this, work on a scheme atlas $S \rightarrow M_{Y_n}$. The pulled-back universal curve $\pi_S : C_S \rightarrow S$ is proper, flat, and finitely presented. Apply [36, Tag 0A1H, Lemma 36.30.1] with the perfect object $f_S^* \mathcal{O}_{Y_n}(\pm H)$ and the flat finitely presented sheaf $\mathcal{O}_{C_S}$. The derived pushforward is perfect and commutes with arbitrary base change. Its geometric fibers have cohomology only in degree 0 for the positive line bundle and only in degree 1 for the negative one. A perfect complex with this property is locally a finite locally free module in that single degree. In a local two-term representative, fiberwise surjectivity or injectivity of the differential makes it locally split, leaving the kernel or cokernel as a vector bundle. The ranks are $m + 1$ and $m - 1$ by Riemann–Roch.

The canonical identifications descend from the atlas and give

$$R\pi_* f^* \mathcal{O}_{Y_n}(H) \simeq E_+[0], \quad R\pi_* f^* \mathcal{O}_{Y_n}(-H) \simeq E_-[-1].$$

For every base change, including a nonreduced closed substack, the derived base-change formula therefore gives the ordinary pullback of these vector bundles in the indicated degree. In particular $E_-|_{M_{F_{n,s}}} = R^1 \pi_{s*} f_s^* \mathcal{O}_{F_{n,s}}(-H) = \mathcal{V}_s$ on the actual, possibly nonreduced, stable-map stack.

The same nonnegative-degree argument applied to $f^*T_{Y_n} = \bigoplus_i f^*\mathcal{O}_{Y_n}(2p_i)$ gives $H^1(C, f^*T_{Y_n}) = 0$. Thus M_{Y_n} is smooth because its relative deformation theory over the smooth Artin stack of prestable marked genus-zero curves is unobstructed. Its dimension is $n - 2 + 2m$. Smoothness here is smoothness of a Deligne–Mumford stack, not a claim that its coarse space is smooth.

C.3.2. The zero stack and its virtual class. — On $M_{Y_n} \times U$, the tautological section defining the universal hypersurface induces a section

$$\sigma \in H^0(M_{Y_n} \times U, \mathrm{pr}_1^* E_+ \otimes \mathrm{pr}_2^* \mathcal{O}_U(1)).$$

Write $\mathcal{M} = Z(\sigma)$. For a family of maps over any scheme, vanishing of this section is precisely vanishing of the pulled-back defining equation on the universal curve. Hence $\mathcal{M}$ is the relative stable-map stack to $\mathcal{F}_U/U$ with degree $\mathbf{d}$, as an equality of moduli problems, including nonreduced bases. The composition with $F_{n,s} \hookrightarrow Y_n$ preserves stability and automorphisms. In particular $\mathcal{M} \rightarrow U$ is proper, and there is a proper evaluation morphism

$$\mathrm{Ev} : \mathcal{M} \longrightarrow \mathcal{F}_U.$$

Put $\mathcal{A} = M_{Y_n} \times U$ and $\mathcal{E} = \mathrm{pr}_1^* E_+ \otimes \mathrm{pr}_2^* \mathcal{O}_U(1)$. The zero section $0_{\mathcal{E}} : \mathcal{A} \rightarrow \mathcal{E}$ is a regular embedding of codimension $m + 1$. Pull its refined Gysin map back by $\sigma : \mathcal{A} \rightarrow \mathcal{E}$, and denote the resulting localized map by $0_{\mathcal{E},\sigma}^!$. Define

$$[\mathcal{M}]^{\mathrm{vir}} := 0_{\mathcal{E},\sigma}^! [\mathcal{A}] \in \mathrm{CH}_{n-3+m+N}(\mathcal{M})_{\mathbb{Q}}. \quad (\text{C.5})$$

Neither σ nor $\mathcal{M} \hookrightarrow \mathcal{A}$ is assumed regular. Only the zero section of the vector bundle is regular, which is sufficient for this definition.

We now prove the specialization identity in two distinct steps.

Step 1: refined Gysin base change. Let $\iota_s : \{s\} \hookrightarrow U$ be the codimension- N regular embedding. Its bivariant refined pullback is defined on any stack over U ; we denote all these maps by $s^!$. In particular it is defined on $\mathcal{M}$, although the fiber inclusion $\mathcal{M}_s \hookrightarrow \mathcal{M}$ need not itself be regular. Commutativity of refined Gysin maps gives

$$\begin{aligned} s^! [\mathcal{M}]^{\mathrm{vir}} &= s^! 0_{\mathcal{E},\sigma}^! [\mathcal{A}] \\ &= 0_{\mathcal{E}_s,\sigma_s}^! s^! [\mathcal{A}] = 0_{E_+,\sigma_s}^! [M_{Y_n}]. \end{aligned} \quad (\text{C.6})$$

The last equality uses smoothness of $\mathcal{A} \rightarrow U$ and the fact that $\mathcal{O}_U(1)|_s$ is a constant line, so that after choosing a nonzero vector in it $\mathcal{E}_s$ is identified with E_+ . The localized class is unchanged by this choice. These are ordinary refined-Gysin compatibilities, as in [55, Theorems 6.2 and 6.4], in the rational Deligne–Mumford intersection theory specified above. No flatness of $\mathcal{M} \rightarrow U$ enters this step.

Step 2: identification with the stable-map virtual class. We apply Kim–Kresch–Pantev [58, Introduction, formula (2), and Theorem 1]. In its convex-bundle application, the refined zero-section pullback of the ambient stable-map

virtual class is the sum of the virtual classes of stable maps into the smooth zero locus, over curve classes with the prescribed ambient image. Here the smooth projective ambient variety is Y_n , the convex bundle is $\mathcal{O}_{Y_n}(H)$, and its section has smooth zero locus $F_{n,s}$. Convexity on all nodal genus-zero maps was proved above. Moreover M_{Y_n} is smooth of expected dimension, so its standard virtual class equals $[M_{Y_n}]$. The curve-class map is injective by integral weak Lefschetz, so the sum in that theorem has only the single possible class β_s . Its conclusion is

$$0_{E_+, \sigma_s}^! [M_{Y_n}] = [M_{F_{n,s}}]^{\text{vir}} \quad \text{in } \text{CH}_*(M_{F_{n,s}})_{\mathbb{Q}}. \quad (\text{C.7})$$

This is the equality on the zero stack itself, not merely an equality after pushforward to M_{Y_n} .

The obstruction-theory compatibility in this application comes from the exact normal sequence

$$0 \longrightarrow T_{F_{n,s}} \longrightarrow T_{Y_n}|_{F_{n,s}} \xrightarrow{ds} \mathcal{O}_{F_{n,s}}(H) \longrightarrow 0.$$

Derived pushforward along the universal curve and dualization give the triangle over the Artin stack of prestable marked curves. The morphism is induced by the derivative of the same defining section, so this is the compatible diagram required by the theorem, not just an identity of K -classes. One may restrict the prestable curve stack to a finite-type smooth open containing the relevant images. The stable-map stacks are finite-type Deligne–Mumford stacks with quotient stratifications, as required for the intersection-theoretic construction. The compatible-triple formalism of [39, Theorem 4.8 and Corollary 4.9] gives the same comparison.

Combining (C.6) and (C.7) proves

$$s^! [\mathcal{M}]^{\text{vir}} = [M_{F_{n,s}}]^{\text{vir}}. \quad (\text{C.8})$$

Remark C.4 (Relative and absolute obstruction theories)

For the zero stack, the relative two-term complex over U is $[\mathcal{E}^\vee|_{\mathcal{M}} \rightarrow \Omega_{\mathcal{A}/U}|_{\mathcal{M}}]$, in degrees $[-1, 0]$. Its fiber gives the zero-locus obstruction theory used in (C.7). The absolute complex instead contains $\Omega_{\mathcal{A}}$ and hence the parameter-space cotangent directions from Ω_U . Its ordinary restriction to a fiber is not the absolute obstruction theory of $M_{F_{n,s}}$. The conclusion needed here is the refined equality (C.8), proved by the two steps above.

C.3.3. Restriction and the proof of ambientness. — Pull E_- and the cotangent-line class from M_{Y_n} to $\mathcal{M}$ and form

$$\mathcal{C}_a^\lambda = \text{Ev}_*(\psi^a e_\lambda(E_-) \cap [\mathcal{M}]^{\text{vir}}) \in \text{CH}^*(\mathcal{F}_U)_{\mathbb{Q}}[\lambda]. \quad (\text{C.9})$$

The target $\mathcal{F}_U$ is smooth, so its Chow homology is identified with its codimension-graded Chow ring. The expression is a polynomial in λ of degree at most $m - 1$.

Proper pushforward commutes with refined fiber pullback. Chern classes and ψ also commute with it. More precisely, the evaluation square over ι_s is cartesian,

and $q : \mathcal{F}_U \rightarrow U$ is smooth, so $j_s : F_{n,s} \hookrightarrow \mathcal{F}_U$ is a regular embedding. Proper-pushforward compatibility identifies its Gysin pullback of (C.9) with the evaluation pushforward of $s^!$ of the capped class on $\mathcal{M}$. Applying (C.8) therefore gives, coefficientwise,

$$j_s^* \mathcal{C}_a^\lambda = C_{a,\beta_s}^\lambda. \quad (\text{C.10})$$

Lemma C.2 now proves (C.2). Applying the cycle-class map proves all cohomological assertions in Theorem C.1. This completes its proof.

Remark C.5 (Relative canonical bundles). — Adjunction on the universal family gives $K_{\mathcal{F}_U/U} = \mathcal{O}_{Y_n}(-H)|_{\mathcal{F}_U} \otimes q^* \mathcal{O}_U(1)$. We deliberately used the globally fixed line bundle $\mathcal{O}_{Y_n}(-H)$ in (C.9). On a fiber the extra factor is a constant one-dimensional vector space, and has zero Chern class; the resulting Euler classes agree with (C.1). Thus no trivialization of $\mathcal{O}_U(1)$ over the parameter space is being assumed.

Remark C.6 (A uniform lift). — Choosing one lift of each coefficient of $\mathcal{C}_a^\lambda$ to $\mathcal{F}$ in Lemma C.2 produces a polynomial $\alpha_{a,d}(\lambda) \in \text{CH}^*(Y_n)_{\mathbb{Q}}[\lambda]$ such that $i_s^* \alpha_{a,d}(\lambda) = C_{a,\beta_s}^\lambda$ for every $s \in U$. The ambient polynomial need not be unique, but the argument works simultaneously for all smooth fibers.

C.4. Monodromy and the full one-point comparison. — There is also a cohomological proof of ambientness, using the global class (C.9) and invariant cycles.

Proposition C.7. — *For every k and $s \in U$,*

$$H^k(F_{n,s}, \mathbb{Q})^{\pi_1(U,s)} = i_s^* H^k(Y_n, \mathbb{Q}).$$

Proof. — Apply the global invariant-cycle theorem to $q : \mathcal{F} \rightarrow B$. Both spaces are smooth projective varieties, B is connected, q is surjective, and its restriction over U is smooth and proper. The theorem identifies the invariant subspace with the image of $H^k(\mathcal{F}, \mathbb{Q})$ in $H^k(F_{n,s}, \mathbb{Q})$; see [59, Theorem 3.3]. The cohomological projective-bundle formula writes every class on $\mathcal{F}$ in powers of ξ with coefficients from Y_n . Restricting to $F_{n,s}$ kills every positive power of ξ . This proves the assertion. $\square$

For each coefficient of λ , the cycle class of $\mathcal{C}_a^\lambda$ is a global cohomology class on $\mathcal{F}_U$. Its restrictions to the fibers form a flat section of $R^k q_* \mathbb{Q}$. Namely, on a contractible analytic trivializing neighborhood, the fiber inclusions are homotopic, so the restricted class is constant under parallel transport. Equation (C.10) therefore makes C_{a,β_s}^λ monodromy invariant. Proposition C.7 proves its cohomological ambientness. The fixed degree tuple ensures that monodromy preserves the individual curve class, not just a sum of classes.

The global invariant-cycle theorem supplies a real step in this argument. It identifies the image on a smooth fiber of cohomology of the open family with the image from its smooth projective compactification. Restriction $H^*(\mathcal{F}) \rightarrow H^*(\mathcal{F}_U)$

need not be surjective; the cohomological argument does not use the surjectivity available for Chow groups in Lemma C.2.

For odd n , let $V_{\text{na}} \subset H^{n-1}(F_{n,s}, \mathbb{Q})$ be the Poincaré-orthogonal complement of the ambient middle part. The ambient pairing is nondegenerate, so Proposition C.7 gives $V_{\text{na}}^{\pi_1(U,s)} = 0$. This includes the 18-dimensional nonambient middle summand for $n = 5$. It does not identify that summand with the entire Lefschetz-primitive cohomology.

For clarity, nondegeneracy on the ambient part follows from weak and hard Lefschetz and Poincaré duality. If $k < n - 1$, weak Lefschetz identifies $H^k(F_n)$ with $H^k(Y_n)$, and hard Lefschetz on F_n supplies the dual ambient classes in degree $2n - 2 - k$. In middle degree, multiplication by H gives the isomorphism $H^{n-1}(Y_n) \rightarrow H^{n+1}(Y_n)$ and hence the nondegenerate form $\int_Y abH$. The odd ambient groups are zero.

Corollary C.8 (Upgrading an ambient comparison)

Suppose an ambient class $E_{a,\beta}^\lambda \in H^(Y_n)[\lambda]$ satisfies*

$$\int_{F_n} i^* b C_{a,\beta}^\lambda = \int_{F_n} i^* b i^* E_{a,\beta}^\lambda \quad \text{for all } b \in H^*(Y_n).$$

Then $C_{a,\beta}^\lambda = i^ E_{a,\beta}^\lambda$ in cohomology. Thus a comparison against ambient insertions determines the full one-point evaluation classes. It also determines every generating series formed from them, in particular the one-point J -series once its normalization is fixed.*

Proof. — The difference is ambient by Theorem C.1, and is orthogonal to the ambient space by the displayed equality. Nondegeneracy makes the difference zero. Apply this coefficientwise in λ and then in the Novikov and descendant variables. $\square$

This conclusion does not remove the need to retain an ambient lift when dividing by $\lambda - H$. In particular a top coefficient that vanishes after restriction to F_n can still contribute after division. Ambientness and that extraction issue are distinct statements.

C.5. Dependencies of the ambientness proof. — The argument above uses ordinary intersection theory and the stable-map virtual class. It uses no mirror theorem, no compact Calabi–Yau interpretation and no differential-equation input. Ambientness follows from the universal family, without assuming that a fixed hypersurface admits coordinate permutations as automorphisms. The separate differential argument for minimality has its own, independent dependencies; see Proposition 15.2.

Appendix D. Extension and Fourier-transform lemmas for minimality

Work with left modules over

$$\mathcal{D} = \mathbb{C}[t]\langle \partial_t \rangle, \quad \mathcal{D}^\times = \mathbb{C}[t, t^{-1}]\langle \partial_t \rangle.$$

Localization at t is the usual exact restriction from the affine line to $\mathbb{G}_m$. All modules in the argument are algebraic modules on these smooth curves. No statement about arbitrary schemes or stacks is used.

We first record a point-module fact in a form that makes the extension argument explicit. Let

$$\delta_0 = \mathcal{D}/\mathcal{D}t = \mathbb{C}[\partial_t]e, \quad te = 0.$$

Its basis $e_k = \partial_t^k e$ satisfies

$$te_k = -ke_{k-1}, \quad \theta e_k = -(k+1)e_k. \quad (\text{D.1})$$

More generally, a module on which t acts locally nilpotently is a direct sum of copies of δ_0 . This particular point-support statement can be proved directly because the map

$$\mathbb{C}[\partial_t] \otimes_{\mathbb{C}} \ker t \longrightarrow M, \quad \partial_t^k \otimes h \longmapsto \partial_t^k h,$$

is surjective by induction on the nilpotence order of an element. If $t^N m = 0$, subtract $(-1)^{N-1} \partial_t^{N-1}(t^{N-1}m)/(N-1)!$ to reduce that order. It is injective by applying the highest relevant power of t to a finite relation. Consequently every nonzero such module has a quotient isomorphic to δ_0 .

Lemma D.1 (The extension used in each transform)

Let $r \geq 1$ and

$$L = \theta^r + \sum_{j \geq 1} t^j p_j(\theta), \quad p_j \in \mathbb{C}[\theta].$$

Suppose $\mathcal{D}^\times/\mathcal{D}^\times L$ is nonzero and simple. Then

$$B = t^{-1}L = \partial_t \theta^{r-1} + \sum_{j \geq 1} t^{j-1} p_j(\theta) \in \mathcal{D}, \quad M = \mathcal{D}/\mathcal{D}B$$

is simple. More precisely, M has neither a nonzero submodule nor a nonzero quotient supported at $t = 0$, and restricts to $\mathcal{D}^\times/\mathcal{D}^\times L$.

Proof. — The order in which the factors are written is important. Let $v = \sum_{k=0}^N c_k e_k \in \delta_0$ with $c_N \neq 0$. The first term of Bv has coefficient

$$c_N(-N-1)^{r-1} \neq 0$$

at e_{N+1} . Each remaining term $t^{j-1} p_j(\theta)v$ has degree at most N in ∂_t . Thus B acts injectively on δ_0 , and

$$\text{Hom}_{\mathcal{D}}(M, \delta_0) = 0. \quad (\text{D.2})$$

Let $*$ denote formal transpose, such that $t^* = t$, $\partial_t^* = -\partial_t$, and $(UV)^* = V^*U^*$. Since $\theta^* = -\theta - 1$, one has

$$B^* = (-1)^r \partial_t \theta^{r-1} + \sum_{j \geq 1} t^{j-1} p_j(-\theta - j). \quad (\text{D.3})$$

The same highest-degree calculation shows that B^* acts injectively on δ_0 .

This also proves directly, without invoking holonomic duality, that multiplication by t is injective on M . Indeed, suppose $t[U] = 0$. Then $tU = VB$ for some $V \in \mathcal{D}$, whence

$$U^*t = B^*V^*.$$

Apply this equality to $e \in \delta_0$. Injectivity of B^* gives $V^*e = 0$, so $V^* = Wt$ for some $W \in \mathcal{D}$. Transposition yields $V = tW^*$. Since the Weyl algebra is a domain, cancellation in $tU = tW^*B$ gives $U = W^*B$ and $[U] = 0$.

It follows that M has no nonzero t -torsion submodule. It has no nonzero quotient supported at zero either. Indeed, such a quotient would be a nonzero module with locally nilpotent t and hence would have a quotient δ_0 , contrary to (D.2).

Finally let $N \subset M$ be a nonzero submodule. Its localization is nonzero, by the absence of t -torsion, and therefore is the entire localization of M by the assumed simplicity. Thus M/N is supported at zero, and must vanish. Hence M is simple. $\square$

Remark D.2. — In the language of holonomic modules, the lemma identifies this presentation with the intermediate extension at zero. Using L itself as a polynomial Weyl-algebra presentation, rather than $t^{-1}L$, would not justify that assertion.

We shall also use a simple localization observation. A simple module stays simple after a nonzero localization. Indeed, the kernel of its localization is a submodule and must be zero if the localization is nonzero. Any nonzero submodule of the localized module meets the original module after clearing a denominator; the intersection is a nonzero submodule and hence is the whole original module. This proves simplicity after localization, both at t and at all nonzero polynomials.

D.1. One Fourier step. — Use the Weyl-algebra isomorphism

$$\mathcal{F} : \mathbb{C}[t]\langle \partial_t \rangle \longrightarrow \mathbb{C}[x]\langle \partial_x \rangle, \quad t \longmapsto -\partial_x, \quad \partial_t \longmapsto x.$$

It preserves the commutation relation, since $[x, -\partial_x] = 1$, and transports simple modules to simple modules. This is the algebraic Fourier transform; no integral convergence or analytic contour assertion is involved.

Lemma D.3 (Explicit transformed operator). — For L as in Lemma D.1, put

$$\mathfrak{T}(L) = \theta_y^{r-1} + \sum_{j \geq 1} y^j p_j(\theta_y) \prod_{a=1}^{j-1} (\theta_y + a). \quad (\text{D.4})$$

If $\mathcal{D}^\times/\mathcal{D}^\times L$ is simple and $\mathfrak{T}(L)$ has positive differential order, then

$$\mathbb{C}[y, y^{-1}]\langle \partial_y \rangle / \mathbb{C}[y, y^{-1}]\langle \partial_y \rangle \mathfrak{T}(L)$$

is simple, as is its differential-field localization over $\mathbb{C}(y)$.

Proof. — By Lemma D.1, $M = \mathcal{D}/\mathcal{D}(t^{-1}L)$ is simple. Its Fourier transform has cyclic presentation by

$$\mathcal{F}(B) = x(-\theta_x - 1)^{r-1} + \sum_{j \geq 1} (-\partial_x)^{j-1} p_j(-\theta_x - 1). \quad (\text{D.5})$$

Restrict to $x \neq 0$ and invert the coordinate, $x = 1/y$. Then

$$\theta_x = -\theta_y, \quad -\partial_x = y\partial_y, \quad (y\theta_y)^k = y^k \prod_{a=0}^{k-1} (\theta_y + a).$$

Thus (D.5) becomes

$$C = y^{-1}(\theta_y - 1)^{r-1} + \sum_{j \geq 1} y^{j-1} \left(\prod_{a=0}^{j-2} (\theta_y + a) \right) p_j(\theta_y - 1).$$

The rational gauge change $f(y) = y g(y)$ replaces C by $y^{-1}Cy$. Multiplying on the left by the unit y then gives precisely

$$y(y^{-1}Cy) = \mathfrak{T}(L).$$

These equalities take place in the localized Weyl algebra, with all products in their stated order. In particular they identify the cyclic modules, not merely a selected formal solution.

Positive differential order implies that the final cyclic module has nonzero generic rank. Thus the restriction of $\mathcal{F}(M)$ to $x \neq 0$ was nonzero. The localization observation, coordinate inversion, and the gauge isomorphism prove simplicity. The same observation applies on passing to $\mathbb{C}(y)$. $\square$

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October 9, 2026— preprint number LAPTH-051/26

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